

COMPLETE STUDY MANUAL

PHY121

General Physics II

Electrostatics

Electric Circuits and Materials

Electricity and Circuit Analysis

Magnetism and Electromagnetic Induction

Maxwell's Equations and Electromagnetic Waves

Built from the five course slide decks and every classwork set in them. Taught from zero: each idea is explained before it is used, every worked example in the slides is reproduced and solved step by step, and every number has been recomputed independently in code rather than copied.

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ORIENTATION

How to use this manual

This manual teaches PHY121 from the ground up. It is written so that you can work through it **without the slides open beside you**. Everything the slides present is here, in the same order, but with the reasoning filled in: the steps that were spoken aloud in class rather than typed onto the slide, and the background a newcomer needs before the first equation makes sense. Physics is not a memory subject with a few formulas bolted on. It is a small number of ideas, each used over and over. The whole of this course rests on roughly a dozen of them. Once you can state those dozen in your own words, the several hundred slides collapse into something you can hold in your head.

The box system

Every box carries a labelled header and always means the same thing, so you can navigate by shape and colour instead of reading everything twice.

BOX	WHAT IT IS FOR
TEACH	One idea, explained in plain language. Never more than one idea per box.
FORMULA	A result worth boxing, with every symbol defined and its unit given.
MUST-MEMORISE	The reserved colour. Magenta appears nowhere else. If it is magenta, it goes into your memory. Each one carries a hook to make that easier.
WORKED EXAMPLE	Numbered steps with the reasoning written on every line, then the answer, then a prompt to close the book and reproduce it.
RECALL AND CHECK	A question with the answer hidden below a dashed line. Cover it, answer, then look.
TRAP	A mistake that is made every single year. Read these twice.
EXAM	Exercises, past classwork, and the practice papers.

How to actually study this

1. **Do not read passively.** Reading physics feels productive and teaches almost nothing. The RECALL boxes exist to force retrieval, which is what actually builds memory. Cover the answer before you look at it, every time. 2. **Redo every worked example on blank paper.** Not "follow along and nod" but close the manual and reproduce it. Where you get stuck is exactly the thing you did not understand, and you will not discover it any other way.

3. **Start with Part I.** If powers of ten, rearranging a formula, or resolving a vector are shaky, then nothing built on top of them will hold. Part I is short and it is the highest-value part of this book. 4. **Work the exercises in Part VI.** That is where marks are won. The lessons make the exercises make sense, but only the exercises make them stick. **A note on the numbers.** Every numerical an-

swer in this manual was recomputed independently rather than copied from a slide. Where a slide rounds differently, or where a slide contains a slip, this is stated openly in the box rather than quietly corrected. You are told what the slide says and what is actually true, so that nothing surprises you in the exam hall.

PART I

Foundations

The mathematics this course actually uses, and nothing more. If you can do everything in this part, no equation later in the book can intimidate you. Skipping this part is the single most common reason capable students lose marks.



Powers of ten, prefixes and calculator discipline

Almost every quantity in this course is either enormous or minuscule. Handling that cleanly is a skill, and it is worth more marks than any single formula.

TEACH · WHY POWERS OF TEN EXIST

The charge on an electron is 0.000000000000000000160 coulombs. Nobody can read that, let alone multiply by it. So we write it as 1.60×10^{-19} C. The rule is simply: **10^n means move the decimal point n places**. Positive n moves it right and makes the number bigger. Negative n moves it left and makes the number smaller. That is all a power of ten is: a shorthand for where the decimal point sits.

● MUST-MEMORISE · THE THREE INDEX LAWS

$$10^a \times 10^b = 10^{a+b} \quad 10^a \div 10^b = 10^{a-b} \quad (10^a)^b = 10^{ab}$$

MEMORY HOOK: multiply then add, divide then subtract, power then multiply. Every single exponent error in this course comes from breaking one of these three.

TEACH · THE PREFIXES YOU WILL MEET

A prefix is just a power of ten wearing a costume. When a question says 5 nC, it is handing you 5×10^{-9} C and hoping you notice.

PREFIX	SYMBOL	MULTIPLIER	WHERE IT TURNS UP HERE
mega	M	10^6	resistance, $M\Omega$
kilo	k	10^3	resistance, $k\Omega$
centi	c	10^{-2}	distances in cm (very common)
milli	m	10^{-3}	currents in mA, energy in mJ
micro	μ	10^{-6}	capacitance in μF , charge in μC
nano	n	10^{-9}	charge in nC
pico	p	10^{-12}	capacitance in pF

MEMORY HOOK: going down in thousands from the unit: milli, micro, nano, pico. Each step is another factor of 10^{-3} .

⚠ TRAP · CENTI IS NOT MILLI

1 cm = 10^{-2} m, not 10^{-3} m. This one line costs more marks in this course than any other. A distance of 5 cm is 0.05 m. A plate separation of 10^{-3} m is 1 mm, not 1 cm. Worse, distance is usually *squared* in this course (Coulomb's law, Gauss's law), so a factor-of-ten slip in the distance becomes a factor of one hundred in the answer.

WORKED EXAMPLE · CONVERT BEFORE YOU CALCULATE

METHOD

A capacitor plate has area $4 \times 10^{-2} \text{ m}^2$ and separation 10^{-3} m . Before touching a formula, note what these are.

- 1 Area $4 \times 10^{-2} \text{ m}^2 = 0.04 \text{ m}^2$. That is a square of side 20 cm, which is a believable plate.
Always ask whether the number is physically sensible. It catches typos.
- 2 Separation $10^{-3} \text{ m} = 1 \text{ mm}$. Believable as a thin dielectric layer.
If your separation came out as 1 metre, you converted wrongly.

Everything is already in SI. No conversion needed. Proceed.

Now redo it: a plate of area 25 cm^2 separated by 2 mm. In SI these are $25 \times 10^{-4} \text{ m}^2$ and $2 \times 10^{-3} \text{ m}$. Note that cm^2 carries a factor of $(10^{-2})^2 = 10^{-4}$, not 10^{-2} .

RECALL AND CHECK

Without a calculator: what is $(2 \times 10^{-6}) \times (3 \times 10^8)$?

ANSWER

Multiply the fronts: $2 \times 3 = 6$. Add the exponents: $-6 + 8 = 2$. So the answer is $6 \times 10^2 = 600$.

FOUNDATIONS F.2

Rearranging formulas

You will almost never be given the formula in the direction you need it. Rearranging is therefore not a side skill: it is the skill.

TEACH · THE ONE RULE

An equation is a balance. **Whatever you do to one side, do to the other.** That is the entire theory. Everything else is practice. To free a symbol, undo what is being done to it, using the opposite operation:

- ▶ It is being *multiplied*? Divide both sides.
- ▶ It is being *divided*? Multiply both sides.
- ▶ It is *added*? Subtract from both sides.
- ▶ It is *squared*? Take the square root of both sides.
- ▶ It is *under* a fraction bar? Multiply both sides by the whole denominator first.

WORKED EXAMPLE · THREE REARRANGEMENTS YOU WILL NEED CONSTANTLY

The task. Each of these three formulas is given to you in the wrong direction. Make the named symbol the subject in each case.

1. $Q = CV$. Make **C** the subject.
2. $V = IR$. Make **I** the subject.
3. $W = \frac{1}{2}CV^2$. Make **C** the subject.

Try all three on paper before reading on. These exact rearrangements appear in Modules 2, 3 and in both CBT tests.

- 1 From $Q = CV$, make C the subject. C is multiplied by V, so divide both sides by V: $C = Q / V$.
This is the definition of capacitance, used in Module 2 all the time.
- 2 From $V = IR$, make I the subject. I is multiplied by R, so divide both sides by R: $I = V / R$.
Ohm's law in the direction you actually need it in Module 3.
- 3 From $W = \frac{1}{2}CV^2$, make C the subject. Multiply both sides by 2: $2W = CV^2$. Then divide by V^2 : $C = 2W / V^2$.
Do the multiply first. Trying to divide by V^2 while the $\frac{1}{2}$ is still attached is where people go wrong.

$$C = Q/V \quad \cdot \quad I = V/R \quad \cdot \quad C = 2W/V^2$$

Now redo it: from $r = mv / (qB)$, make B the subject. Answer: $B = mv / (qr)$. You will need exactly this in Module 4.

⚠ TRAP · REARRANGE FIRST, SUBSTITUTE SECOND

Put the numbers in *last*. If you substitute first, you carry long decimals through every line, you cannot see the structure any more, and a single slip is unrecoverable. Rearrange in symbols until the unknown is alone, and only then put numbers in. It is faster and it is far easier to check.

FILL IN THE GAP · REARRANGING FORMULAS

1. To free a symbol you apply the _____ operation to *both* sides of the equation.
2. You should rearrange in _____ first, and substitute the numbers _____.
3. To make C the subject of $W = \frac{1}{2}CV^2$, you must multiply by 2 _____ you divide by V^2 .

ANSWERS

1. **opposite** (or inverse). Undo a multiplication by dividing, a square by rooting.
2. **symbols** and **last**. Substituting first drags long decimals through every line and hides the structure.
3. **before**. Trying to divide by V^2 while the $\frac{1}{2}$ is still attached is where people go wrong.

FOUNDATIONS F.3

Scalars, vectors and components

Force, electric field and magnetic field are all vectors. Charge, potential and energy are not. Knowing which is which prevents the most expensive class of error in this course.

TEACH · THE DISTINCTION

A **scalar** has size only. Charge, mass, potential, energy, capacitance and resistance are scalars. To combine them you simply add the numbers. A **vector** has size *and direction*. Force, electric field, and magnetic field are vectors. To combine them you cannot just add the numbers, because direction decides whether they reinforce or cancel. Charge is a scalar even though it carries a plus or minus sign. The sign says *what kind* of charge it is, not *which way* it points. This confuses people every year.

● MUST-MEMORISE · ADDING TWO VECTORS

If two vectors act along the **same line**, add or subtract the sizes directly. If they act at **right angles**, use Pythagoras for the size and the tangent for the angle:

$$R = \sqrt{A^2 + B^2} \quad \tan \theta = B / A$$

MEMORY HOOK: square, add, root. It is Pythagoras, because a right-angled triangle is exactly what two perpendicular vectors make.

TEACH · RESOLVING A VECTOR INTO COMPONENTS

A vector at an angle can always be replaced by two vectors at right angles, one horizontal and one vertical, that together do exactly the same job. For a vector of size A at angle θ to the horizontal:

$$A_x = A \cos \theta \quad A_y = A \sin \theta$$

Cosine goes with the side *adjacent* to the angle, sine with the side *opposite*. This is used directly in Module 4, where a velocity at an angle to the magnetic field is split into a part along the field and a part across it.

RECALL AND CHECK

Two forces on a charge: 3 N to the right and 4 N upward. What is the size of the net force?

ANSWER

They are perpendicular, so square, add and root: $\sqrt{3^2 + 4^2} = \sqrt{25} = 5 \text{ N}$. Not 7 N. Adding the numbers directly is only valid along the same line.

FOUNDATIONS F.4

Trigonometry, and the angles that matter

You need surprisingly little trigonometry here, but you need it reliably.

● MUST-MEMORISE · THE FOUR VALUES THAT KEEP APPEARING

θ	SIN θ	COS θ	WHAT IT MEANS PHYSICALLY
0°	0	1	Aligned. Magnetic force vanishes, flux is maximum.
30°	0.5	0.866	Turns up in the tutorial questions.
60°	0.866	0.5	Turns up in the helical-path question.
90°	1	0	Perpendicular. Magnetic force is maximum, flux is zero.

MEMORY HOOK: $\sin 0^\circ = 0$ and $\sin 90^\circ = 1$. Those two alone decide most of Module 4. If a charge moves *along* the field it feels no force, because $\sin 0^\circ = 0$.

⚠ TRAP · SIN $0^\circ = 0$ IS DOING REAL WORK

When the slides say a conductor parallel to the field feels no force, that is not a separate fact to memorise. It is just $F = BIL \sin \theta$ with $\theta = 0$. Likewise the maximum force at 90° is the same formula with $\sin 90^\circ = 1$. One formula, two special cases. Do not memorise three things where one will do.

FILL IN THE GAP · TRIGONOMETRY AND THE ANGLES THAT MATTER

1. $\sin 0^\circ = \underline{\hspace{2cm}}$, which is why a charge moving *along* a magnetic field feels no force.
2. $\sin 90^\circ = \underline{\hspace{2cm}}$, which gives the *maximum* magnetic force.
3. Magnetic flux is maximum when the field is at $\underline{\hspace{2cm}}$ degrees to the normal, because flux uses $\underline{\hspace{2cm}}$, not sine.

ANSWERS

1. zero. 2. one. Those two values alone decide most of Module 4.

3. 0 degrees, and it uses **cosine**.

Watch the pairing carefully: magnetic force uses sine (zero when aligned), but flux uses cosine (maximum when aligned). Swapping them is a classic and expensive error.

FOUNDATIONS F.5

Simultaneous equations

Kirchhoff's laws hand you two or three equations in two or three unknowns. This is the machinery for solving them, and it is the whole difficulty of that topic.

TEACH · SUBSTITUTION, THE ONLY METHOD YOU NEED

If you have as many equations as unknowns, you can solve. The reliable method is **substitution**:

1. Take the simplest equation and make one unknown the subject.
2. Substitute that expression into the other equation. You now have one equation in one unknown, which you can solve.
3. Put that value back into the first equation to get the other unknown.
4. Check both original equations with your two answers. If either fails, you have slipped.

Module 3 produces this system. Solve it now, so that when it reappears inside a circuit problem the algebra is not also new.

$$I_1(18) + I_3(10) = 50 \quad \text{and} \quad I_1 = 1.25 - I_3$$

- 1 The second equation already has I_1 as the subject. Substitute it into the first.

Always substitute the one that is already solved. Do not create work.

- 2 $(1.25 - I_3)(18) + 10 I_3 = 50$

Now there is only one unknown left, which is the point of the method.

- 3 Expand the bracket: $22.5 - 18 I_3 + 10 I_3 = 50$

Multiply both terms inside the bracket by 18. Forgetting the second is the classic slip.

- 4 Collect: $22.5 - 8 I_3 = 50$, so $-8 I_3 = 27.5$

$-18 + 10 = -8$. Watch that sign.

- 5 $I_3 = 27.5 / (-8) = -3.44 \text{ A}$

A negative current is not an error. It means the real current flows opposite to the direction you assumed. The size is still right.

$$I_3 = -3.44 \text{ A, and then } I_1 = 1.25 - (-3.44) = 4.69 \text{ A}$$

Now redo it: check by substituting both values back into $I_1(18) + I_3(10)$: $4.69(18) + (-3.44)(10) = 84.4 - 34.4 = 50$. It balances. Always run this check. It costs ten seconds and it catches sign errors.

TEACH · WHAT A NEGATIVE CURRENT MEANS

When you apply Kirchhoff's laws you must *assume* a direction for each current before you know the answer. If your assumption was backwards, the algebra tells you so by handing back a negative number. It does not mean you have failed. Report the size, and state that the current runs opposite to the assumed arrow.

FILL IN THE GAP · SIMULTANEOUS EQUATIONS

- To solve, make one unknown the _____ of the simplest equation, then _____ it into the other.
- You need at least as many _____ as you have unknowns.
- A _____ current is not an error. It means the true direction is _____ to the one you assumed.

ANSWERS

1. **subject**, then **substitute**.

2. **equations**. Three unknowns need three equations, or the system cannot be solved.

3. **negative**, and **opposite**. Report the magnitude and say the direction is reversed. Do not re-derive everything in a panic.

FOUNDATIONS F.6

Reciprocal arithmetic

Parallel resistors and series capacitors both need this. It is easy, and it is also where a startling number of marks quietly disappear.

TEACH · THE STEP EVERYONE FORGETS

When you compute something of the form

$$1 / R_{\text{eq}} = 1/R_1 + 1/R_2 + 1/R_3$$

the arithmetic gives you $1/R_{\text{eq}}$, not R_{eq} . You must **flip it at the end**. Stopping one step early is the single most common mistake in circuit questions.

● MUST-MEMORISE · THE TWO-ITEM SHORTCUT

For exactly **two** items, you can skip the reciprocals entirely:

$$R_{\text{eq}} = (R_1 \times R_2) / (R_1 + R_2)$$

MEMORY HOOK: product over sum. It only works for two. With three or more, go back to the reciprocals. This shortcut also works for two capacitors in *series*, since they follow the same reciprocal rule.

WORKED EXAMPLE · BOTH ROUTES, SAME ANSWER

Two resistors, 250 Ω and 150 Ω , in parallel.

- 1 Product over sum: $(250 \times 150) / (250 + 150) = 37500 / 400 = 93.75 \Omega$.
Two items, so the shortcut is legal.
- 2 The long way, as a check: $1/250 + 1/150 = 0.004 + 0.006667 = 0.010667$.
This is $1/R$, not R . Not finished yet.
- 3 Flip it: $R = 1 / 0.010667 = 93.75 \Omega$. Same answer.
This flip is the step people forget.

$$R_{\text{eq}} = 93.75 \Omega$$

Sanity check that always works: a parallel combination is always *smaller than the smallest* resistor present. Here $93.75 < 150$. If your parallel answer ever comes out bigger than the smallest resistor, you have made an error. This check is free and it never fails.

⚠ TRAP · SERIES AND PARALLEL SWAP BETWEEN R AND C

Resistors *add* in series and use *reciprocals* in parallel. Capacitors do the exact opposite: they *add* in parallel and use *reciprocals* in series. This is not arbitrary and it is explained in Module 2. But be warned now, because applying the resistor rule to capacitors is a guaranteed loss of marks.

FILL IN THE GAP · RECIPROCAL ARITHMETIC

1. The parallel formula gives you $1/R_{\text{eq}}$, so you must _____ it at the end.
2. The product-over-sum shortcut only works for exactly _____ items.
3. A parallel combination is always _____ than the _____ resistor in the group.

ANSWERS

1. **flip** (invert). Stopping one step early is the single most common error in the course.
2. **two**. With three or more, go back to reciprocals.
3. **smaller**, and **smallest**. This is a free check on every parallel answer you ever produce.

Proportional reasoning

Many exam questions never give you a number. They ask what happens to one quantity when another doubles. These are free marks if you can read a formula properly.

TEACH · READING A FORMULA AS A STATEMENT ABOUT CHANGE

Look at where a symbol sits and what power it is raised to.

- ▶ On the **top**: if it doubles, the answer doubles. Directly proportional.
- ▶ On the **bottom**: if it doubles, the answer halves. Inversely proportional.
- ▶ **Squared on the bottom**: if it doubles, the answer drops to a *quarter*. This is the inverse square law, and it governs Coulomb's law.

So in $F = kq_1q_2 / r^2$, tripling the separation r does not cut the force to a third. It cuts it to a **ninth**, because r is squared.

RECALL AND CHECK

In a parallel plate capacitor, $C = \epsilon_0 A / d$. If you double the plate separation d and also double the area A , what happens to C ?

ANSWER

A is on top, so doubling it doubles C . d is on the bottom, so doubling it halves C . The two effects cancel exactly: **C is unchanged.**

The small amount of calculus you need

Two places in this course use calculus. You do not need to be fluent, but you should not be startled by the notation.

TEACH · WHAT THE NOTATION IS TELLING YOU

A derivative, $d\Phi/dt$, is a rate of change. It means: how fast is Φ changing per second. That is all. When Faraday's law says the induced emf equals $-N d\Phi/dt$, it is saying that a *faster* change of flux induces a bigger voltage. A flux that sits still induces nothing, however large it is. **An integral is a total, accumulated in small pieces.** When Module 2 charges a capacitor, each extra scrap of charge dq is harder to add than the last, because the capacitor is already charged and pushes back. The integral adds up all those unequal scraps of work. That is where the factor of $1/2$ in the energy formula comes from, and it is why the energy is not simply QV .

● MUST-MEMORISE · THE ONE INTEGRAL USED IN THIS COURSE

$$\int_0^Q q \, dq = Q^2 / 2$$

MEMORY HOOK: raise the power by one and divide by the new power. Here q^1 becomes $q^2/2$. This single line is what produces $W = Q^2/2C$ for the energy in a capacitor.

FILL IN THE GAP · THE CALCULUS YOU NEED

1. A derivative $d\Phi/dt$ is a _____ of change.
2. Faraday's law responds to the _____ of change of flux, not to the size of the flux.
3. The factor of $\frac{1}{2}$ in $W = \frac{1}{2}CV^2$ appears because the voltage _____ steadily from zero as the capacitor charged.

ANSWERS

1. **rate.** How fast Φ changes per second.
2. **rate.** A huge *steady* flux induces exactly nothing.
3. **rose** (increased). So the *average* voltage you worked against was only $V/2$, which is where the half comes from.

FOUNDATIONS F.9

Circles, areas and the geometry you need

Gauss's law and Ampere's law both work by wrapping a surface or a loop around something. That means areas and circumferences.

● MUST-MEMORISE · THE FOUR GEOMETRIC FACTS THIS COURSE USES

SHAPE	FORMULA	WHERE IT IS USED
Circle, circumference	$2\pi r$	Ampere's law around a wire
Circle, area	πr^2	Flux through a loop
Sphere, surface area	$4\pi r^2$	Gauss's law around a point charge
Cylinder, curved area	$2\pi rL$	Gauss's law around a long wire

MEMORY HOOK: the $4\pi r^2$ in the sphere's area is precisely the 4π that appears in Coulomb's law.

That is not a coincidence: it is Gauss's law telling you that the field spreads over the surface of a sphere.

TEACH · WHY THIS MATTERS MORE THAN IT LOOKS

When Module 1 derives the field of a point charge from Gauss's law, the entire derivation is: the flux is E multiplied by the area of a sphere, that equals the enclosed charge over ϵ_0 , therefore $E(4\pi r^2) = Q/\epsilon_0$. Rearranging gives Coulomb's law. If you know the surface area of a sphere, that derivation is three lines. If you do not, it is impossible. That is the whole reason this page exists.

FILL IN THE GAP · CIRCLES, AREAS AND GEOMETRY

1. The surface area of a sphere is _____, and this is the source of the 4π in Coulomb's law.
2. The circumference of a circle is _____, which appears underneath the field of a straight wire.

ANSWERS

1. $4\pi r^2$. Gauss's law spreads the field over the surface of a sphere, and that is *why* Coulomb's law has a 4π in it. Not a coincidence.
2. $2\pi r$. It is the Amperian loop you walk around the wire.

FOUNDATIONS F.10

Rate of change, and the flux that changes

Module 5 rests on one new idea: a quantity that changes in time can do the same job as a current. To read it, you only need to be comfortable with what "rate of change" means. You already met this in F.7 and F.8; here it is, aimed straight at Maxwell.

Teach · What d/dt actually asks

The symbol $d\Phi/dt$ reads "the rate at which Φ is changing, per second." It is a slope: how fast a quantity climbs or falls as the clock ticks. If a flux of 0.030 Wb falls to 0.010 Wb in 0.050 s, then

$$d\Phi/dt = (\text{change in } \Phi) / (\text{time taken}) = (0.010 - 0.030) / 0.050 = -0.40 \text{ Wb/s.}$$

The minus sign says Φ is falling. The *size* of the rate, 0.40 Wb/s, is what drives the physics. This is the exact quantity inside Faraday's law, and (with E in place of B) inside Maxwell's new term.

Teach · The curly ∂ is the same idea, hedged

In the differential form of Maxwell's equations you will see $\partial E/\partial t$ instead of dE/dt . The curly ∂ is a *partial* derivative. It means exactly "rate of change with time" but adds one promise: **hold position fixed while you differentiate**. A wave depends on both where you are (x) and when it is (t); the partial symbol just says which one you are varying. For this course you may read $\partial/\partial t$ as "how fast it changes in time, at a fixed point." Nothing more is required of you.

⚠ Trap · A steady quantity has zero rate

If E or Φ is constant, then $dE/dt = 0$ and the whole new term vanishes. This is why the old Ampere's law worked perfectly for steady currents: nothing was changing, so Maxwell's extra piece was silently zero. The new physics only appears when something *changes*. Keep this in your pocket. It answers half the conceptual questions on this module.

Fill in the gap · Rate of change

1. The quantity $d\Phi/dt$ is the _____ at which the flux Φ changes each second.
2. The curly symbol $\partial/\partial t$ means rate of change in time at a _____ point.
3. If E does not change, then $dE/dt =$ _____, and Maxwell's new term contributes nothing.

ANSWERS

1. rate. 2. fixed. 3. zero. A steady field induces nothing, the same lesson as Faraday's law.

Divergence and curl, just enough to read Maxwell

This one page is optional. The exam works almost entirely with the integral form. Read this only so that the differential form of Maxwell's equations, when you meet it, is a picture and not a wall of symbols.

Teach · Divergence asks: is this point a source?

The divergence, written $\nabla \cdot \mathbf{E}$ (the symbol ∇ , "del", is just shorthand for these operations; you never have to compute it by hand here), measures whether field lines are being *created* at a point. Positive divergence means lines spray outward from there, like water from a tap, a source. Zero divergence means every line that enters also leaves. Gauss's law for electricity, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$, simply says: **electric field lines are born on charge** (here ρ , the Greek letter rho, is the charge density, meaning charge per unit volume). Gauss's law for magnetism, $\nabla \cdot \mathbf{B} = 0$, says: **magnetic field lines are never born anywhere**, there are no magnetic charges, so every line is a closed loop.

Teach · Curl asks: does this point stir?

The curl, written $\nabla \times \mathbf{E}$, measures whether the field *circulates* around a point, like a tiny whirlpool. Faraday's law $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ says a changing magnetic field stirs the electric field into loops. The Ampere-Maxwell law $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t$ says both a current and a changing electric field stir the magnetic field into loops (here $\mathbf{J}$, the current density, means current per unit area). That is the whole content of the differential form: **divergence for the two Gauss laws, curl for the two circulation laws.**

Recall and check

Which Maxwell equation is written with $\nabla \cdot = 0$, and what does that zero mean physically?

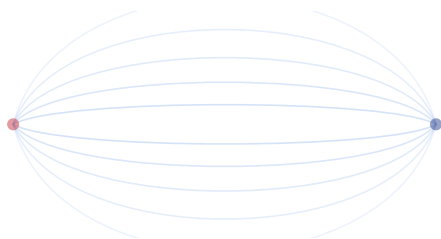
ANSWER

$\nabla \cdot \mathbf{B} = 0$, Gauss's law for magnetism. The zero means no point is ever a source of magnetic field lines: there are no magnetic monopoles, so field lines always close on themselves.

PART II

Module 1: Electrostatics

Charge at rest: what it is, the force between charges, the field and potential it creates, and Gauss's law, which is the elegant shortcut for finding fields when the situation is symmetric.



Forces in nature and the idea of charge

Learning outcomes: identify the four fundamental forces; explain electric charge and its quantisation; describe charging by friction, conduction and induction.

TEACH · THE FOUR FUNDAMENTAL FORCES

Every interaction in the universe, from a balloon sticking to a wall to a star holding itself together, is produced by just four forces.

FORCE	RELATIVE STRENGTH	WHAT IT DOES
Strong nuclear	Strongest	Holds the nucleus together against the protons repelling each other.
Electromagnetic	Very strong	Electricity, magnetism, light, chemistry, and everything in this course.
Weak nuclear	Weak	Governs radioactive decay.
Gravity	Weakest by far	Attracts all mass. Only noticeable because it never cancels out.

The **electromagnetic force** is the one we meet most in daily life. It is responsible for electricity, magnetism and light, and it is what PHY121 studies. Gravity is the *weakest* of the four, which sounds absurd when it holds you to the Earth. The reason it feels dominant is that it never cancels: mass only attracts. Electric forces are vastly stronger but positive and negative charge almost perfectly balance in ordinary matter, so their effects mostly cancel and we barely notice them.

TEACH · WHAT ELECTROSTATICS IS

Electrostatics is the study of electric charges at rest, and of the forces and fields they produce. The charges are not moving, yet they still push and pull on each other across empty space. That interaction is the electric force. At the heart of it is **electric charge**, a basic property of matter that comes in two kinds, positive and negative.

● MUST-MEMORISE · THE FIRST LAW OF CHARGE

Like charges repel. Unlike charges attract.

MEMORY HOOK: opposites attract. Everything electrostatic follows from this one sentence plus an inverse square law.

TEACH · CHARGE, SYMBOL AND UNIT

Electric charge is the property of matter that causes it to experience a force in an electric field.

- ▶ **Symbol:** Q or q
- ▶ **Unit:** the coulomb, C
- ▶ **Positive charge** is carried by protons.
- ▶ **Negative charge** is carried by electrons.

The six properties of charge

TEACH · PROPERTIES 1 AND 2: ADDITIVITY AND CONSERVATION

1. Additivity. Total charge is just the algebraic sum of the individual charges, signs included:

$$Q_{\text{total}} = q_1 + q_2 + q_3 + \dots$$

So +5 C combined with −3 C gives +2 C. You add them like ordinary signed numbers, because charge is a scalar.

2. Conservation. Charge cannot be created or destroyed, only *transferred* from one body to another. The total charge of an isolated system never changes.

When you rub a balloon on your hair, you do not manufacture charge. Electrons move from your hair to the balloon. The balloon gains exactly as much negative charge as the hair loses. The books always balance.

● MUST-MEMORISE · PROPERTY 3: QUANTISATION OF CHARGE

Charge does not come in arbitrary amounts. It comes in whole multiples of one fundamental lump:

$$q = n e \quad \text{where} \quad e = 1.602 \times 10^{-19} \text{ C}$$

n is an integer (... , −2, −1, 0, 1, 2, ...) and **e** is the elementary charge, the magnitude of the charge on one electron or one proton.

MEMORY HOOK: charge comes in coins, not in liquid. You can have 3 coins or 4 coins, never 3.7 coins. A charge of 1.5e simply cannot exist.

TEACH · PROPERTIES 4, 5 AND 6

4. Invariance. The magnitude of a charge does not depend on its speed, on the frame you observe it from, or on temperature and pressure. A charge is the same whether it is moving or at rest. (Mass, by contrast, does change with speed at high velocities. Charge does not.)

5. Charge is a scalar. It has magnitude only, and no direction. The plus or minus sign indicates the *type* of charge, not a direction in space.

6. Transferability. Charge can be moved between bodies by friction, by conduction, or by induction. These are the three methods of charging, covered next.

⚠ TRAP · THE SIGN ON A CHARGE IS NOT A DIRECTION

Because charge is a scalar, you add charges as signed numbers. But *force* is a vector, so you cannot add forces as signed numbers unless they lie on the same line. Students routinely add up the forces from several charges as if they were numbers. They are not. See Unit 1.2 on superposition.

The three ways to charge a body

TEACH · FRICTION, CONDUCTION AND INDUCTION

(a) Charging by friction. Rub two different materials together and electrons transfer from one to the other. One becomes positive, the other equally negative. This is the balloon-and-hair case. **(b) Charging by conduction.** Touch a charged object to a neutral one. Charge flows across the contact, and both objects end up with the *same sign* of charge, sharing what was there. **(c) Charging by induction.** Bring a charged object *near* a neutral one, without touching. The charges inside the neutral object rearrange: the opposite kind is pulled towards the near side, the same kind is pushed to the far side. If you now earth the far side and remove it, the object is left with a net charge *opposite* to the one you brought near, and you never touched it.

RECALL AND CHECK

A body carries a charge of 4.8×10^{-19} C. How many elementary charges is that, and is such a charge possible?

ANSWER

$n = q/e = (4.8 \times 10^{-19}) / (1.602 \times 10^{-19}) = 3.0$. That is a whole number, so yes: this is **3 elementary charges**. A charge of, say, $2.4 e$ would be impossible, because charge is quantised.

FILL IN THE GAP · CHARGE AND ITS PROPERTIES

1. Charge is _____, meaning it exists only in whole multiples of $e = 1.602 \times 10^{-19}$ C.
2. Charge is _____: it cannot be created or destroyed, only transferred.
3. Charge is a _____ quantity, so the plus or minus sign indicates the *type*, not a direction.
4. The three methods of charging are friction, _____ and _____.

ANSWERS

1. **quantised.** A charge of $2.4e$ simply cannot exist.
2. **conserved.**
3. **scalar.** This is why you add charges as signed numbers but cannot add forces that way.
4. **conduction** and **induction.** Conduction needs contact; induction does not.

UNIT 1.2

Coulomb's law and the electric field

Learning outcomes: state and apply Coulomb's law; calculate forces between multiple charges using superposition; define and interpret electric field lines.

● MUST-MEMORISE · COULOMB'S LAW

The force between two point charges is proportional to the product of the charges and inversely proportional to the square of the distance between them:

$$F = k \cdot q_1 q_2 / r^2$$

F is the force in newtons (N) **q₁, q₂** are the two charges in coulombs (C) **r** is the separation in metres (m) **k** = $8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$, the Coulomb constant, where $k = 1 / (4\pi\epsilon_0)$ **ε₀** = $8.85 \times 10^{-12} \text{ F m}^{-1}$, the permittivity of free space

MEMORY HOOK: product of charges on top, distance squared underneath. It has exactly the same shape as Newton's law of gravitation, with charge in place of mass.

TEACH · READING THE INVERSE SQUARE

Because *r* is *squared* on the bottom, distance matters enormously:

- ▶ Double the separation, and the force falls to a **quarter**.
- ▶ Triple the separation, and the force falls to a **ninth**.
- ▶ Halve the separation, and the force becomes **four times** larger.

The sign of the answer tells you the nature of the force: a positive product $q_1 q_2$ means repulsion, a negative product means attraction.

WORKED EXAMPLE · FORCE BETWEEN TWO PROTONS 5 M APART

SLIDE

The question. Two protons are 5.0 m apart. Calculate the electrostatic force between them, and state whether it is attractive or repulsive.

- 1 Write down what you know. Each proton carries $q = 1.6 \times 10^{-19} \text{ C}$, and $r = 5.0 \text{ m}$.
Both charges are identical, so $q_1 q_2 = (1.6 \times 10^{-19})^2$.
- 2 Substitute into Coulomb's law: $F = (8.99 \times 10^9)(1.6 \times 10^{-19})^2 / (5.0)^2$
Units are already SI: metres and coulombs. No conversion needed.
- 3 Square the charge: $(1.6 \times 10^{-19})^2 = 2.56 \times 10^{-38}$. Square the distance: $5.0^2 = 25$.
Squaring doubles the exponent: -19 becomes -38 .
- 4 $F = (8.99 \times 10^9)(2.56 \times 10^{-38}) / 25$
Now it is just arithmetic.

$$F \approx 9.2 \times 10^{-30} \text{ N} \quad (\text{repulsive, since both are positive})$$

Now redo it: I recomputed this independently and obtained $9.206 \times 10^{-30} \text{ N}$, so $9.2 \times 10^{-30} \text{ N}$ is correct to two significant figures. Note how tiny it is: this is why the electric force between individual particles is irrelevant at everyday distances, yet decisive inside an atom.

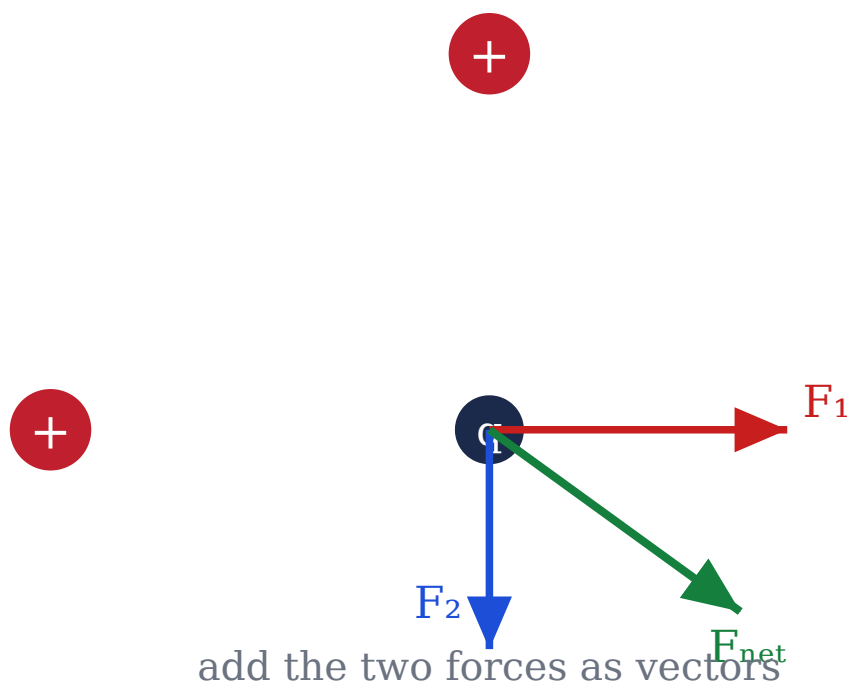
TEACH · SUPERPOSITION: MORE THAN TWO CHARGES

Coulomb's law handles exactly two charges. For three or more, use the **principle of superposition**: Work out the force from each other charge *separately*, one at a time, as if the others were not there. Then add those forces **as vectors**. That is the entire method. The only difficulty is the vector addition, which is exactly why Foundations F.3 exists.

● MUST-MEMORISE · THE PRINCIPLE OF SUPERPOSITION

$$F_{\text{net}} = F_1 + F_2 + F_3 + \dots \quad (\text{added as vectors})$$

MEMORY HOOK: one at a time, then add. Superposition is the most reusable idea in this course. It works for forces, for electric fields, and for potentials. The only difference is that potentials are scalars, so for those you add plain numbers.



Work out each force separately, then add them as vectors.

The electric field

TEACH · WHY WE NEED A FIELD AT ALL

Coulomb's law says two charges push on each other across empty space. But how does one charge *know* the other is there? The field is the answer, and it is a genuinely useful change of viewpoint. We say a charge alters the space around it, creating an **electric field**. Any other charge placed in that space then feels a force from the *field at its own location*. The field is the middleman, and it exists whether or not there is a second charge there to feel it.

● MUST-MEMORISE · ELECTRIC FIELD: DEFINITION AND THE FIELD OF A POINT CHARGE

The electric field is the force per unit positive charge:

$$E = F / q \quad \text{so} \quad F = qE$$

For a single point charge Q , the field at distance r is:

$$E = kQ / r^2 = Q / (4\pi\epsilon_0 r^2)$$

E is the electric field, in newtons per coulomb (N/C) or volts per metre (V/m). These two units are identical.

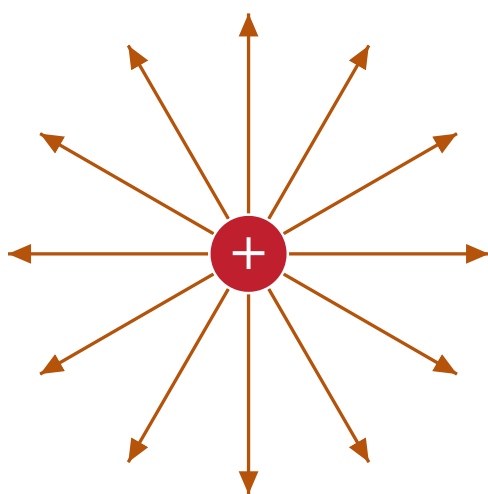
MEMORY HOOK: the field is Coulomb's law with one of the charges removed. Take $F = kq_1q_2/r^2$, divide out one charge, and you have the field the other one makes.

TEACH · ELECTRIC FIELD LINES AND HOW TO READ THEM

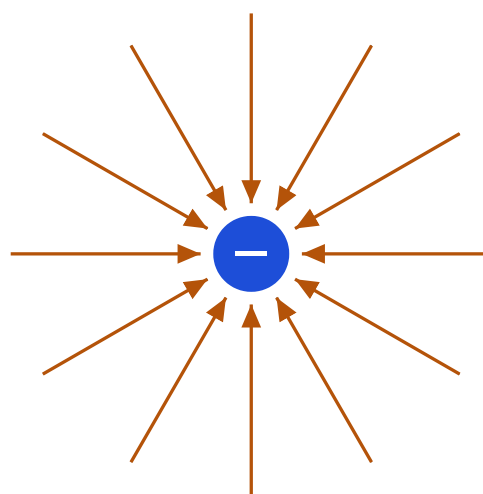
Field lines are a picture of the field. They obey strict rules:

- ▶ They point in the direction of the force on a **positive** test charge.
- ▶ They **begin on positive** charges and **end on negative** charges. Positive charges are sources, negative charges are sinks.
- ▶ Where the lines are **close together the field is strong**; where they spread out it is weak.
- ▶ They **never cross**. If they did, the field would have two directions at one point, which is meaningless.

So the field points *away* from an isolated positive charge and *towards* an isolated negative charge.



field points OUT of +



field points IN to -

Field lines point out of a positive charge and in to a negative charge.

Question (slides): What is the direction of the field 1 m away from a 1 C positive charge, in or out? What force would a 1.4×10^{-5} C *negative* charge experience at 1 m from it, and is that force towards or away?

- 1 Direction of the field: the source charge is **positive**, and field lines leave positive charges. So the field points **outward**, away from the 1 C charge.

The field direction depends only on the charge making it, never on what you put in it.

- 2 Field strength at $r = 1$ m: $E = kQ/r^2 = (8.99 \times 10^9)(1) / 1^2 = 8.99 \times 10^9$ N/C.

A colossal field. A charge of one full coulomb is enormous.

- 3 Force on the test charge: $F = qE = (1.4 \times 10^{-5})(8.99 \times 10^9)$

Use the magnitude of the charge for the size, then reason separately about direction.

- 4 $F = 1.259 \times 10^5$ N.

Multiply the fronts ($1.4 \times 8.99 = 12.59$), then add exponents ($-5 + 9 = 4$), giving $12.59 \times 10^4 = 1.259 \times 10^5$.

- 5 Direction of the force: the test charge is **negative** and the source is **positive**, so they *attract*. The force is **towards** the 1 C charge.

Note this is opposite to the field direction, exactly as expected for a negative charge: $F = qE$ with q negative flips the direction.

Field: outward. Force $\approx 1.26 \times 10^5$ N, directed toward the 1 C charge.

Now redo it: the key idea to carry away is that the field points outward, but the force on a *negative* charge points inward. Field direction and force direction agree only for positive charges.

WORKED EXAMPLE · TWO POINT CHARGES, 1.0 CM APART

CLASSWORK

Question (slides): Point charge A has charge -1.0 nC and point charge B has charge 4.0 nC . They are separated by 1.0 cm . Find the magnitude and direction of the electric force on each.

- 1 Convert everything to SI *first*. $q_A = -1.0 \times 10^{-9} \text{ C}$, $q_B = 4.0 \times 10^{-9} \text{ C}$, $r = 1.0 \text{ cm} = \mathbf{0.01 \text{ m}}$.

This is the trap. 1.0 cm is 10^{-2} m , not 10^{-3} m . And r gets squared, so an error here is magnified a hundredfold.

- 2 Substitute magnitudes into Coulomb's law: $F = (8.99 \times 10^9)(1.0 \times 10^{-9})(4.0 \times 10^{-9}) / (0.01)^2$

Use magnitudes for the size. Decide the direction afterwards from the signs.

- 3 Numerator: $(8.99 \times 10^9)(4.0 \times 10^{-18}) = 3.596 \times 10^{-8}$.

$(1.0 \times 10^{-9})(4.0 \times 10^{-9}) = 4.0 \times 10^{-18}$: multiply fronts, add exponents.

- 4 Denominator: $(0.01)^2 = 1 \times 10^{-4}$.

Squaring a small number makes it much smaller, which makes the force much larger.

- 5 $F = (3.596 \times 10^{-8}) / (1 \times 10^{-4}) = 3.6 \times 10^{-4} \text{ N}$.

Dividing, so subtract exponents: $-8 - (-4) = -4$.

- 6 Direction: one charge is negative and the other positive, so the force is **attractive**.

Each charge is pulled towards the other.

$F = 3.6 \times 10^{-4} \text{ N}$ on each charge, attractive.

Now redo it, and notice the physics: the forces on A and B are *equal in magnitude* and opposite in direction, even though B carries four times the charge of A. That is Newton's third law, and it holds no matter how unequal the charges are. Students very often expect the bigger charge to feel the bigger force. It does not.

⚠ TRAP · THE BIGGER CHARGE DOES NOT FEEL A BIGGER FORCE

In Coulomb's law the force depends on the *product* q_1q_2 , which is the same number for both charges. So both charges always feel exactly the same size of force, in opposite directions, regardless of how lopsided the charges are. Only the *accelerations* differ, because the masses differ.

FILL IN THE GAP · COULOMB'S LAW AND THE ELECTRIC FIELD

1. Coulomb's law is an _____ law, so tripling the distance divides the force by _____.
2. Electric field lines begin on _____ charges and end on _____ charges.
3. Field lines never _____, because the field cannot have two directions at one point.
4. To handle three or more charges you use the principle of _____, adding the forces as _____.

ANSWERS

1. inverse square, and **nine** (not three).

2. positive, then **negative**.

3. cross (intersect).

4. superposition, added as **vectors**.

Vectors, not numbers. Charge is a scalar, but force is not.

Electric potential and energy

Learning outcomes: compute the electric potential due to point charges; relate the electric field to the potential gradient; calculate work done in moving charges.

TEACH · THE GRAVITATIONAL ANALOGY, WHICH IS EXACT ENOUGH TO TRUST

Lift a book onto a shelf and you do work against gravity. The book now stores *gravitational potential energy*: let it go and the energy comes back as motion. Electric potential energy works identically. Push a positive charge towards another positive charge and you do work against the repulsion. The charge now stores **electric potential energy**: let it go and it flies away, converting that energy into kinetic energy. The height of the shelf is the analogue of *potential*. The book on the shelf is the analogue of *potential energy*. Note the difference carefully: the shelf has a height whether or not a book sits on it.

● MUST-MEMORISE · POTENTIAL ENERGY, POTENTIAL, AND THE DIFFERENCE BETWEEN THEM

Electric potential energy is the work done in bringing a charge from infinity to a point in the field:

$$U = kQq / r \quad (\text{joules, J})$$

Electric potential is the work done *per unit positive charge* in bringing it from infinity to that point:

$$V = kQ / r \quad (\text{volts, V})$$

and therefore

$$U = qV \quad \text{and} \quad W = qV$$

MEMORY HOOK: potential is *per unit charge*. It belongs to the *point in space*, and exists even with no charge there. Potential *energy* needs an actual charge to be sitting at that point. Shelf height versus book on shelf.

⚠ TRAP · POTENTIAL HAS R, NOT R SQUARED

Compare carefully:

$$E = kQ / r^2 \quad \text{but} \quad V = kQ / r$$

The **field** has r squared. The **potential** has plain r . Mixing these up is one of the most common errors in this module. Also remember that **E is a vector but V is a scalar**: to combine potentials from several charges you simply add the numbers, signs included, with no vector work at all. That makes potential much easier to handle than field, which is precisely why it is worth having.

● MUST-MEMORISE · POTENTIAL DIFFERENCE AND WORK

The potential difference between two points is the work done per unit charge in moving a charge from one to the other. The work done in moving a charge q through a potential difference V is:

$$W = qV$$

MEMORY HOOK: work equals charge times voltage. This one formula reappears in Module 3 as the work done by a battery, and in Module 4 for emf. It is the same idea each time.

TEACH · THE LINK BETWEEN FIELD AND POTENTIAL

The electric field is the **negative gradient** of the potential:

$$E = - dV / dr$$

In words: the field points in the direction in which the potential falls most steeply, and its strength is the steepness of that fall. Water flows downhill fastest where the hill is steepest. A positive charge is pushed "downhill" in potential in exactly the same way. The minus sign encodes "downhill". For a uniform field between parallel plates separated by distance d this simplifies to the very useful:

$$E = V / d$$

This is why the electric field can be measured in volts per metre.

WORKED EXAMPLE · POTENTIAL 2 M FROM A 5×10^{-6} C CHARGE

CLASSWORK

The question. Calculate the electric potential at a point 2 m from a charge of 5×10^{-6} C.

- 1 Use the potential formula, which has plain r on the bottom: $V = kQ / r$.
Not r squared. That would be the field.
- 2 $V = (8.99 \times 10^9)(5 \times 10^{-6}) / 2$
Everything is already in SI.
- 3 Numerator: $(8.99 \times 10^9)(5 \times 10^{-6}) = 4.495 \times 10^4$.
Multiply fronts: $8.99 \times 5 = 44.95$. Add exponents: $9 + (-6) = 3$. So $44.95 \times 10^3 = 4.495 \times 10^4$.
- 4 Divide by 2: $V = 2.2475 \times 10^4$ V.

$$V \approx 2.25 \times 10^4 \text{ V (about 22.5 kV)}$$

Now redo it: then ask what the *field* would be at the same point. $E = kQ/r^2 = 4.495 \times 10^4 / 4 = 1.12 \times 10^4$ V/m. Note that V and E are numerically different because of that extra factor of r . Confirm you can produce both without hesitating.

The question. A charge of 3 C moves through a potential difference of 12 V. Calculate the work done.

- 1 Use $W = qV$. The definition of potential difference is work per unit charge, so the work is charge times potential difference.

No need for distances or fields. The potential difference already contains all of that.

- 2 $W = (3)(12) = 36 \text{ J}$.

Coulombs times volts gives joules directly.

$$W = 36 \text{ J}$$

Now redo it: this is why the volt is defined as a joule per coulomb. Driving one coulomb through one volt takes exactly one joule of work.

Equipotential surfaces

● MUST-MEMORISE · THE THREE PROPERTIES OF AN EQUIPOTENTIAL SURFACE

An **equipotential surface** is a surface on which every point has the same electric potential.

1. **No work is done** when a charge moves along an equipotential surface.
2. Electric field lines are always **perpendicular** to equipotential surfaces.
3. Equipotential surfaces **never intersect** one another.

MEMORY HOOK: contour lines on a map. Walking along a contour costs no climbing, and the steepest slope is always at right angles to the contour.

RECALL AND CHECK

Explain, in one line, why no work is done when a charge moves along an equipotential surface.

ANSWER

Because $W = qV$, and along an equipotential surface the potential difference V between start and finish is **zero**. So $W = q \times 0 = 0$. Equivalently: the force is perpendicular to the motion, and a force at right angles to the displacement does no work.

FILL IN THE GAP · POTENTIAL AND ENERGY

1. The electric field goes as $1/r^2$, but the electric potential goes as $1/\underline{\hspace{2cm}}$.
2. Electric field is a $\underline{\hspace{2cm}}$ but potential is a $\underline{\hspace{2cm}}$, which is why potentials are much easier to add.
3. The work done moving a charge q through a potential difference V is $W = \underline{\hspace{2cm}}$.
4. No work is done along an equipotential surface because the potential difference is $\underline{\hspace{2cm}}$.

ANSWERS

1. **r** (plain r, not r squared). Mixing these up is the most common error in Module 1.
2. **vector**, and **scalar**. To combine potentials you just add the numbers with their signs.
3. **qV** .
4. **zero**. $W = q \times 0 = 0$.

Gauss's law

Gauss's law is a shortcut. Where the situation is symmetric, it produces in three lines a field that would otherwise take an unpleasant integral.

TEACH · ELECTRIC FLUX, IN PLAIN LANGUAGE

Electric flux is a measure of how many field lines pass through a surface. Think of the field as rain falling and the surface as a hoop: the flux is how much rain goes through the hoop.

$$\Phi_E = E A \cos \theta$$

Hold the hoop face-on to the rain and you catch the most ($\theta = 0$, $\cos = 1$). Turn it edge-on and you catch none ($\theta = 90^\circ$, $\cos = 0$). The angle θ is measured between the field and the *normal* to the surface, that is, the line sticking straight out of it.

Unit: $\text{N m}^2/\text{C}$.

● MUST-MEMORISE · GAUSS'S LAW

The total electric flux through any *closed* surface is proportional to the total charge enclosed within that surface:

$$\Phi_E = Q_{\text{enclosed}} / \epsilon_0$$

Φ_E is the total flux out through the closed surface Q_{enclosed} is the net charge *inside* the surface only $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$

MEMORY HOOK: flux out equals charge in, divided by epsilon nought. Charges *outside* the surface contribute **nothing** to the net flux: whatever field lines enter one side must leave the other.

TEACH · THE PHYSICAL PICTURE, AND WHY EXTERNAL CHARGES DO NOT MATTER

- ▶ Field lines **start** on positive charges and **end** on negative charges.
- ▶ The number of lines crossing a closed surface depends only on the charge **enclosed** by it.
- ▶ A charge **outside** the surface sends lines in one side and straight out the other. They cancel exactly, contributing zero *net* flux.

This is the whole content of the law, and it is why it is such a powerful shortcut: you can ignore everything outside your chosen surface.

TEACH · WHAT A GAUSSIAN SURFACE IS, AND HOW TO CHOOSE IT

A **Gaussian surface** is an imaginary closed surface that *you* invent, purely to make the calculation easy. It has no physical existence. The art is choosing it so that E is constant over it and perpendicular to it, because then the flux integral collapses to the simple product $E \times A$.

CHARGE DISTRIBUTION	CHOOSE THIS SURFACE	BECAUSE
Point charge	Sphere, centred on the charge	E has the same size everywhere on it and points straight out.
Long straight wire	Cylinder, coaxial with the wire	E points radially out of the curved side; the flat ends catch nothing.
Infinite plane sheet	Cylinder (or box) through the sheet	E is perpendicular to the sheet, so only the two flat ends catch flux.

The standard results, derived

CASE 1 · FIELD OF A POINT CHARGE

SLIDE

The task. Use Gauss's law to find the electric field at a distance r from a point charge Q .

- 1 Draw a sphere of radius r centred on the charge Q .
By symmetry, E has the same magnitude everywhere on this sphere and points straight out.
- 2 The flux is then simply E multiplied by the sphere's area: $\Phi = E(4\pi r^2)$.
This uses the surface area of a sphere from Foundations F.9. The $\cos \theta$ is 1 because E is perpendicular to the surface everywhere.
- 3 Set that equal to Q/ϵ_0 by Gauss's law: $E(4\pi r^2) = Q / \epsilon_0$.
- 4 Rearrange for E .

$$E = Q / (4\pi\epsilon_0 r^2) = kQ / r^2$$

Look at what just happened: this is Coulomb's law, recovered in four lines from Gauss's law. The two are not separate facts. They are the same physics stated two ways, and the 4π in Coulomb's law is precisely the 4π from the surface of a sphere.

CASE 2 · INFINITE LINE CHARGE (LONG STRAIGHT WIRE)

A wire carrying a uniform **linear charge density** λ (charge per metre, C/m). Wrap a cylinder of radius r and length L around it. The charge enclosed is λL , and the curved area is $2\pi rL$. So $E(2\pi rL) = \lambda L/\epsilon_0$, and the L cancels:

$$E = \lambda / (2\pi\epsilon_0 r)$$

Note the field falls off as **$1/r$** , not $1/r^2$. A line of charge spreads its influence more slowly than a point does.

CASE 3 · INFINITE PLANE SHEET

A sheet carrying a uniform **surface charge density** σ (charge per square metre, C/m²). Push a cylinder through the sheet so it sticks out both sides. Flux escapes only through the two flat ends, giving $2A$ of area:

$$E = \sigma / (2\epsilon_0)$$

Remarkably, **r does not appear**. The field of an infinite sheet is the same at every distance. Move away from it and it does not weaken.

● MUST-MEMORISE · CASE 4: THE FIELD INSIDE A HOLLOW CONDUCTOR IS ZERO

$E = 0$ everywhere inside a hollow conducting shell

Draw a Gaussian surface inside the cavity. It **encloses no charge**, because all the charge on a conductor sits on its outer surface. So the flux through it is zero, and therefore $E = 0$. *Outside* the shell ($r > R$), the Gaussian sphere encloses the whole charge Q , and the shell behaves exactly like a point charge at its centre: $E = kQ/r^2$.

MEMORY HOOK: no charge inside means no field inside. This is the principle of the Faraday cage, and it is why you are safe inside a car during a lightning strike and why sensitive electronics are shielded in metal boxes.

WORKED EXAMPLE · FIELD AT THE SURFACE OF A GAUSSIAN SPHERE

CLASSWORK

Question (slides): A point charge of 5×10^{-6} C sits at the centre of a spherical Gaussian surface of radius 0.20 m. Find the electric field at the surface.

- 1 The field of a point charge at distance r is $E = kQ/r^2$.

The Gaussian surface is imaginary. It does not alter the field; it just gives us a convenient place to evaluate it.

- 2 $E = (8.99 \times 10^9)(5 \times 10^{-6}) / (0.20)^2$

Note r is squared here, because this is a field and not a potential.

- 3 Numerator: 4.495×10^4 . Denominator: $(0.20)^2 = 0.04$.

0.2 squared is 0.04, not 0.4. This is a very common slip.

- 4 $E = 4.495 \times 10^4 / 0.04 = 1.124 \times 10^6$ N/C.

$E \approx 1.12 \times 10^6$ N/C, directed radially outward

Now redo it: and note that we never actually needed Gauss's law to compute this, only the point charge result that Gauss's law produces. The mention of a "Gaussian surface" in the question is really just telling you the distance.

WORKED EXAMPLE · TOTAL FLUX THROUGH A SURFACE ENCLOSING 3 μC
CLASSWORK

The question. A closed spherical surface encloses a charge of 3 μC . Calculate the total electric flux through the surface.

- 1 Gauss's law directly: $\Phi = Q_{\text{enclosed}} / \epsilon_0$.

The shape of the surface is irrelevant. Only the enclosed charge matters. This is the whole beauty of the law.

- 2 $\Phi = (3 \times 10^{-6}) / (8.85 \times 10^{-12})$

$3 \mu\text{C} = 3 \times 10^{-6} \text{ C}$.

- 3 Divide the fronts: $3 / 8.85 = 0.339$. Subtract exponents: $-6 - (-12) = 6$.

So $0.339 \times 10^6 = 3.39 \times 10^5$.

$$\Phi \approx 3.39 \times 10^5 \text{ N m}^2/\text{C}$$

Now redo it: notice you were given no radius and no shape, and you did not need them. If a question gives you the enclosed charge and asks for total flux, the answer is one division. Do not go looking for the geometry.

WORKED EXAMPLE · FIELD FROM A LONG CHARGED WIRE
CLASSWORK

Question (slides): An infinitely long straight wire carries a uniform linear charge density of $4 \times 10^{-6} \text{ C/m}$. Using a cylindrical Gaussian surface of radius 0.05 m, find the field at that distance.

- 1 Use the line charge result: $E = \lambda / (2\pi\epsilon_0 r)$.

Falls off as $1/r$, since this is a line and not a point.

- 2 Denominator: $2\pi(8.85 \times 10^{-12})(0.05) = 2.781 \times 10^{-12}$.

$2\pi \times 8.85 = 55.6$, times 0.05 gives 2.78, keeping the 10^{-12} .

- 3 $E = (4 \times 10^{-6}) / (2.781 \times 10^{-12}) = 1.44 \times 10^6 \text{ N/C}$.

Divide fronts: $4/2.781 = 1.44$. Subtract exponents: $-6 - (-12) = 6$.

$$E \approx 1.44 \times 10^6 \text{ N/C, pointing radially away from the wire}$$

Now redo it: the length L of the cylinder never appeared in the answer, because it cancelled. That cancellation is the signature of a correct Gauss's law calculation for a line charge. If L survives into your final answer, you have gone wrong.

RECALL AND CHECK

A closed surface encloses no charge, but there is a large charge sitting just outside it. What is the net flux through the surface?

ANSWER

Zero. Gauss's law depends only on the *enclosed* charge. The external charge sends field lines in through one side and out through the other, and they cancel exactly. Note that the field on the surface is *not* zero, but the net flux is.

FILL IN THE GAP · GAUSS'S LAW

1. Gauss's law says the flux out equals the charge _____, divided by ϵ_0 .
2. A charge sitting *outside* the surface contributes _____ net flux.
3. The electric field inside a hollow conductor is _____, because the enclosed charge is _____.
4. For an infinite plane sheet, $E = \sigma/2\epsilon_0$, which does *not* depend on the _____.

ANSWERS

1. **enclosed.** Only what is inside the surface counts.
2. **zero.** Its lines enter one side and leave the other, cancelling exactly.
3. **zero**, because the enclosed charge is **zero** (all charge sits on the outer surface). This is the Faraday cage.
4. **distance.** Move away from an infinite sheet and the field does not weaken at all.

PART III

Module 2: Electric Circuits and Materials

How materials respond to a field: conductors let charge move, insulators do not but polarise instead. That leads to the dipole, then to the capacitor, and finally to energy stored in the electric field itself.



Conductors, insulators and the electric dipole

The slides set out the logic of this unit explicitly: material response (conductors and insulators), then microscopic behaviour (dipoles), then devices (capacitors), then energy storage. Follow that chain and the unit is straightforward.

TEACH · WHY MATERIALS MATTER

Electrostatics deals with charges at rest and the fields they produce. But fields exist *inside matter* too, and how a material responds to an applied field determines everything about it: whether it stores energy, whether it conducts, whether it can be used to build a device. Understanding this response explains energy storage in capacitors, the behaviour of insulating materials, and the operation of electronic devices.

Conductors

● MUST-MEMORISE · THE FOUR PROPERTIES OF A CONDUCTOR IN EQUILIBRIUM

1. **Free charge movement.** Electrons respond immediately to an applied field, and charges redistribute until equilibrium is reached.
2. **Electrostatic equilibrium.** Once settled, there is no net motion of charge, and the **internal electric field becomes zero**.
3. **Charge distribution.** All excess charge resides on the **surface**, with a higher concentration at sharp points.
4. **Field direction.** At the surface, the electric field is **perpendicular** to that surface.

MEMORY HOOK: inside a conductor, $E = 0$. If it were not zero, the free charges would still be moving, and by definition we would not be at equilibrium yet. The field being zero *is* what equilibrium means.

TEACH · WHY CHARGE SITS ON THE SURFACE, AND CROWDS AT SHARP POINTS

Put excess charge on a conductor and every bit of it repels every other bit. They push each other as far apart as they can get, and the furthest apart they can get is the outer surface. So that is where they end up. Nothing is left in the interior, which is also why the internal field is zero, exactly as Gauss's law told us in Unit 1.4. At a **sharp point**, a charge has fewer neighbours crowding it from the sides, so charge packs more densely there. This produces very strong fields at points and edges, and it is the entire principle of the lightning conductor.

TEACH · OHM'S LAW, AND WHERE IT FAILS

Conduction of charge in a conductor is governed by **Ohm's law**: the current flowing through a conductor is directly proportional to the potential difference across it, *provided temperature and other physical conditions remain constant*.

$$V \propto I \quad \text{so, introducing a constant} \quad V = IR$$

V = potential difference, in volts (V) I = current, in amperes (A) R = resistance, in ohms (Ω)
From $V = IR$ it follows that increasing the voltage increases the current, provided R is constant, and that R measures the opposition to the flow of current. **The law does not apply to everything.** It works for *ohmic* materials such as metals. It fails for **semiconductors with a negative temperature coefficient**, where resistance *decreases* as temperature increases, so conductivity increases. That proviso about constant temperature is doing real work.

From the slides: a material carries 2 mA at 2 V. At 350 K, the same 2 V drives 5 mA. Does the material obey Ohm's law?

- 1 First condition: $V = 2 \text{ V}$, $I = 2 \text{ mA} = 2 \times 10^{-3} \text{ A}$.
Convert milliamps to amps immediately, or the resistance will be out by a factor of 1000.
- 2 $R = V/I = 2 / (2 \times 10^{-3}) = \mathbf{1000 \Omega}$.
Dividing by 10^{-3} is the same as multiplying by 1000.
- 3 Second condition at 350 K: $V = 2 \text{ V}$, $I = 5 \text{ mA} = 5 \times 10^{-3} \text{ A}$.
- 4 $R = V/I = 2 / (5 \times 10^{-3}) = \mathbf{400 \Omega}$.
- 5 Compare. The resistance *fell* from 1000Ω to 400Ω when the temperature rose. **R drops from 1000Ω to 400Ω . Resistance decreases as temperature increases, so the material does not obey Ohm's law. It behaves as a semiconductor.**
For an ohmic material R must stay constant. It did not.

Now redo it: the whole question turns on the phrase "provided temperature remains constant". A material that changes its resistance when heated is not ohmic, and this is the signature of a semiconductor with a negative temperature coefficient.

Insulators and polarisation

TEACH · WHAT AN INSULATOR DOES INSTEAD

An **insulator** is a material in which there is **no free charge movement**. Common examples are glass, rubber and plastic. But an insulator is not inert. Apply a field to it and, although no charge can travel through it, something still happens:

1. The charges **shift slightly within each atom**. The electron cloud is pulled one way, the nucleus the other.
2. An **induced dipole** is created.

Polarisation is the process whereby an induced dipole is created in an insulator by an applied electric field.

● MUST-MEMORISE · CONDUCTOR VERSUS INSULATOR

PROPERTY	CONDUCTOR	INSULATOR
Charge mobility	Free	Bound
Internal field	Zero	Non-zero
Response to a field	Redistribution	Polarisation

MEMORY HOOK: conductors *move* charge, insulators *stretch* it. That single distinction generates every row of this table, and it is a very likely exam question.

The electric dipole

TEACH · WHAT A DIPOLE IS

An atom in a field has its electron cloud centre pulled away from its nucleus. Those two displaced centres of charge, equal in size and opposite in sign, form an **electric dipole**. Formally: a system of two equal and opposite charges (+q and -q) separated by a small distance d constitutes an electric dipole. Polarisation in an insulator is simply what you get when a vast number of these dipoles all line up together.

● MUST-MEMORISE · DIPOLE MOMENT

$$p = qd$$

q = magnitude of one of the charges, in coulombs (C) **d** = separation distance, in metres (m) **p** = dipole moment, measured in **coulomb metres (C m)**

MEMORY HOOK: the unit follows straight from the formula. Charge times distance gives coulomb metres. The slides pose exactly this as a class drill, and the unit is the answer.

FORMULA · A DIPOLE IN A UNIFORM ELECTRIC FIELD

Place a dipole in a uniform field E. The positive end is pushed one way and the negative end the other, so the field **twists** it. The forces cancel, so the dipole does not move off, but it rotates, tending to align with the field.

$$\tau = pE \sin \theta$$

and the potential energy stored is

$$U = -pE \cos \theta$$

τ = torque (the twisting effect), **θ** = angle between the dipole and the field.

TEACH · READING THE ENERGY EQUATION

The slides state that the dipole is **aligned with the field at minimum energy**, and **opposite to the field at maximum energy**. Check this against the torque formula: when $\theta = 0$ the dipole is aligned, $\sin 0 = 0$, and the torque vanishes. There is nothing left to twist it, so it stays put. That is exactly what an equilibrium position means. When the dipole is *opposite* to the field, it is unstable: the slightest nudge and it flips round.

A compass needle behaves in precisely this way in a magnetic field. It swings until it lines up, then stops.

RECALL AND CHECK

Two charges of +2 nC and -2 nC are separated by 3 mm. What is the dipole moment?

ANSWER

$p = qd = (2 \times 10^{-9} \text{ C})(3 \times 10^{-3} \text{ m}) = 6 \times 10^{-12} \text{ C m}$. Use the magnitude of one charge, not the sum of both, and remember that 3 mm is $3 \times 10^{-3} \text{ m}$.

FILL IN THE GAP · CONDUCTORS, INSULATORS AND DIPOLES

1. In electrostatic equilibrium the field inside a conductor is _____, and all excess charge sits on the _____.
2. An insulator cannot conduct, but it responds to a field by _____.
3. Conductors *move* charge; insulators _____ it.
4. The dipole moment is $p = qd$, measured in _____.

ANSWERS

1. **zero**, and the **surface**. If the internal field were not zero the charges would still be moving, so by definition it would not be equilibrium yet.
2. **polarising** (polarisation).
3. **stretch** (or distort). That one distinction generates the whole comparison table.
4. **coulomb metres** (C m). The unit follows straight from charge times distance.

UNIT 2.2

Capacitance and dielectrics

A capacitor stores energy in an electric field. This unit builds it from the definition, through the geometry, to the energy, and then to combinations.

TEACH · WHAT A CAPACITOR IS

A **capacitor** is a device for storing electrical energy by allowing charge to accumulate on metal plates. Apply a voltage across it and one plate becomes positively charged while the other becomes equally negatively charged. The **electric field between the plates stores electrostatic energy**. That energy is recovered when the charges are allowed to flow back off the plates into the circuit. Generally, *any* device that can store charge is a capacitor.

● MUST-MEMORISE · THE DEFINITION OF CAPACITANCE

The capacitance C of a capacitor is the ratio of the magnitude of the charge Q stored on either conductor to the potential difference V between them:

$$C = Q / V \quad (\text{so } Q = CV)$$

C = capacitance, in **farads (F)** 1 farad = 1 coulomb per volt. Common submultiples: microfarad ($\mu\text{F} = 10^{-6} \text{ F}$) and picofarad ($\text{pF} = 10^{-12} \text{ F}$).

MEMORY HOOK: capacitance is *charge per volt*. It measures how much charge the device will hold for each volt you push across it. A big capacitance holds a lot of charge cheaply.

△ TRAP · CAPACITANCE DOES NOT DEPEND ON Q OR V

$C = Q/V$ looks as though capacitance depends on charge and voltage. It does not. **Capacitance is fixed by the geometry:** the size and shape of the conductors and the material between them. Double the voltage and you simply double the charge stored, and the ratio C stays exactly the same. The slides state this explicitly: potential is geometrically dependent, so in determining capacitance, the geometry of the surface storing the charge is what matters.

Capacitance depends on geometry

WORKED EXAMPLE 1 · CAPACITANCE OF AN ISOLATED CONDUCTING SPHERE

SLIDE

An insulated conducting sphere of radius R carries charge Q . Find (1) the potential of its outer surface, and (2) its capacitance.

- 1 All the charge on a conductor sits on its surface, and for external points the sphere behaves exactly like a point charge at its centre.

This is the Gauss's law result from Unit 1.4, being put to work.

- 2 So the potential at the surface ($r = R$) is $V = Q / (4\pi\epsilon_0 R)$.

Potential of a point charge, with plain R on the bottom, not R squared. This is constant over the surface, which is why it is an equipotential surface.

- 3 Now apply the definition of capacitance, $C = Q/V$, and substitute that V : $C = Q \div [Q / (4\pi\epsilon_0 R)]$.

Dividing by a fraction means multiplying by its reciprocal.

- 4 The Q cancels top and bottom.

And this is the whole point: the charge disappears, leaving capacitance as a function of geometry alone, exactly as promised.

$$V = Q / (4\pi\epsilon_0 R) \quad \text{and} \quad C = 4\pi\epsilon_0 R$$

Read the result: capacitance depends only on R . A larger sphere has a larger capacitance. Zero charge is mentioned nowhere in the answer. Note also that the potential goes to zero as R tends to infinity, which is why we take earth (ground) as the zero of potential.

WORKED EXAMPLE · CAPACITANCE OF A SHELL OF RADIUS 100 CM

CLASSWORK

Find the capacitance if $R = 100$ cm. Take $\epsilon_0 = 8.85 \times 10^{-12}$ F/m.

- 1 Convert first: $R = 100$ cm = **1.0 m**.

The single most likely place to lose this mark.

- 2 $C = 4\pi\epsilon_0 R = 4\pi(8.85 \times 10^{-12})(1.0)$.

- 3 $4\pi = 12.566$, and $12.566 \times 8.85 = 111.2$.

So $C = 111.2 \times 10^{-12}$ F.

- 4 Write in standard form: $111.2 \times 10^{-12} = 1.11 \times 10^{-10}$ F.

$$C \approx 1.1 \times 10^{-10} \text{ F} \quad (\text{about } 111 \text{ pF})$$

Now redo it: I recomputed this independently as 1.1121×10^{-10} F, so the slide's 1.1×10^{-10} F is correct. Note how small a farad is as a unit: a sphere a metre across has a capacitance of only about 100 pF. Real farad-sized capacitors are an engineering achievement.

The parallel plate capacitor

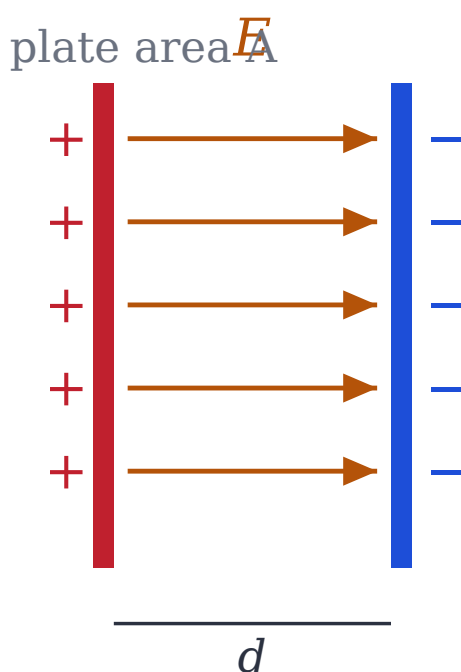
● MUST-MEMORISE · THE PARALLEL PLATE CAPACITOR

This is the most commonly used capacitor: two metal plates held close together with an insulator between them.

$$C = \epsilon_0 A / d$$

A = area of one plate, in m^2 d = separation between the plates, in m $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$

MEMORY HOOK: area on top, distance underneath. Bigger plates hold more charge, so capacitance rises with A . Push the plates closer and the attraction between the opposite charges is stronger, so more charge is held for the same voltage, so capacitance rises as d falls.



Two plates, area A , separation d , with a uniform field E between them.

TEACH · THE THREE FACTORS THAT SET THE CAPACITANCE

Reading $C = \epsilon_0 A / d$ straight off, the factors affecting a parallel plate capacitor are:

1. The **area** of the plates.
2. The **separation** between the plates.
3. The **dielectric material** insulating the plates.

The third does not appear in the formula above, because that formula assumes a vacuum. Putting a real insulator in changes things, and that is the next box.

● MUST-MEMORISE · DIELECTRICS

A **dielectric** is an insulating material placed between the plates. It *increases* the capacitance by a factor called the **dielectric constant** (or relative permittivity) ϵ_r :

$$C = \epsilon_r C_0 \quad \text{and so} \quad C = \epsilon_r \epsilon_0 A / d$$

C_0 = the capacitance *in vacuum*, that is, with no dielectric present. ϵ_r has no units. It is a pure multiplying factor, always greater than 1.

MEMORY HOOK: the dielectric *helps*. It always multiplies the capacitance *up*, never down. The reason is polarisation from Unit 2.1: the dielectric's dipoles line up and partly cancel the field between the plates, so the voltage falls for the same charge, and since $C = Q/V$ a smaller V means a larger C .

WORKED EXAMPLE 2 · INSERTING A DIELECTRIC

SLIDE

A parallel plate capacitor has capacitance 8.85×10^{-11} F. What is the capacitance if a dielectric of constant 3 is inserted between the plates?

- 1 The stated capacitance is the value without a dielectric, so $C_0 = 8.85 \times 10^{-11}$ F.

The subscript nought means "in vacuum".

- 2 Apply $C = \epsilon_r C_0 = 3 \times 8.85 \times 10^{-11}$ F.

A single multiplication. The geometry has not changed, only the material.

- 3 $= 26.55 \times 10^{-11}$ F $= 2.655 \times 10^{-10}$ F.

$$C \approx 2.7 \times 10^{-10} \text{ F}$$

Sanity check: the answer must be *bigger* than the original, because a dielectric always increases capacitance. It went from 0.885×10^{-10} to 2.655×10^{-10} , exactly three times larger. If your answer ever comes out smaller after adding a dielectric, you have divided when you should have multiplied.

Combining capacitors

● MUST-MEMORISE · CAPACITORS IN PARALLEL AND IN SERIES

	PARALLEL	SERIES
Equivalent	$C_{\text{eq}} = C_1 + C_2 + \dots + C_n$	$1/C_{\text{eq}} = 1/C_1 + 1/C_2 + \dots + 1/C_n$
Voltage	Same across each: $V_1 = V_2 = V$	Divides: $V_i = Q/C_i$, and $V = \sum V_i$
Charge	Adds: $Q_{\text{tot}} = \sum C_i V$	Same on each: $Q_1 = Q_2 = Q$
Result	Increases total capacitance	C_{eq} is <i>less than the smallest</i> capacitor present

MEMORY HOOK: capacitors behave *backwards* compared with resistors. Capacitors **add in parallel**; resistors add in series. If you remember one, you get the other by flipping it.

TEACH · WHY THE RULES ARE THAT WAY ROUND

Parallel: putting capacitors side by side is effectively making one big capacitor with more plate area. Since C is proportional to A , capacitance simply adds. **Series:** connecting them end to end is like increasing the total separation d . Since C is proportional to $1/d$, capacitance *falls*. That is why the series result is always smaller than the smallest capacitor in the chain, which is a free sanity check on every series answer you produce. For exactly **two** capacitors in series you may use the product-over-sum shortcut from Foundations F.6: $C_{eq} = C_1 C_2 / (C_1 + C_2)$.

Energy stored in a capacitor

TEACH · WHERE THE FACTOR OF ONE HALF COMES FROM

Charge a capacitor from zero up to a final charge Q . At an intermediate moment it already carries charge q , and the potential difference across it is $v = q/C$. To add one more infinitesimal scrap of charge dq , you must do work against that existing potential:

$$dW = v dq = (q/C) dq$$

The point is that **each scrap is harder to add than the last**, because the capacitor is already partly charged and pushes back. So the total work is *not* simply QV . Adding up all those unequal scraps, using the one integral from Foundations F.8:

$$W = \int_0^Q (q/C) dq = (1/C) \cdot [q^2/2]_0^Q = Q^2 / 2C$$

That is where the one half is born: from the fact that the voltage rose steadily from zero to V as you charged it, so the *average* voltage you worked against was only $V/2$.

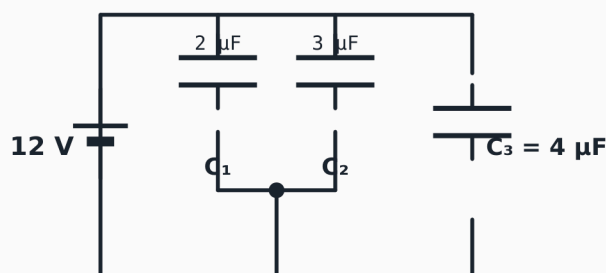
● MUST-MEMORISE · THE THREE FORMS OF THE ENERGY STORED

Starting from $W = Q^2/2C$ and using $Q = CV$, all three of these are the same equation wearing different clothes:

$$W = Q^2 / 2C = \frac{1}{2}CV^2 = \frac{1}{2}QV$$

MEMORY HOOK: memorise one, derive the other two by substituting $Q = CV$. Use whichever form matches the quantities you were given. If you know C and V , use $\frac{1}{2}CV^2$. If you know Q and C , use $Q^2/2C$.

A circuit has $C_1 = 2.0 \mu\text{F}$ and $C_2 = 3.0 \mu\text{F}$ in parallel. That combination is in series with $C_3 = 4.0 \mu\text{F}$. The whole thing is connected to a 12 V battery. Find (a) the equivalent capacitance, (b) the total charge, (c) the total energy stored.



C_1 and C_2 are in parallel; that pair is in series with C_3

C_1 and C_2 add in parallel to give $5 \mu\text{F}$. That is then in series with C_3 .

- 1 Deal with the parallel pair first, because it is self-contained. Capacitors in parallel simply add: $C_p = C_1 + C_2 = 2 + 3 = 5 \mu\text{F}$.
Always collapse the innermost group first, then treat it as a single component.
- 2 Now $C_p = 5 \mu\text{F}$ is in series with $C_3 = 4 \mu\text{F}$. Two items, so use product over sum: $C_{eq} = (5 \times 4) / (5 + 4) = 20/9$.
Series means reciprocals, and product over sum is the two-item shortcut.
- 3 $C_{eq} = 20/9 = 2.22 \mu\text{F}$.
Sanity check: smaller than the smallest in the series pair ($4 \mu\text{F}$). Correct.
- 4 Total charge: $Q = C_{eq}V = (20/9 \times 10^{-6})(12) = 2.67 \times 10^{-5} \text{ C}$.
Convert microfarads to farads before multiplying, or the answer will be out by 10^6 . Keep C_{eq} exact as $20/9$ here: the rounded 2.22 gives $2.66 \times 10^{-5} \text{ C}$, wrong in the third figure. That is this manual's own rounding trap, in the wild.
- 5 Total energy: $W = \frac{1}{2}CV^2 = \frac{1}{2}(2.22 \times 10^{-6})(12)^2 = \frac{1}{2}(2.22 \times 10^{-6})(144)$.
Square the voltage before multiplying. $12^2 = 144$.

(a) $C_{eq} \approx 2.22 \mu\text{F}$ · (b) $Q \approx 26.7 \mu\text{C}$ · (c) $W \approx 1.60 \times 10^{-4} \text{ J}$

Now redo it: I recomputed each independently: $C_{eq} = 2.2222 \mu\text{F}$, $Q = 2.6667 \times 10^{-5} \text{ C}$, $W = 1.600 \times 10^{-4} \text{ J}$. All three slide values are correct. The method to carry away is *collapse the innermost group first*. It works for every network you will meet.

RECALL AND CHECK

Three capacitors, $2 \mu\text{F}$, $4 \mu\text{F}$ and $8 \mu\text{F}$, are connected in series. Without calculating, what can you immediately say about the answer?

ANSWER

It must be **less than $2 \mu\text{F}$** , the smallest one present, because a series combination is always smaller than its smallest member. (For the record, the actual value is $1/C = 1/2 + 1/4 + 1/8 = 0.875$, so $C = 1.14 \mu\text{F}$, which is indeed under $2 \mu\text{F}$.) Use this check on every series answer.

FILL IN THE GAP · CAPACITANCE AND DIELECTRICS

1. Capacitance is fixed by the _____ of the conductors, not by Q or V .
2. Capacitors _____ in parallel, which is the _____ of how resistors behave.
3. A series combination of capacitors is always _____ than the smallest one present.
4. A dielectric always _____ the capacitance, by a factor ϵ_r .

ANSWERS

1. **geometry.** Double the voltage and the charge doubles too, so the *ratio* C is unchanged.
2. **add**, and this is the **opposite** (reverse) of resistors.
3. **smaller.** A free sanity check.
4. **increases** (never decreases). If your answer shrinks after adding a dielectric, you divided when you should have multiplied.

PART IV

Module 3: Electricity and Circuit Analysis

Charge in motion. Current, resistance and the semiconductor devices built from it, then how to analyse any circuit at all: series, parallel, dividers, and finally Kirchhoff's two laws, which will solve circuits that nothing else can touch.



Electric current, resistance and semiconductors

Everything in Modules 1 and 2 held charges still. Now we let them move.

TEACH · WHAT A CURRENT IS

All electric circuits have one thing in common: they depend on the flow of electric charge. When charge flows from one place to another, we say it forms an **electric current**. The more charge that flows, and the faster it flows, the greater the current. In general, the charge is carried through a circuit by electrons.

MUST-MEMORISE · DEFINITION OF ELECTRIC CURRENT

The current is the amount of charge passing a point, divided by the time taken:

$$I = \Delta Q / \Delta t$$

I = current, in amperes (A) **ΔQ** = charge that flows past the point, in coulombs (C) **Δt** = time taken, in seconds (s) SI unit: coulombs per second (C/s), which is the ampere (A).

MEMORY HOOK: 1 A = 1 C/S. One amp is one coulomb of charge passing per second. The unit is named after the French physicist Andre-Marie Ampere.

WORKED EXAMPLE · COMPARING CURRENTS

SLIDE, CONCEPTUAL

Which has the greatest current, and which the smallest? (a) the flow of 6 C in 3 s, (b) the flow of 3 C in 1 s, (c) a person walking 8 m in 4 s.

1 (a) $I = 6 \text{ C} / 3 \text{ s} = 2 \text{ A}$.

Current is charge per time. Just divide.

2 (b) $I = 3 \text{ C} / 1 \text{ s} = 3 \text{ A}$.

Less charge than (a), but it moves in a third of the time, so the current is higher.

3 (c) The current is **zero**.

A person's net charge is essentially zero. Moving a neutral object from one place to another transports no net charge, so there is no current. This is the whole point of the question.

Greatest: (b) at 3 A. Smallest: (c) at zero.

The lesson: current is not about things moving. It is about *net charge* moving. Something can move a great deal and carry no current at all.

WORKED EXAMPLE · HOW MANY ELECTRONS?

SLIDE, GUIDED

A disk drive draws a current of 0.22 A for 4.5 s. How many electrons pass through it? (Charge on an electron, $e = 1.60 \times 10^{-19}$ C.)

- 1 First find the total charge. Rearrange $I = \Delta Q / \Delta t$ to give $\Delta Q = I \Delta t$.

We know the current and the time, so the charge follows immediately.

- 2 $\Delta Q = (0.22 \text{ A})(4.5 \text{ s}) = \mathbf{0.99 \text{ C}}$.

Amps times seconds gives coulombs, straight from the definition.

- 3 Now divide that total charge by the charge on *one* electron: $N = \Delta Q / e$.

If each electron carries e , then the number of them is the total divided by the individual amount. This is just counting.

- 4 $N = 0.99 / (1.60 \times 10^{-19}) = 6.2 \times 10^{18}$ electrons.

Dividing by 10^{-19} is multiplying by 10^{19} , and $0.99/1.60 = 0.62$, giving $0.62 \times 10^{19} = 6.2 \times 10^{18}$.

$$N \approx 6.2 \times 10^{18} \text{ electrons}$$

Now redo it: I recomputed this as 6.1875×10^{18} , so the slide's 6.2×10^{18} is correct. Note the insight the slides draw: even a modest current flowing briefly moves an *extraordinary* number of electrons.

TEACH · CIRCUITS, DC AND AC

When charge flows through a closed path and returns to its starting point, that closed path is an **electric circuit**.

- ▶ In a **direct current (DC)** circuit, the current always flows in the same direction. Battery-powered circuits are typically DC.
- ▶ In an **alternating current (AC)** circuit, the current periodically reverses direction. The electricity from a wall socket is AC.

TEACH · THE BATTERY, AND WHY ANYTHING FLOWS AT ALL

Electrons move freely in a metal, but something must *push* them. The slides use a garden hose: hold both ends level and no water flows, because there is no difference in potential energy. Raise one end and water flows from high potential energy to low. A battery does exactly this job in a circuit. A battery uses **chemical reactions** to produce a difference in electric potential between its two terminals. The positive terminal is at high potential, the negative terminal at low potential. The difference in potential between the terminals is the **electromotive force**, or **emf**, given the symbol ϵ . Despite the name it is *not* a force: its unit is the volt, the same as potential.

● MUST-MEMORISE · WORK DONE BY A BATTERY

$$W = (\Delta Q) \epsilon$$

W = work done, in joules (J) **ΔQ** = charge moved, in coulombs (C) **ϵ** = emf of the battery, in volts (V)

MEMORY HOOK: this is the same $W = qV$ from Module 1. Charge times voltage gives energy, every single time. The emf determines how much work the battery does to move a given amount of charge around the circuit.

WORKED EXAMPLE · CHARGE AND WORK IN A FLASHLIGHT
SLIDE, ACTIVE

A battery of emf 1.5 V delivers a current of 0.44 A to a bulb for 64 s. Find (a) the charge that passes through the circuit and (b) the work done by the battery.

1 (a) Rearrange the definition of current: $\Delta Q = I \Delta t = (0.44)(64) = \mathbf{28\ C}$.
28.16 C exactly, which the slides round to 28 C.

2 (b) Now use $W = \Delta Q \varepsilon = (28)(1.5) = \mathbf{42\ J}$.
Coulombs times volts gives joules.

$$\Delta Q \approx 28\ \text{C} \quad \cdot \quad W \approx 42\ \text{J}$$

Insight from the slides: the more charge a battery moves, the more work it does; and the greater the emf, the more work. So a 12 V car battery delivering several amps does vastly more work than this 1.5 V flashlight cell, exactly as you would expect.

⚠ TRAP · CONVENTIONAL CURRENT RUNS BACKWARDS TO THE ELECTRONS

By convention, the **direction of current** in a circuit is the direction in which a *positive* test charge would move: out of the positive terminal, round the circuit, and back into the negative terminal. But the charges that actually move in a metal wire are **negatively charged electrons**, which leave the *negative* terminal and return to the positive one. So **the electron flow and the current arrow point in opposite directions**. This is a historical accident, not a mistake. Every formula in this course uses conventional current, so draw your arrows that way and do not worry about the electrons.

Resistance and Ohm's law
● MUST-MEMORISE · OHM'S LAW

$$V = IR \quad I = V/R \quad R = V/I$$

V = applied voltage, in volts (V) **I** = current, in amperes (A) **R** = resistance, in **ohms (Ω)**, where **1 Ω = 1 V/A**

MEMORY HOOK: $V = IR$. Cover the one you want with your thumb and the formula rearranges itself. Named for the German physicist Georg Simon Ohm. A device that measures resistance is an ohmmeter.

WORKED EXAMPLE · CURRENT THROUGH A RESISTOR
SLIDE, QUICK

A potential difference of 24 V is applied to a 150 Ω resistor. How much current flows?

1 Solve Ohm's law for current: $I = V/R$.

2 $I = 24\ \text{V} / 150\ \Omega = \mathbf{0.16\ A}$.
Volts divided by ohms gives amps, since 1 Ω = 1 V/A.

$$I = 0.16\ \text{A}$$

TEACH · THE CIRCUIT ELEMENTS AND THEIR SYMBOLS

ELEMENT	PHYSICAL CHARACTERISTICS
Resistor	Resists the flow of current. Converts electrical energy into thermal energy. Drawn as a zigzag line.
Ideal wire	Has <i>zero</i> resistance. Used to connect the elements together.
Battery	Produces a constant difference in electrical potential between its terminals.
Switch	Controls whether current is allowed to flow. Open means no charge can flow; closed means current flows.
Diode	Allows current to flow in one direction only.
Incandescent bulb	Contains a resistor that gets hot enough to give off visible light.

● MUST-MEMORISE · RESISTIVITY: WHAT A WIRE'S RESISTANCE DEPENDS ON

$$R = \rho L / A$$

ρ (rho) = resistivity of the material, in ohm metres ($\Omega \text{ m}$) L = length of the wire, in m A = cross-sectional area of the wire, in m^2

MEMORY HOOK: long and thin is high resistance; short and fat is low. Think of water in a hose: a long narrow hose resists flow, a short wide one does not. Insulators have *high* resistivity, conductors have *low* resistivity.

TEACH · THE FOUR FACTORS AFFECTING THE RESISTANCE OF A WIRE

FACTOR	EFFECT	REASON
Material	Copper is lower than lead	Different materials have different resistivity ρ .
Length	Longer means more resistance	L is on the top of $R = \rho L / A$.
Area	Thinner means more resistance	A is on the bottom, so a small area means a large R .
Temperature	Hotter usually means more resistance	Atoms jiggle faster, collide with electrons more, and slow their progress.

Note the temperature row applies to *metals*. Semiconductors do the opposite, which is exactly the exception discussed in Unit 2.1.

Semiconductors

TEACH · DOPING, AND THE TWO TYPES OF SEMICONDUCTOR

Ohm's law is an excellent approximation for metal wires and ordinary resistors, but it does not apply to all materials. **Semiconductors are an important exception.** Elements such as germanium and silicon are *insulators* in their pure form. But adding a tiny quantity of impurity, a process called **doping**, turns them into conductors. Doping produces two types of semiconductor.

- ▶ **n-type:** add arsenic to silicon (roughly one arsenic atom per million silicon atoms) and electrons break free from the arsenic and move through the material. Current is carried by **negative** electrons, hence the name.
- ▶ **p-type:** doped the other way, so that current is carried by positive charge carriers.

MUST-MEMORISE · DIODE AND TRANSISTOR

Diode. The simplest semiconductor device: a p-type joined to an n-type. Its basic property is that it **allows current to flow in one direction only**. **Transistor.** A sandwich of three layers. The common type has n-type on either side of a thin p-type middle: an **npn transistor**. Reverse the sequence and you have a **pn transistor**. Its three terminals are the collector, the base and the emitter. The basic function of a transistor is to act as an **electronic switch** that controls the flow of current in a circuit.

MEMORY HOOK: the diode is a one-way valve; the transistor is a switch. Those two devices, made from doped silicon, are the foundation of every computer ever built.

RECALL AND CHECK

A wire is stretched so that it becomes twice as long and half as thick in cross-sectional area. What happens to its resistance?

ANSWER

$R = \rho L/A$. Doubling L doubles R . Halving A doubles R again. So the resistance becomes **four times** as large. Both changes push in the same direction, so they multiply rather than cancel.

FILL IN THE GAP · CURRENT, RESISTANCE AND SEMICONDUCTORS

1. One ampere is one _____ per second.
2. Conventional current runs in the _____ direction to the electron flow.
3. In $R = \rho L/A$, a long thin wire has a _____ resistance.
4. A _____ allows current in one direction only, while a _____ acts as an electronic switch.

ANSWERS

1. **coulomb**. $1 \text{ A} = 1 \text{ C/s}$.
2. **opposite**. A historical accident, but every formula in this course uses conventional current.
3. **high** (large). L is on top, A is on the bottom.
4. **diode**, and a **transistor**. A one-way valve and a switch, which together build every computer.

Series, parallel and combination circuits

Nearly every circuit question reduces to this: replace groups of resistors with a single equivalent resistor until only one remains, then apply Ohm's law.

● MUST-MEMORISE · RESISTORS IN SERIES

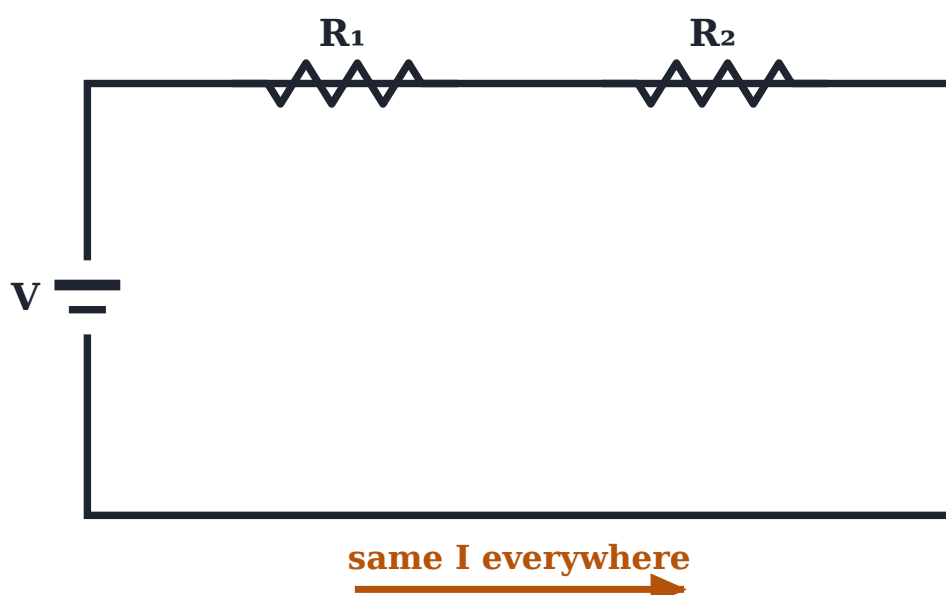
Resistors connected end to end form a **series circuit**. The equivalent resistance is simply the sum:

$$R_{\text{eq}} = R_1 + R_2 + R_3 + \dots$$

The defining features of a series circuit:

- ▶ The **same current** flows through every resistor.
- ▶ The voltages across them **add up** to the supply voltage.
- ▶ R_{eq} is **greater** than the largest individual resistance.

MEMORY HOOK: series adds. More resistors in a line means a longer, harder path, so more resistance. Two identical resistors R in series give $2R$.



In series the same current flows through every resistor.

● MUST-MEMORISE · RESISTORS IN PARALLEL

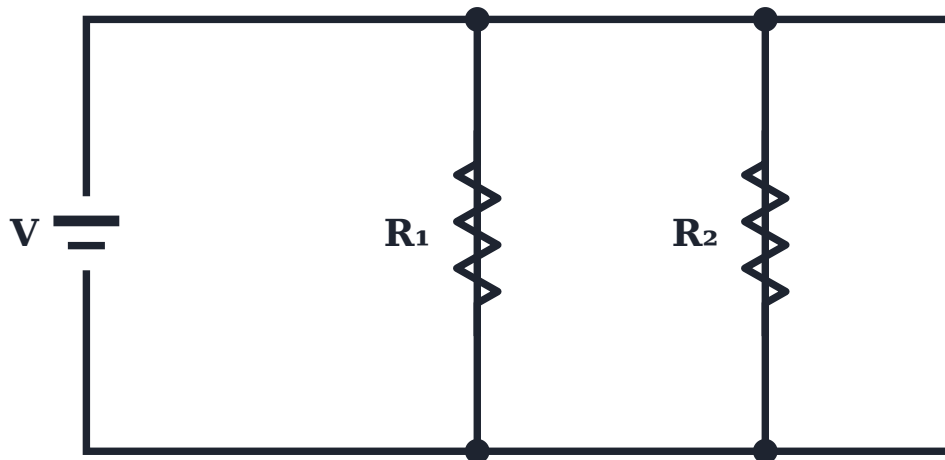
Resistors connected across the **same potential difference** form a **parallel circuit**. The reciprocals add:

$$1/R_{\text{eq}} = 1/R_1 + 1/R_2 + 1/R_3 + \dots$$

The defining features of a parallel circuit:

- ▶ The **same voltage** is across every resistor.
- ▶ The currents through the branches **add up** to the total current.
- ▶ R_{eq} is **less than the smallest** individual resistance.

MEMORY HOOK: parallel lanes on a motorway. Adding another lane makes it *easier* for traffic to get through, so resistance *falls*. Two identical resistors R in parallel give $R/2$; three give $R/3$.



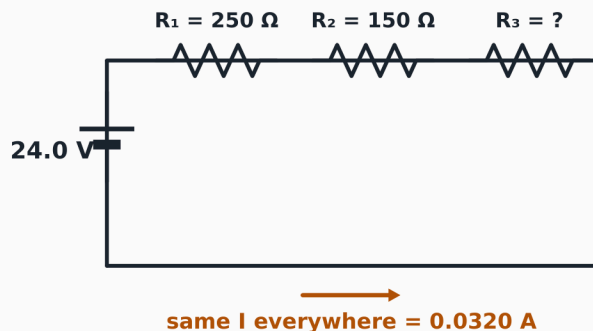
same V across each branch

In parallel every branch has the same voltage across it.

⚠ TRAP · DO NOT FORGET TO FLIP THE RECIPROCAL

The parallel formula gives you $1/R_{eq}$, not R_{eq} . You must invert at the end. Stopping one step early is the most common error in the entire course. Free check, every time: **a parallel answer must be smaller than the smallest resistor in the group.** If it is not, you forgot to flip.

Three resistors are in series across a 24.0 V battery. The current is 0.0320 A. Given $R_1 = 250.0 \, \Omega$ and $R_2 = 150.0 \, \Omega$, find (a) R_3 and (b) the potential difference across each resistor.



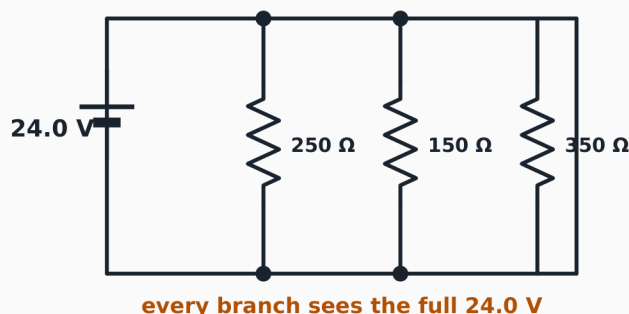
Series: one path, so the same current flows through all three.

- 1 Find the total resistance from Ohm's law: $R_{\text{eq}} = \mathcal{E}/I = 24.0 / 0.0320 = \mathbf{750 \, \Omega}$.
The same current flows through all three, which is the key property of series.
- 2 In series the resistances simply add, so $R_3 = R_{\text{eq}} - R_1 - R_2 = 750 - 250 - 150 = \mathbf{350 \, \Omega}$.
One unknown, one equation. Straightforward subtraction.
- 3 Now the voltage across each, using $V = IR$ with the *same* I each time. $V_1 = (0.0320)(250.0) = \mathbf{8.00 \, \text{V}}$.
This is the whole reason series circuits are easy: I is common to all.
- 4 $V_2 = (0.0320)(150.0) = \mathbf{4.80 \, \text{V}}$.
- 5 $V_3 = (0.0320)(350) = \mathbf{11.2 \, \text{V}}$.

$$R_3 = 350 \, \Omega \quad \cdot \quad V_1 = 8.00 \, \text{V}, \quad V_2 = 4.80 \, \text{V}, \quad V_3 = 11.2 \, \text{V}$$

The check that proves it: the individual voltages must add up to the battery voltage. $8.00 + 4.80 + 11.2 = \mathbf{24.0 \, \text{V}}$. Exactly the emf. This check costs five seconds and catches almost every arithmetic slip. Note also that the largest resistance takes the largest share of the voltage.

$R_1 = 250.0 \, \Omega$, $R_2 = 150.0 \, \Omega$ and $R_3 = 350.0 \, \Omega$ are connected in parallel across a $24.0 \, \text{V}$ battery. Find (a) the total current from the battery and (b) the current through each resistor.



Parallel: every branch is connected across the same two nodes, so each sees the full battery voltage.

- 1 Add the reciprocals: $1/R_{\text{eq}} = 1/250.0 + 1/150.0 + 1/350.0 = 0.004 + 0.006667 + 0.002857 = 0.01352 \, \Omega^{-1}$.
Three items, so the product-over-sum shortcut is not available. Use reciprocals.
- 2 Flip it: $R_{\text{eq}} = 1 / 0.01352 = \mathbf{73.96 \, \Omega}$.
Do not stop at the previous line. Sanity check: 73.96 is less than 150, the smallest resistor. Correct.
- 3 Total current: $I = \mathcal{E}/R_{\text{eq}} = 24.0 / 73.96 = \mathbf{0.325 \, \text{A}}$.
- 4 Each branch has the full 24.0 V across it, so use $I = \mathcal{E}/R$ for each. $I_1 = 24.0/250.0 = \mathbf{0.0960 \, \text{A}}$.
This is the key property of parallel: every branch sees the same voltage.
- 5 $I_2 = 24.0/150.0 = \mathbf{0.160 \, \text{A}}$, and $I_3 = 24.0/350.0 = \mathbf{0.0686 \, \text{A}}$.

$$I = 0.325 \, \text{A total} \quad \cdot \quad I_1 = 0.0960 \, \text{A}, \quad I_2 = 0.160 \, \text{A}, \quad I_3 = 0.0686 \, \text{A}$$

The check that proves it: the branch currents must add to the total. $0.0960 + 0.160 + 0.0686 = \mathbf{0.325 \, \text{A}}$. It matches. Note also that the *smallest* resistor ($150 \, \Omega$) carries the *largest* current, as it must.

Short circuits

TEACH · WHAT HAPPENS WHEN ONE BRANCH HAS ZERO RESISTANCE

In general the equivalent resistance of a parallel circuit is less than or equal to the smallest individual resistance. So what if one of them is **zero**? Then R_{eq} is also zero, because it cannot be larger than the smallest, and it cannot be negative. This is a **short circuit**: *all* the current flows through the path of zero resistance, and none flows through the other branches. They are by-passed completely.

Combination circuits

● MUST-MEMORISE · THE METHOD FOR ANY CIRCUIT

Most real circuits are neither purely series nor purely parallel. The method is always the same:

1. **Find a group** that is purely series or purely parallel.
2. **Replace it** with its single equivalent resistance.
3. **Redraw** the simpler circuit.
4. **Repeat** until one resistor remains, then use Ohm's law.

MEMORY HOOK: collapse from the inside out. Never try to see the whole circuit at once. Take one group at a time and redraw after each step. Redrawing is not optional: it is what stops you making mistakes.

WORKED EXAMPLE · FOUR RESISTORS, EACH R

SLIDE

A circuit has four resistors each of resistance R , combining series and parallel features. Two are in parallel on the right; those sit in series with two others.

- 1 Start with the two vertical resistors on the right, which are in *parallel*. Two identical resistors in parallel give $R/2$.

$$1/R_{eq} = 1/R + 1/R = 2/R, \text{ so } R_{eq} = R/2.$$

- 2 Replace them with a single $R/2$. The circuit is now three resistors in *series*: R , then $R/2$, then R .

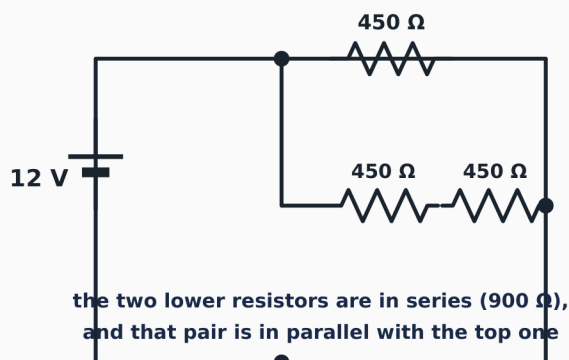
This is why redrawing matters. Once replaced, the structure is obvious.

- 3 Series resistances add: $R_{eq} = R + R/2 + R = \mathbf{2.5R}$.

$$R_{eq} = 2.5R$$

Now redo it: the whole technique is visible here. Collapse the parallel pair, redraw, then add the series chain. Any combination circuit yields to this.

A 12 V battery drives three resistors, each $450\ \Omega$. Two are in series with one another, and that pair is in parallel with the third. Find the current supplied by the battery.



Two $450\ \Omega$ in series ($900\ \Omega$), in parallel with a third $450\ \Omega$.

- 1 The two lower resistors are in series: $R_{\text{lower}} = R + R = 2R = 900\ \Omega$.
Collapse the innermost group first.
- 2 That $2R$ is now in *parallel* with the single upper R . Two items, so use product over sum: $R_{\text{eq}} = (R \times 2R) / (R + 2R) = 2R^2/3R = (2/3)R$.
Working in symbols first keeps it clean. Substitute numbers at the end.
- 3 $R_{\text{eq}} = (2/3)(450) = \mathbf{300\ \Omega}$.
Sanity check: $300\ \Omega$ is less than $450\ \Omega$, the smallest branch. Correct.
- 4 Ohm's law for the current from the battery: $I = \varepsilon/R_{\text{eq}} = 12 / 300 = \mathbf{0.040\ A}$.

$$I = 0.040\ \text{A}$$

Note the insight from the slides: the total resistance of three $450\ \Omega$ resistors is only $300\ \Omega$, which is *less* than any single one of them. That feels wrong until you remember the parallel connection provides an extra path, and extra paths always reduce total resistance.

RECALL AND CHECK

Two $100\ \Omega$ resistors. What is the equivalent resistance in series, and in parallel?

ANSWER

Series: $100 + 100 = \mathbf{200\ \Omega}$ (twice the individual value). Parallel: $R/2 = \mathbf{50\ \Omega}$ (half the individual value). In series it goes up; in parallel it goes down. Always.

FILL IN THE GAP · SERIES, PARALLEL AND COMBINATION CIRCUITS

1. In series, the _____ is the same through every resistor, and the _____ divide.
2. In parallel, the _____ is the same across every branch, and the _____ divide.
3. To solve a combination circuit you collapse the _____ group first, then _____ the circuit.

ANSWERS

1. current is common; the **voltages** divide.

2. voltage is common; the **currents** divide.

These two lines are the whole of circuit analysis. Everything else follows.

3. innermost (or simplest) group, then **redraw**. Redrawing is not optional; it is what stops you making mistakes.

UNIT 3.3

Voltage dividers and current dividers

Two shortcuts that let you find a voltage or a current directly, without computing the whole circuit first. Worth knowing well: they turn several lines of work into one.

● MUST-MEMORISE · THE VOLTAGE DIVIDER RULE

The voltage drop across any given resistor in a **series** circuit equals the ratio of that resistor to the total resistance, multiplied by the source voltage.

$$V_1 = [R_1 / (R_1 + R_2)] \times V$$

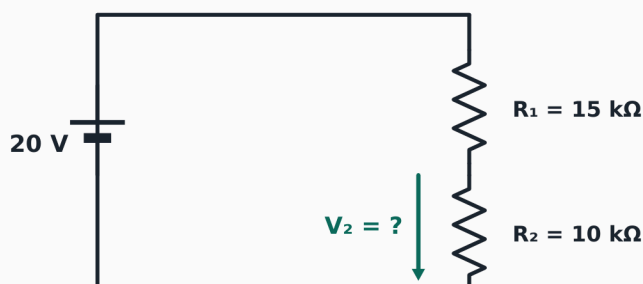
and correspondingly $V_2 = [R_2 / (R_1 + R_2)] \times V$. In general, replace the denominator with the *total* series resistance R_T .

MEMORY HOOK: the resistor you want goes on *top*; the total goes underneath. The bigger resistor takes the bigger share of the voltage, which is exactly what you would expect.

WORKED EXAMPLE · VOLTAGE ACROSS R 2

SLIDE

A 20 V source drives $R_1 = 15 \text{ k}\Omega$ in series with $R_2 = 10 \text{ k}\Omega$. What is the voltage across R_2 ?



The two resistors share the 20 V in proportion to their size.

- 1 Total series resistance: $R_T = 15 + 10 = 25 \text{ k}\Omega$.
Series, so simply add.
- 2 Apply the divider, putting the resistor we want (R_2) on top: $V_2 = (10 \text{ k}\Omega / 25 \text{ k}\Omega) \times 20 \text{ V}$.
The kilo-ohms cancel top and bottom, so no conversion is needed. This is one of the nice features of the divider rule.
- 3 $V_2 = (0.4)(20) = 8.0 \text{ V}$.

$$V_2 = 8.0 \text{ V}$$

Check it: R_1 should then take $(15/25) \times 20 = 12 \text{ V}$. And $8 + 12 = 20 \text{ V}$, the source voltage. It balances. R_1 is larger, and it took the larger share. That is the sanity check to run every time.

WORKED EXAMPLE · DIVIDER WITH 4 Ω AND 10 Ω

CLASSWORK

A 12 V source drives $R_1 = 4 \Omega$ in series with $R_2 = 10 \Omega$. Find the voltage drop across each.

- 1 $R_T = 4 + 10 = 14 \Omega$.
- 2 $V_1 = (4/14) \times 12 = 48/14 = 3.43 \text{ V}$.
- 3 $V_2 = (10/14) \times 12 = 120/14 = 8.57 \text{ V}$.

$$V_1 \approx 3.43 \text{ V} \quad \cdot \quad V_2 \approx 8.57 \text{ V}$$

Check: $3.43 + 8.57 = 12.0 \text{ V}$ exactly. And the 10Ω resistor took the larger share, as it must.

● MUST-MEMORISE · THE CURRENT DIVIDER RULE

When current enters a node (junction) it divides into currents that are **inversely proportional** to the resistance values. For two resistors in parallel:

$$I_1 = [R_2 / (R_1 + R_2)] \times I_T$$

Note the subscripts carefully. To find the current in R_1 , the resistor in the *numerator* is R_2 , the *other* one.

MEMORY HOOK: for current, the *opposite* resistor goes on top. This is the reverse of the voltage divider, and it is deliberately so: current takes the path of *least* resistance, so the smaller resistor gets the larger current. Putting the other resistor on top is what produces that inversion.

⚠ TRAP · THE TWO DIVIDERS USE OPPOSITE RESISTORS

This is the single most confusing pair of formulas in the module, so be deliberate:

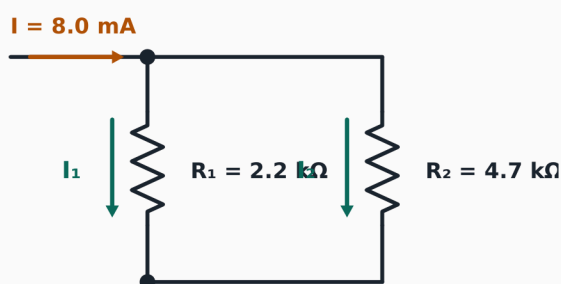
- ▶ **Voltage** divider: the resistor you *want* goes on top. Bigger R means bigger V.
- ▶ **Current** divider: the *other* resistor goes on top. Bigger R means *smaller* I.

If you ever forget which is which, reason it out rather than guessing: current prefers the easy path, so the small resistor must get the big current. Only one of the two arrangements delivers that.

WORKED EXAMPLE · CURRENT DIVIDER, 2.2 kΩ AND 4.7 kΩ

SLIDE

$R_1 = 2.2 \text{ k}\Omega$ is in parallel with $R_2 = 4.7 \text{ k}\Omega$. The total current into the pair is 8.0 mA. Find the current in each resistor.



Current divides inversely with resistance at the junction.

- 1 Sum of the resistances: $2.2 + 4.7 = 6.9 \text{ k}\Omega$.
Both in kΩ, so the units will cancel in the ratio.
- 2 Current through R_1 , with the *other* resistor on top: $I_1 = (4.7 / 6.9) \times 8.0 \text{ mA} = (0.681)(8.0) = 5.45 \text{ mA}$.
 R_2 on top because we want the current in R_1 .
- 3 Current through R_2 : $I_2 = (2.2 / 6.9) \times 8.0 \text{ mA} = (0.319)(8.0) = 2.55 \text{ mA}$.

$$I_1 \approx 5.45 \text{ mA (through } 2.2 \text{ k}\Omega) \quad \cdot \quad I_2 \approx 2.55 \text{ mA (through } 4.7 \text{ k}\Omega)$$

Two checks: first, $5.45 + 2.55 = 8.0 \text{ mA}$, the total. Second, and as the slides point out, **the larger resistor has the smaller current**. The $4.7 \text{ k}\Omega$ carries only 2.55 mA while the $2.2 \text{ k}\Omega$ carries 5.45 mA. If your answer has the big resistor carrying the big current, you have put the wrong resistor on top.

A source drives $R_1 = 100\ \Omega$ in parallel with $R_2 = 200\ \Omega$. Q1: what is the current in R_2 ? Options: 6.7 mA, 13.3 mA, 20 mA, 26.7 mA. Q2: what is the voltage across R_2 ? Options: 0 V, 0.67 V, 1.33 V, 4.0 V.

- 1** Work backwards from the options to identify the source. If the total current is 20 mA, the current divider gives $I_2 = (100/300) \times 20 = 6.67\text{ mA}$.

The other resistor, 100 Ω , goes on top when finding the current in R_2 .

- 2** So $I_2 \approx \mathbf{6.7\text{ mA}}$, which is option (a).

Note the larger resistor (200 Ω) gets the smaller current, exactly as required.

- 3** For the voltage: in parallel, both resistors have the *same* voltage across them. $V = I_2 R_2 = (6.67 \times 10^{-3})(200) = \mathbf{1.33\text{ V}}$.

Which is option (c). Check with the other branch: $I_1 = (200/300) \times 20 = 13.3\text{ mA}$, and $V = (13.3 \times 10^{-3})(100) = 1.33\text{ V}$. Identical, as it must be for parallel.

Q1: 6.7 mA · Q2: 1.33 V

The check that confirms both: the two branch currents must sum to the total, $6.7 + 13.3 = 20\text{ mA}$.

And both branches must show the same voltage, which they do at 1.33 V. Two independent confirmations that the answers are consistent.

FILL IN THE GAP · VOLTAGE AND CURRENT DIVIDERS

1. In the *voltage* divider, the resistor you _____ goes on top.
2. In the *current* divider, the _____ resistor goes on top.
3. Current takes the path of least resistance, so the _____ resistor carries the larger current.

ANSWERS

1. want. Bigger R means a bigger share of the voltage.

2. other (the opposite one). This inversion is deliberate.

3. smaller.

If you ever forget which formula is which, reason it out from this fact rather than guessing. Only one arrangement of the ratio gives the small resistor the big current.

UNIT 3.4

Kirchhoff's laws

When a circuit cannot be reduced by series and parallel rules alone, Kirchhoff's two laws will always solve it. They are the most powerful tool in this module and the most heavily examined.

● MUST-MEMORISE · THE THREE DEFINITIONS YOU MUST HAVE FIRST

- ▶ **Junction** (also called a node or branch point): a point in a circuit where two or more conductors meet.
- ▶ **Branch**: a single element in the circuit, such as a resistor, a current source or a voltage source.
- ▶ **Loop**: a collection of branches forming a *closed* conducting path.

MEMORY HOOK: junctions are where wires *meet*; loops are closed paths you can walk round and return to the start. Kirchhoff's first law is about junctions, the second is about loops.

● MUST-MEMORISE · KIRCHHOFF'S CURRENT LAW (KCL)

The algebraic sum of the currents entering any junction equals the sum of the currents leaving it.

$$\sum I = 0 \quad \text{or} \quad \sum I_{\text{in}} = \sum I_{\text{out}}$$

MEMORY HOOK: KCL is a statement of the **conservation of charge**. Charge cannot pile up at a junction or vanish there. Whatever flows in must flow out. Think of water pipes meeting at a T-joint.

● MUST-MEMORISE · KIRCHHOFF'S VOLTAGE LAW (KVL)

The algebraic sum of the potential differences across all the elements around any closed loop must be zero.

$$\sum V = 0 \quad \text{or} \quad \sum(\text{voltage drops}) = \sum(\text{voltage rises})$$

MEMORY HOOK: KVL is a statement of the **conservation of energy**. Walk all the way round a loop and return to where you started, and you must be back at the same potential. Like walking round a hill and returning to your front door: your net change in height is zero, however much you climbed on the way.

The sign conventions, which are where marks are lost

● MUST-MEMORISE · THE FOUR SIGN RULES FOR TRAVERSING A LOOP

First choose a **travel direction** around the loop. It may be clockwise or anticlockwise; it genuinely does not matter, as long as you are consistent. Then:

WHAT YOU PASS THROUGH	SIGN OF THE TERM
A source , travelling from $-$ to $+$	emf is positive $(+\varepsilon)$
A source , travelling from $+$ to $-$	emf is negative $(-\varepsilon)$
A resistor , travelling <i>with</i> the assumed current	the IR term is negative $(-IR)$
A resistor , travelling <i>against</i> the assumed current	the IR term is positive $(+IR)$

MEMORY HOOK: going *with* the current through a resistor is going *downhill* in potential, so the term is negative. Going from minus to plus through a battery is climbing *uphill*, so it is positive. Voltage is like height, and you are walking a landscape.

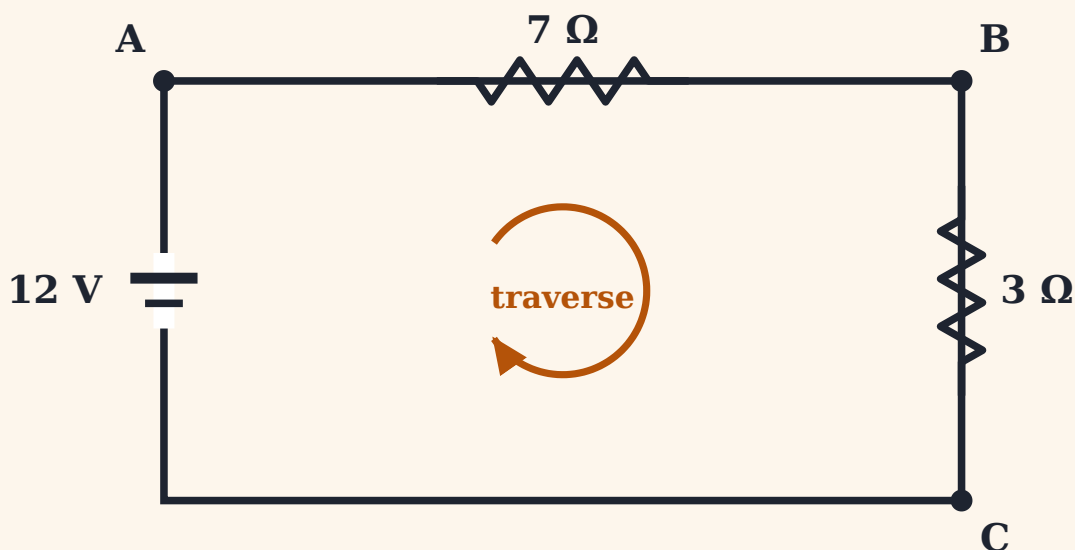
⚠ TRAP · A NEGATIVE CURRENT IS NOT A MISTAKE

You must *assume* a current direction before you can write the equations, and you have no way of knowing in advance whether you guessed right. If your assumption was backwards, the algebra returns a **negative** value. This is not an error, and you should not go back and redo it. It means: the magnitude is correct, but the real current flows in the direction *opposite* to your arrow. Simply state that. Losing your nerve and re-deriving everything is the actual mistake here.

WORKED EXAMPLE · A SINGLE LOOP, WORKED COMPLETELY

SLIDE

A 12 V source drives a 7 Ω resistor and a 3 Ω resistor in a single loop. Find the current, then the potential differences V_{AB} across the 7 Ω and V_{BC} across the 3 Ω .

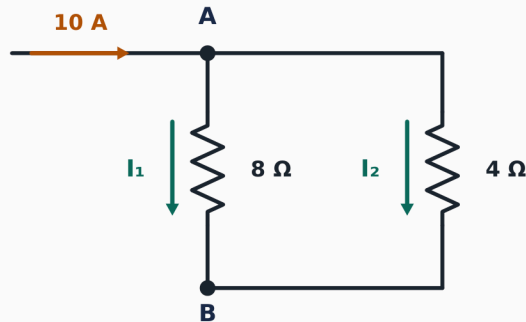


- 1 Assume a current I flows, and choose a travel direction (the arrow shows the one used here; the slides take anticlockwise, which gives the same answer).
Commit to a direction and stick with it. The answer will tell you if you guessed wrong.
- 2 Apply KVL around the loop. Travelling with the current through both resistors gives two drops, and the battery gives a rise: $I(7) + I(3) - 12 = 0$.
Sum of all potential changes round a closed loop is zero. That is the whole law.
- 3 Collect: $10I = 12$, so $I = \mathbf{1.2\text{ A}}$.
Positive, so the assumed direction was correct.
- 4 Now Ohm's law for each resistor, using that current: $V_{AB} = I \times 7 = (1.2)(7) = \mathbf{8.4\text{ V}}$.
- 5 $V_{BC} = I \times 3 = (1.2)(3) = \mathbf{3.6\text{ V}}$.

$$I = 1.2\text{ A} \quad \cdot \quad V_{AB} = 8.4\text{ V} \quad \cdot \quad V_{BC} = 3.6\text{ V}$$

The check: $8.4 + 3.6 = 12.0\text{ V}$, precisely the source voltage. The voltages around the loop balance, which is KVL confirming itself. Note this circuit is a simple series one, so you could have found $I = 12/10$ directly. Kirchhoff agrees with the simple method, which is exactly what should happen.

A 10 A source feeds a junction. One branch has $8\ \Omega$, the other has $4\ \Omega$. Find the current in each branch and the voltage V_{AB} .



At node A the 10 A splits. KCL: whatever goes in must come out.

- 1 By KVL, the two parallel branches must have the same voltage across them, so the loop containing both gives $8I_1 = 4I_2$, hence $I_2 = 2I_1$.
The $4\ \Omega$ branch is half the resistance, so it takes twice the current.
- 2 By KCL at the junction, the total current in must equal the total out: $10 = I_1 + I_2$.
Conservation of charge. Nothing accumulates at the node.
- 3 Substitute the first equation into the second: $10 = I_1 + 2I_1 = 3I_1$.
Two equations, two unknowns, solved by substitution, exactly as in Foundations F.5.
- 4 $I_1 = 10/3 = \mathbf{3.33\ A}$, and therefore $I_2 = 2(3.33) = \mathbf{6.67\ A}$.
The smaller resistor carries the larger current, as always in parallel.
- 5 Voltage across the pair: $V_{AB} = I_1 \times 8 = (3.33)(8) = \mathbf{26.67\ V}$.

$$I_1 = 3.33\ \text{A} \quad \cdot \quad I_2 = 6.67\ \text{A} \quad \cdot \quad V_{AB} = 26.67\ \text{V}$$

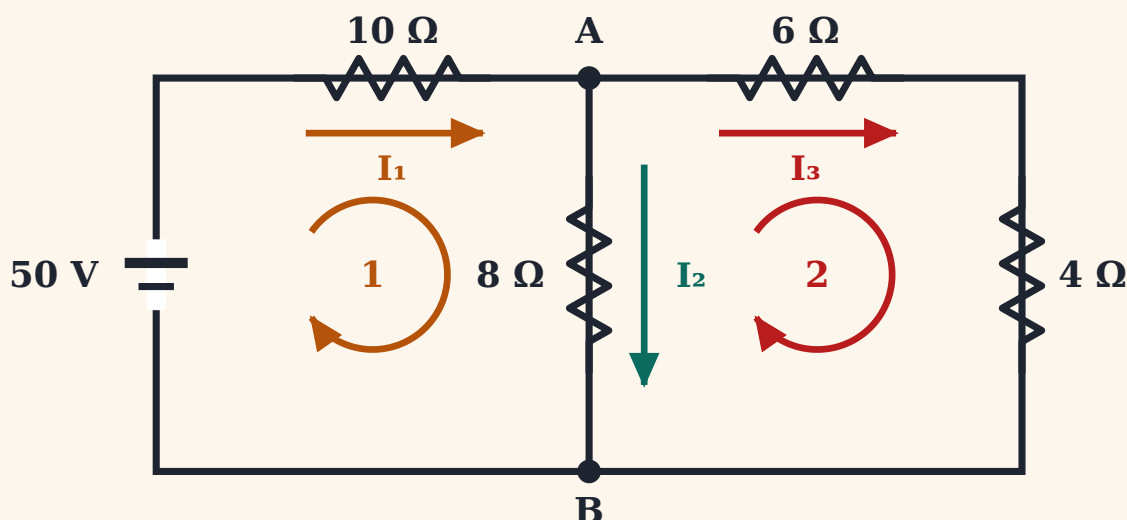
Two checks: the currents sum to $3.33 + 6.67 = 10\ \text{A}$, matching the source. And the other branch gives $V = (6.67)(4) = 26.67\ \text{V}$, the same voltage. Both branches agree, which is exactly what parallel demands.

The full two-loop problem

WORKED EXAMPLE · THE COMPLETE KIRCHHOFF CIRCUIT

SLIDE

A 50 V source drives a network with a 10 Ω , an 8 Ω , a 6 Ω and a 4 Ω resistor arranged in two loops. Solve for the current in each resistor and the voltage across each. This is the flagship example of the module, so follow every line.



- 1 Label the currents. I_1 is the current from the source, I_2 and I_3 are the currents in the two branches. Assume a clockwise traverse.
Three unknowns, so we will need three equations.
- 2 **First loop (KVL):** $-50 + I_1(10) + I_2(8) = 0$.
The battery is a rise, so it enters negative when we traverse it this way; the resistors are drops. This is equation (a).
- 3 **Second loop (KVL):** $-I_2(8) + I_3(6) + I_3(4) = 0$.
We pass through the 8 Ω against the assumed current, so it is positive there and negative here. This is equation (b), and it gives $10I_3 = 8I_2$.
- 4 **At the junction (KCL):** $I_1 = I_2 + I_3$.
Conservation of charge. This is equation (c). Now three equations, three unknowns, so the system is solvable.
- 5 Substitute (c) into (a) to eliminate I_1 : $-50 + (I_2 + I_3)(10) + I_2(8) = 0$, which tidies to $18I_2 + 10I_3 = 50$.
Expand the bracket: $10I_2 + 10I_3$, then add the $8I_2$.
- 6 From (b), $10I_3 = 8I_2$, so $I_2 = 1.25 I_3$. Substitute this into the previous line: $18(1.25 I_3) + 10I_3 = 50$.
Reduce to one unknown, which is always the goal.
- 7 $22.5 I_3 + 10I_3 = 32.5 I_3 = 50$, so $I_3 = 50/32.5 = \mathbf{1.538 \text{ A}}$.
- 8 Back-substitute: $I_2 = 1.25(1.538) = \mathbf{1.923 \text{ A}}$.
Always go back and get the others. Do not stop at one current.
- 9 And from (c): $I_1 = I_2 + I_3 = 1.923 + 1.538 = \mathbf{3.461 \text{ A}}$.
- 10 Finally the voltages, using $V = IR$ on each resistor:
10 Ω carries I_1 ; 8 Ω carries I_2 ; 6 Ω and 4 Ω carry I_3 .

$I_1 = 3.461 \text{ A}$, $I_2 = 1.923 \text{ A}$, $I_3 = 1.538 \text{ A}$ $V(10 \Omega) = 34.61 \text{ V}$ · $V(8 \Omega) = 15.38 \text{ V}$ · $V(6 \Omega) = 9.23 \text{ V}$ · $V(4 \Omega) = 6.15 \text{ V}$ **The check that proves the whole thing:** go round the first loop. $34.61 + 15.38 = 49.99 \approx 50 \text{ V}$, the source. Now the second loop: $9.23 + 6.15 = 15.38 \text{ V}$, which is exactly the voltage across the 8Ω . Both loops balance, so every number is consistent. I recomputed all of these independently and they match the slides. *Always run these two loop checks.* They will catch any sign error you have made.

● MUST-MEMORISE · THE KIRCHHOFF RECIPE

1. **Label** every current and assume a direction for each.
2. **Choose** a travel direction for each loop and stick to it.
3. Write **one KVL equation per loop**, using the four sign rules.
4. Write **KCL at the junctions** to link the currents.
5. **Count:** you need as many equations as unknowns.
6. **Solve** by substitution.
7. **Check** by walking each loop and confirming the voltages sum to zero.

MEMORY HOOK: label, loop, junction, solve, check. Every Kirchhoff question in every exam yields to exactly this sequence. The physics is in steps 3 and 4; the rest is bookkeeping and algebra.

RECALL AND CHECK

Which conservation law does each of Kirchhoff's laws express?

ANSWER

KCL (junctions) expresses **conservation of charge**: charge cannot accumulate at a point. **KVL** (loops) expresses **conservation of energy**: return to your starting point and you are back at the same potential. They are not arbitrary circuit rules. They are two of the deepest laws in physics, applied to wires.

FILL IN THE GAP · KIRCHHOFF'S LAWS

1. Kirchhoff's current law expresses the conservation of _____.
2. Kirchhoff's voltage law expresses the conservation of _____.
3. Travelling *with* the assumed current through a resistor, the IR term is _____.
4. Travelling through a source from minus to plus, the emf is _____.

ANSWERS

1. **charge.** Nothing accumulates at a junction.
2. **energy.** Walk a loop, return to the start, and you are back at the same potential.
3. **negative.** Going with the current is going *downhill* in potential.
4. **positive.** Minus to plus is climbing *uphill*.

Voltage behaves like height, and you are walking a landscape. That picture gets all four sign rules right without memorising a table.

PART V

Module 4: Magnetism and EM Induction

Moving charges make magnetic fields, and magnetic fields push moving charges. Then the discovery that changed the world: a changing magnetic field creates electricity. That single fact powers every generator on the planet.



Magnetic fields and the Lorentz force

Everything here follows from one formula and one hand rule. Get those two secure and the unit is done.

TEACH · WHAT A MAGNETIC FIELD IS

A **magnetic field** is a vector field describing the magnetic influence on moving electric charges, electric currents, and magnetic materials. Its symbol is **B**. The crucial sentence, which contains the whole unit: **a moving charge in a magnetic field experiences a force perpendicular to both its velocity and the field.** Unit: the tesla (T), where $1 \text{ T} = 1 \text{ N s} / (\text{C m})$.

● MUST-MEMORISE · PROPERTIES OF MAGNETIC FIELD LINES

- ▶ They form **closed loops**.
- ▶ Outside a magnet they run from the **north pole to the south pole**.
- ▶ They **never intersect**.
- ▶ Their **density** shows the strength: closer lines mean a stronger field.

MEMORY HOOK: the key difference from electric field lines. Electric lines *begin* on positive charges and *end* on negative ones. Magnetic lines **neither begin nor end**: they always close on themselves. This is because there is no such thing as an isolated magnetic pole.

TEACH · TWO MORE DIFFERENCES FROM THE ELECTRIC FIELD

1. Electric field lines lie *along* the direction of the force on a positive charge. Magnetic field lines are **perpendicular** to the magnetic force on a moving charge. So the field does not tell you the force direction directly, which is why we need a hand rule.
2. Electric field lines begin and end on charges; magnetic field lines never do.

Sources of magnetic fields: moving charges (that is, currents), permanent magnets, and changing electric fields.

● MUST-MEMORISE · THE LORENTZ FORCE ON A MOVING CHARGE

$$F = qvB \sin \theta$$

F = magnetic force, in newtons (N) **q** = charge, in coulombs (C) **v** = speed of the charge, in m/s
B = magnetic flux density, in tesla (T) **θ** = angle between the velocity **v** and the field **B**

MEMORY HOOK: the $\sin \theta$ does all the work. Move *along* the field ($\theta = 0$) and $\sin 0 = 0$, so **there is no force at all**. Move *across* it ($\theta = 90^\circ$) and $\sin 90^\circ = 1$, giving the **maximum force**, $F = qvB$.

⚠ TRAP · A STATIONARY CHARGE FEELS NO MAGNETIC FORCE

Look at $F = qvB \sin \theta$. If $v = 0$, then $F = 0$. A charge sitting still in a magnetic field feels **nothing**, no matter how strong the field. This is completely unlike an electric field, which pushes a charge whether it moves or not. **Magnetic fields influence only moving charges.** Remember this and half the conceptual questions in this module answer themselves.

● MUST-MEMORISE · THE RIGHT-HAND RULE

To find the direction of the force on a **positive** charge:

- ▶ Point your **fingers** along the velocity **v**.
- ▶ Curl or point your **palm** in the direction of the field **B**.
- ▶ Your **thumb** now points along the force **F**.

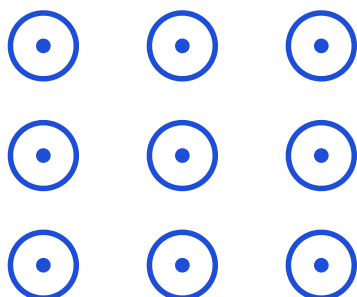
MEMORY HOOK, AND A WARNING THE SLIDES GIVE IN CAPITALS: the right-hand rule is for a **POSITIVE** charge. For an **electron** or any negative charge, work out the direction for a positive charge and then **reverse it**. Forgetting this flip is the most common error in the module.

TEACH · FIELD INTO AND OUT OF THE PAGE

Because this is a three-dimensional problem drawn on flat paper, two symbols are used:

- ▶ $\times$ means the field points **into** the page (think of the tail feathers of an arrow flying away from you).
- ▶ $\odot$ (a dot) means the field points **out of** the page (the tip of an arrow coming towards you).

B out of page



B into page



Dots show the field out of the page; crosses show it into the page.

Motion of charged particles in a magnetic field

● MUST-MEMORISE · A MAGNETIC FIELD DOES NO WORK

The magnetic force is always *perpendicular* to the motion. Work is force times displacement times $\cos \theta$, and here $\theta = 90^\circ$, so:

$$W = F d \cos 90^\circ = 0$$

Therefore, in a magnetic field:

- ▶ Kinetic energy **remains constant**.
- ▶ Speed **remains constant**.
- ▶ Only the **direction** changes.

MEMORY HOOK: a magnetic field *steers* but never *speeds up*. It cannot change the speed or the kinetic energy of a particle, only bend its path. An electric field, by contrast, can accelerate a charge and change its energy.

TEACH · THE THREE CASES OF MOTION

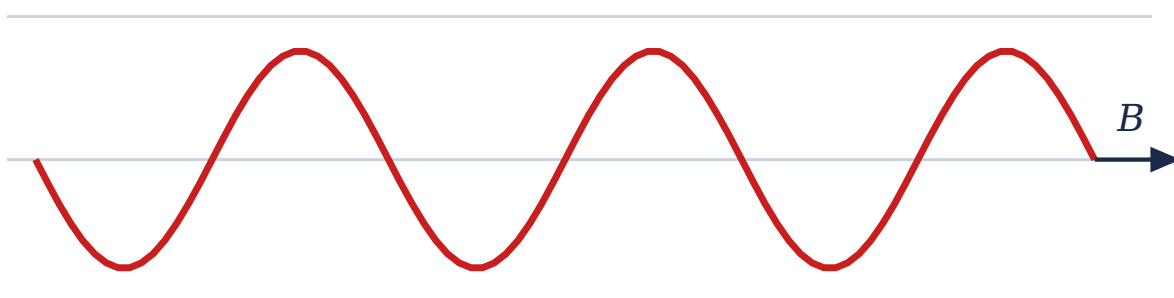
1. Motion parallel to the field ($\theta = 0$). $\sin 0 = 0$, so $F = 0$. The particle carries straight on, undeflected.

2. Motion perpendicular to the field ($\theta = 90^\circ$). The force is maximum and always at right angles to the velocity, so it acts as a **centripetal force**. The particle moves in a **circle**.

3. Motion at an angle θ . Split the velocity into components (Foundations F.3):

$$v_{\text{parallel}} = v \cos \theta \quad v_{\text{perpendicular}} = v \sin \theta$$

The perpendicular component produces circular motion; the parallel component carries the particle steadily forward. Combine a circle with a forward drift and you get a **helix**, that is, a spiral. The forward distance covered in one full revolution is called the **pitch**.



velocity at an angle to B gives a helix

A velocity at an angle to B produces a helical path.

● MUST-MEMORISE · RADIUS, PERIOD AND CYCLOTRON FREQUENCY

For circular motion, the magnetic force supplies the centripetal force: $qvB = mv^2/r$. Cancel one v and rearrange:

$$r = mv / (qB)$$

The angular velocity is $\omega = v/r$, and substituting the radius gives $\omega = qB/m$. The frequency $f = \omega/2\pi$ is therefore:

$$f = qB / (2\pi m)$$

MEMORY HOOK, AND THE REASON CYCLOTRONS WORK: look hard at that frequency. **It does not contain v .** The frequency is completely independent of the particle's speed. Speed it up and the radius grows, but it still completes each circle in the same time. This is the **cyclotron frequency**, and it is the entire principle behind particle accelerators.

TEACH · READING THE RADIUS FORMULA

From $r = mv/(qB)$, the radius depends on the particle's mass, velocity, charge and the field strength. Specifically:

- ▶ **Increasing the velocity increases the radius** (v is on the top).
- ▶ **Increasing the magnetic field decreases the radius** (B is on the bottom).

A stronger field bends the particle more tightly. That is exactly what you would expect, and it is how a mass spectrometer separates ions by mass.

WORKED EXAMPLE · RADIUS OF AN ELECTRON'S PATH

SLIDE

An electron enters a uniform field of 0.40 T perpendicular to the field with a speed of 4.0×10^6 m/s. Find the radius of its motion. ($m = 9.11 \times 10^{-31}$ kg, $q = 1.6 \times 10^{-19}$ C.)

- 1 Perpendicular entry means circular motion, so use $r = mv/(qB)$.

Take the magnitude of the electron's charge. The negative sign affects the direction of circulation, not the size of the radius.

- 2 Numerator: $mv = (9.11 \times 10^{-31})(4.0 \times 10^6) = 3.644 \times 10^{-24}$.

Multiply fronts: $9.11 \times 4 = 36.44$. Add exponents: $-31 + 6 = -25$. So $36.44 \times 10^{-25} = 3.644 \times 10^{-24}$.

- 3 Denominator: $qB = (1.6 \times 10^{-19})(0.40) = 6.4 \times 10^{-20}$.

- 4 $r = (3.644 \times 10^{-24}) / (6.4 \times 10^{-20}) = 5.69 \times 10^{-5}$ m.

Divide fronts: $3.644/6.4 = 0.569$. Subtract exponents: $-24 - (-20) = -4$. So $0.569 \times 10^{-4} = 5.69 \times 10^{-5}$.

$$r \approx 5.69 \times 10^{-5} \text{ m (about 0.057 mm)}$$

Now redo it: I recomputed this independently as 5.694×10^{-5} m, matching the slide. Note how tight the circle is: an electron is so light that even a modest field curls it into a path about a tenth of a millimetre across.

WORKED EXAMPLE · FORCE AT AN ANGLE OF 30 DEGREES

SLIDE

A charge of 2×10^{-6} C moves at 4×10^3 m/s in a field of 0.2 T. The angle between v and B is 30° . Find the force.

- 1 Use the full Lorentz expression, since the angle is neither 0 nor 90° : $F = qvB \sin \theta$.

- 2 $F = (2 \times 10^{-6})(4 \times 10^3)(0.2) \sin 30^\circ$.

Everything is already in SI units.

- 3 First the product qvB : $(2 \times 10^{-6})(4 \times 10^3) = 8 \times 10^{-3}$, then times 0.2 gives 1.6×10^{-3} .

- 4 Now apply $\sin 30^\circ = 0.5$: $F = (1.6 \times 10^{-3})(0.5) = 8.0 \times 10^{-4}$ N.

$\sin 30^\circ = 0.5$ exactly. This is one of the four values from Foundations F.4.

$$F = 8.0 \times 10^{-4} \text{ N}$$

Now redo it: note that at 30° the particle feels only *half* the force it would feel at 90° , purely because $\sin 30^\circ = 0.5$. The angle is not a detail.

FORMULA · COMBINED ELECTRIC AND MAGNETIC FIELDS

When *both* fields are present, the total force on the charge is the sum of the electric force and the magnetic force. This is the complete Lorentz force equation:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Applications include the **velocity selector**, the **mass spectrometer**, and **plasma confinement**. In a velocity selector the electric and magnetic forces are set to cancel exactly, so that only particles at one particular speed pass straight through undeflected.

RECALL AND CHECK

A proton travels exactly *along* a magnetic field line. What force does it feel, and what path does it take?

ANSWER

$\theta = 0$, so $\sin \theta = 0$, so $\mathbf{F} = \mathbf{0}$. It feels no force at all and travels in a **straight line**, completely undeflected. A magnetic field only acts on the component of velocity that is *across* it.

FILL IN THE GAP · MAGNETIC FIELDS AND THE LORENTZ FORCE

1. Magnetic field lines form closed _____: unlike electric field lines, they neither begin nor _____.
2. A _____ charge feels no magnetic force at all, however strong the field.
3. A magnetic field does no _____, so the _____ of the particle never changes.
4. The cyclotron frequency $f = qB/2\pi m$ does not contain _____, which is the whole principle of the cyclotron.

ANSWERS

1. **loops**, and they never **end**. There is no such thing as an isolated magnetic pole.
2. **stationary** ($v = 0$). Magnetic fields act only on *moving* charges.
3. **work**, so the **speed** (or kinetic energy) is constant. A magnetic field steers but never speeds up.
4. **v** (the speed). Speed it up and the radius grows, but each circle still takes the same time.

UNIT 4.2

Sources of magnetic fields

Currents make magnetic fields. Two laws compute them: Biot-Savart for any shape, Ampere's law for symmetric ones.

TEACH · MAGNETIC FORCE ON A CURRENT-CARRYING CONDUCTOR

A wire carrying a current, placed in a magnetic field, experiences a force. This is not a new phenomenon: the wire is full of *moving charges*, and each one feels the Lorentz force. Add up the effect on all of them and you get a net force on the whole wire. This is the operating principle of **electric motors, loudspeakers, galvanometers, moving-coil meters** and rail guns.

● MUST-MEMORISE · FORCE ON A CURRENT-CARRYING CONDUCTOR

$$F = BIL \sin \theta$$

F = magnetic force (N) **B** = magnetic flux density (T) **I** = current (A) **L** = length of conductor inside the field (m) **θ** = angle between the current and the field

MEMORY HOOK: this is $F = qvB \sin \theta$ for a whole wire, with the charge and speed replaced by current and length. Same physics, tidier packaging.

TEACH · THE TWO SPECIAL CASES

(a) Conductor parallel to the field ($\theta = 0^\circ$): $F = BIL \sin 0 = 0$. No force at all. **(b) Conductor perpendicular to the field** ($\theta = 90^\circ$): $F = BIL$, which is the **maximum** force. These are not two extra formulas to memorise. They are the one formula with $\sin 0 = 0$ and $\sin 90^\circ = 1$ substituted in. The force increases with the field strength *B*, the current *I*, the length *L*, and the angle, reaching its maximum at 90° and vanishing at 0° .

● MUST-MEMORISE · FLEMING'S LEFT-HAND RULE

For the direction of the force on a *current-carrying conductor*, hold the **left** hand so that:

- ▶ **F**irst **f**inger points along the magnetic **F**ield (*B*)
- ▶ **S**e**C**ond **f**inger points along the **C**urrent (*I*)
- ▶ **T**hu**M**b points along the **M**otion or force (*F*)

MEMORY HOOK: the initials line up with the fingers themselves. First is Field, seCond is Current, thuMb is Motion. Note this is the **LEFT** hand for the force on a current, whereas the **right** hand is used for the field *produced* by a current. Confusing the two hands is a classic error.

The Biot-Savart law

TEACH · WHERE THE LAW CAME FROM

In 1820 the Danish physicist Hans Christian Oersted noticed that a compass needle deflected when placed near a current-carrying wire. That was the discovery that electricity and magnetism are connected: **a current produces a magnetic field**. In the same year the French physicists Jean-Baptiste Biot and Felix Savart worked out the quantitative relationship. It lets us calculate the magnetic field produced by *any* current distribution, which makes it fundamental to electromagnetism.

FORMULA · THE BIOT-SAVART LAW

The field **dB** produced by a small current element **I dl** is directly proportional to the current and the length of the element, proportional to the sine of the angle between the element and the line joining it to the point, and **inversely proportional to the square of the distance**:

$$dB = (\mu_0 / 4\pi) \cdot (I dl \times \hat{r}) / r^2$$

μ₀ = permeability of free space = $4\pi \times 10^{-7}$ H/m **I** = current (A), **dl** = length element in the direction of the current **r** = distance from the element to the point of interest

Note this is directly analogous to Coulomb's law for the electric field of a point charge: an inverse square law, with a constant out the front. The direction of **dB** is perpendicular to *both* *r* and *dl*.

● MUST-MEMORISE · THE TWO STANDARD BIOT-SAVART RESULTS

1. A long straight wire, at distance r from it:

$$B = \mu_0 I / (2\pi r)$$

2. At the centre of a circular loop of radius R :

$$B = \mu_0 I / (2R) \quad \text{and for } N \text{ turns} \quad B = \mu_0 NI / (2R)$$

MEMORY HOOK: the straight wire has $2\pi r$ underneath (the circumference of the circle you would walk around it). The loop has just $2R$. The field *increases* with current and *decreases* with distance.

WORKED EXAMPLE · FIELD 5 CM FROM A WIRE CARRYING 12 A

SLIDE

The question. A long straight conductor carries a current of 12 A. Calculate the magnetic field at a point 5 cm from the conductor.

1 Use the straight-wire result: $B = \mu_0 I / (2\pi r)$.

2 Convert: $r = 5 \text{ cm} = 0.05 \text{ m}$.

The usual trap. Centimetres are 10^{-2} m .

3 Numerator: $\mu_0 I = (4\pi \times 10^{-7})(12) = 1.508 \times 10^{-5}$.

$4\pi = 12.566$, times 12 gives 150.8, so $150.8 \times 10^{-7} = 1.508 \times 10^{-5}$.

4 Denominator: $2\pi r = 2\pi(0.05) = 0.3142$.

5 $B = (1.508 \times 10^{-5}) / 0.3142 = 4.8 \times 10^{-5} \text{ T}$.

$$B = 4.8 \times 10^{-5} \text{ T}$$

Now redo it: I recomputed this as $4.80 \times 10^{-5} \text{ T}$, matching the slide. For scale, the Earth's magnetic field is about $5 \times 10^{-5} \text{ T}$, so a 12 A current at 5 cm produces roughly the same field as the entire planet. That is why a compass needle jumps near a wire, which is precisely what Oersted saw.

Ampere's law

TEACH · WHY WE NEED A SECOND LAW

Biot-Savart can compute the field of *any* current distribution, but for complicated geometries the integral becomes very unpleasant. In 1826 Andre-Marie Ampere found a shortcut for situations with high **symmetry**. The relationship between the two laws is exactly the relationship between Coulomb's law and Gauss's law in Module 1. One is general but laborious; the other is effortless when the symmetry is right. Ampere's law is one of the four **Maxwell equations**.

● MUST-MEMORISE · AMPERE'S CIRCUITAL LAW

The line integral of the magnetic field around any closed path equals the permeability of free space multiplied by the total current enclosed by that path:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}}$$

The circle on the integral sign means integration **around a closed path**, called an **Amperian loop**. This is an imaginary closed path that you choose to make the calculation easy, exactly like a Gaussian surface.

MEMORY HOOK: the stronger the enclosed current, the stronger the magnetic field circulating around it. Compare with Gauss's law: flux out equals charge enclosed. Here: circulation around equals current enclosed. Same idea, different geometry.

WORKED EXAMPLE · DERIVING THE STRAIGHT-WIRE FIELD FROM AMPERE'S LAW

SLIDE

The task. Use Ampere's law to derive the magnetic field at a distance r from a long straight conductor carrying current I . (Biot-Savart gives the same result, but takes an integral to get there.)

- 1 Take a long straight conductor carrying current I . Choose a **circular** Amperian loop of radius r centred on the wire.
Chosen because by symmetry B has the same magnitude everywhere on this circle and points along it.
- 2 Since B is constant on the loop, the integral collapses to a simple product: $\oint \mathbf{B} \cdot d\mathbf{l} = B(2\pi r)$.
 $2\pi r$ is the circumference, from Foundations F.9.
- 3 Set that equal to $\mu_0 I$: $B(2\pi r) = \mu_0 I$.
- 4 Rearrange for B .

$$B = \mu_0 I / (2\pi r)$$

Look what happened: this is the same result Biot-Savart gives, obtained in four lines instead of an integral. That is what Ampere's law is *for*. Use it whenever the symmetry allows.

● MUST-MEMORISE · THE RIGHT-HAND GRIP RULE

For the *direction* of the field around a current:

- ▶ Point your **thumb** in the direction of the **current**.
- ▶ **Curl your fingers**.
- ▶ Your fingers now show the direction the **magnetic field** circulates.

MEMORY HOOK: thumb is current, fingers are field. The field wraps around the wire in circles, like spaghetti coiled around it. It does not point along the wire or away from it.

● MUST-MEMORISE · SOLENOID AND TOROID

A **solenoid** is a long coil of wire. Inside a long solenoid the field is nearly uniform, and outside it is very small. Applying Ampere's law with a rectangular loop gives $BL = \mu_0 NI$, and with $N = nL$ this becomes:

$$B = \mu_0 n I$$

n = number of turns *per metre*. A **toroid** is a solenoid bent round into a ring. Ampere's law gives $B(2\pi r) = \mu_0 NI$, so:

$$B = \mu_0 NI / (2\pi r)$$

N = total number of turns.

MEMORY HOOK: for the solenoid, note that **r does not appear**. The field inside is uniform: the same everywhere. That is exactly why solenoids are used whenever a controlled uniform field is needed, as in MRI scanners.

FORMULA · THE AMPERE-MAXWELL LAW

Ampere's law as stated is incomplete. Maxwell noticed that a *changing electric field* also creates a magnetic field, and added a term:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (I + \epsilon_0 d\Phi_E/dt)$$

The additional term is called the **displacement current**. In differential form, the original law reads $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$, where $\mathbf{J}$ is the current density. This completed set of equations predicted the existence of electromagnetic waves, and hence light itself.

RECALL AND CHECK

The current in a long straight wire is doubled, and you also move twice as far away from it. What happens to the magnetic field?

ANSWER

$B = \mu_0 I / (2\pi r)$. Doubling I doubles B . Doubling r halves B . The two effects cancel exactly, so **B is unchanged**. Note this is a $1/r$ law, not an inverse square, so doubling the distance halves the field rather than quartering it.

FILL IN THE GAP · SOURCES OF MAGNETIC FIELDS

1. Ampere's law is most useful for current distributions with high _____.
2. The Biot-Savart law falls off with the square of the _____.
3. Inside a long solenoid, $B = \mu_0 nI$, where n is the number of turns _____.
4. The field inside a solenoid does not depend on _____, so it is uniform.

ANSWERS

1. **symmetry**. This is Test 2 Q5, verbatim. The whole trick is choosing a loop on which B does not vary.
2. **distance**. This is Test 2 Q12, verbatim. An inverse-square law, exactly like Coulomb's.
3. **per metre**. Not the total turns. Converting N to $n = N/L$ is exactly what Test 2 Q9 tests.
4. **r** (the radial position). That uniformity is why solenoids are used in MRI scanners.

Electromagnetic induction

Oersted showed that a current makes a magnetic field. Faraday asked the reverse question: can a magnetic field make a current? The answer, found in 1831, is the basis of the entire electrical power industry.

TEACH · WHAT INDUCTION IS

Electromagnetic induction is the production of an emf, and in a closed circuit an electric current, whenever the **magnetic flux linking a conductor changes**. Michael Faraday discovered in 1831 that a *changing* magnetic field could induce a current in a nearby conductor with no direct electrical contact at all. A current is induced only if *something changes*:

- ▶ The magnetic field changes.
- ▶ The conductor moves relative to the field.
- ▶ The area of the conducting loop changes.
- ▶ The orientation of the conductor changes.

It is the underlying principle of **electric generators, transformers, induction motors, wireless charging, microphones, metal detectors and induction cookers**.

● MUST-MEMORISE · MAGNETIC FLUX

Magnetic flux measures the amount of magnetic field passing through a surface. It is the number of field lines through a given area:

$$\Phi = BA \cos \theta$$

Φ = magnetic flux, in **webers (Wb)**, where $1 \text{ Wb} = 1 \text{ T} \times 1 \text{ m}^2$ $\mathbf{B}$ = magnetic flux density (T) $\mathbf{A}$ = area (m^2) θ = angle between the field and the *normal* to the surface If the coil has $\mathbf{N}$ **turns**, the **flux linkage** is $N\Phi$.

MEMORY HOOK: at $\theta = 0$ the field is straight through the loop and $\cos 0 = 1$, so the flux is **maximum** at $\Phi = BA$. At $\theta = 90^\circ$ the field skims along the surface and $\cos 90^\circ = 0$, so the flux is **zero**. Same hoop-and-rain picture as the electric flux in Module 1.

● MUST-MEMORISE · FARADAY'S LAWS OF INDUCTION

Faraday's first law. Whenever the magnetic flux linking a conductor changes, an emf is induced in the conductor. If the circuit is closed, an induced current flows. **Faraday's second law.** The magnitude of the induced emf is directly proportional to the *rate of change* of magnetic flux:

$$\varepsilon = -N \cdot d\Phi/dt$$

$\mathbf{N}$ = number of turns in the coil $d\Phi/dt$ = rate of change of flux, in Wb/s The **minus sign** is Lenz's law, explained next.

MEMORY HOOK: the word to underline is **CHANGE**. A huge steady flux induces *nothing*. A small but rapidly changing flux induces a large emf. It is the *rate* that matters, not the size.

● MUST-MEMORISE · LENZ'S LAW AND THE MINUS SIGN

The **direction** of the induced current is such that it **opposes the change in magnetic flux that produced it**.

MEMORY HOOK: Lenz's law is a consequence of the **conservation of energy**. If the induced current *helped* the change instead of opposing it, you would get a runaway: a tiny push would produce ever-growing current and energy from nothing. Nature forbids that. The induced effect always fights back, which is why you must do *work* to turn a generator.

FORMULA · MOTIONAL EMF AND INDUCED CURRENT

When a conductor of length L moves with velocity v through a field B :

$$\varepsilon = BLv$$

If the conductor forms part of a closed circuit, Ohm's law then gives the induced current:

$$I = \varepsilon / R$$

WORKED EXAMPLE · A COIL OF 200 TURNS

SLIDE

A coil of 200 turns experiences a change in magnetic flux from 0.08 Wb to 0.02 Wb in 0.10 s. Calculate the induced emf.

- 1 Find the *change* in flux: $\Delta\Phi = 0.02 - 0.08 = -0.06 \text{ Wb}$.

It is the change that matters, not either individual value.

- 2 Apply Faraday's law: $\varepsilon = -N \cdot \Delta\Phi/\Delta t = -(200)(-0.06) / 0.10$.

$N = 200$ turns, and $\Delta t = 0.10$ s.

- 3 Numerator: $-(200)(-0.06) = +12$. Divide by 0.10: $12 / 0.10 = 120$.

Dividing by 0.10 is multiplying by 10.

- 4 So $\varepsilon = +120 \text{ V}$.

*The **magnitude** is 120 V. The sign came out **positive** because the flux **fell**: Lenz's law is the minus in Faraday's law, and here it multiplied a negative $\Delta\Phi$. Read the plus as the emf opposing that fall, driving current in the direction that props the flux back up. Had the flux risen instead, the same law would have returned -120 V .*

Magnitude of the induced emf = 120 V

Now redo it: note that the flux *fell*, and the coil has 200 turns multiplying the effect. If the same change happened over 1.0 s instead of 0.10 s, the emf would be only 12 V. Ten times slower gives ten times less. The rate is everything.

TEACH · EDDY CURRENTS, SELF-INDUCTION AND MUTUAL INDUCTION

Eddy currents. When a *solid* conductor experiences a changing flux, closed loops of current are set up inside it. These are eddy currents. They are used in **magnetic braking, induction heating, metal detectors** and speedometers. **Self-induction.** When the current changes in a coil, it induces an emf *in that same coil*. The constant of proportionality is the **inductance L** . **Mutual induction.** When the current changes in one coil, it induces an emf in *another nearby coil*. This is the principle of the **transformer**, of wireless charging, and of induction cookers.

● MUST-MEMORISE · THE TRANSFORMER EQUATION

A transformer transfers electrical energy between two coils by mutual induction:

$$V_s / V_p = N_s / N_p$$

V_p , V_s = primary and secondary voltages N_p , N_s = primary and secondary numbers of turns

MEMORY HOOK: the voltage ratio equals the turns ratio. More turns on the secondary means a higher voltage out (a step-up transformer); fewer turns means a lower voltage (step-down). This is why transformers work only on **AC**: DC produces no *change* of flux, so it induces nothing at all.

RECALL AND CHECK

A strong bar magnet is held perfectly still inside a coil of 1000 turns. What emf is induced?

ANSWER

Zero. The flux is large, but it is not *changing*, so $d\Phi/dt = 0$ and therefore $\varepsilon = 0$. Faraday's law responds only to change. To induce anything you must move the magnet, move the coil, or vary the field.

FILL IN THE GAP · ELECTROMAGNETIC INDUCTION

1. A changing magnetic _____ induces an emf.
2. Lenz's law is a consequence of the conservation of _____.
3. A transformer works only on _____, because _____ produces no change of flux.
4. A magnet held _____ inside a coil induces exactly zero emf, however strong it is.

ANSWERS

1. **flux.** *This is Test 2 Q11, verbatim.* Be precise: saying "field" is *not* good enough, because $\Phi = BA \cos \theta$ can also change through the area or the angle. That is how a rotating coil in a *constant* field still generates electricity.
2. **energy.** If the induced current assisted the change you would get energy from nothing.
3. **AC**, because **DC** gives a steady flux and therefore no induction.
4. **still** (stationary). Faraday's law responds only to *change*.

UNIT 4.4

emf, internal resistance and terminal voltage

A real battery is not ideal. It has resistance of its own, and that single fact explains why your headlights dim when you start the engine.

TEACH · WHAT EMF REALLY IS

Electromotive force is the energy per unit electric charge imparted by an energy source such as a generator or a battery. Energy is converted from one form to another inside the device as it does work on the charge passing through it, leaving one terminal positive and the other negative. The work done per unit charge is the emf. Its unit is the **volt**: driving one coulomb through a potential difference of one volt requires one joule of work. Despite the name, *emf is not a force*. It is an energy per unit charge.

● MUST-MEMORISE · EMF VERSUS TERMINAL VOLTAGE

emf (E) is the potential difference of a source when **no current is flowing**. It is what the battery *would* give you if nothing were connected. **Terminal voltage (V)** is the voltage actually measured across the terminals *while current flows*. It is always **less** than the emf.

$$E = I(R + r) \quad \text{so} \quad I = E / (R + r)$$

E = emf of the cell (V) **R** = external resistance (Ω) **r** = **internal resistance** of the cell (Ω) **I** = current (A)

MEMORY HOOK: the total resistance is $R + r$. The battery has to push current through *itself* as well as through the circuit, and that wastes some voltage internally. Terminal voltage $V = E - Ir$: the emf, minus what is lost inside.

WORKED EXAMPLE 1 · FIND THE EMF

SLIDE

An external resistor of 5Ω is connected to a cell of internal resistance 0.5Ω . The current flowing is 1 A. Calculate the emf of the cell.

- 1 Use $I = E / (R + r)$, rearranged as $E = I(R + r)$.

The current must pass through both the external resistor and the cell's own internal resistance, so the two add in series.

- 2 $E = 1 \times (5 + 0.5) = 1 \times 5.5$.

- 3 $E = 5.5 \text{ V}$.

$$E = 5.5 \text{ V}$$

Now redo it: the terminal voltage here would be $V = IR = (1)(5) = 5.0 \text{ V}$, which is *less* than the 5.5 V emf. The missing 0.5 V is dropped across the internal resistance and wasted as heat inside the battery.

WORKED EXAMPLE 2 · FROM TERMINAL P.D. TO EMF

SLIDE

A cell of unknown emf E and internal resistance 2Ω is connected to a 5Ω resistance. If the terminal p.d. is 1 V, calculate E .

- 1 The terminal p.d. is the voltage across the *external* 5Ω resistor, so $V = 1 \text{ V}$ across $R = 5 \Omega$.

Terminal voltage is what appears across the external circuit.

- 2 Find the current: $I = V/R = 1/5 = 0.2 \text{ A}$.

Ohm's law on the external resistor alone.

- 3 The p.d. lost across the internal resistance: $Ir = (0.2)(2) = 0.4 \text{ V}$.

This is the voltage wasted inside the cell.

- 4 The emf is the sum of the two: $E = (\text{p.d. across } R) + (\text{p.d. across } r) = 1 + 0.4$.

$$E = 1.4 \text{ V}$$

Check it the other way: using $E = I(R + r) = 0.2(5 + 2) = 0.2 \times 7 = 1.4 \text{ V}$. Both routes agree, which is exactly the alternative solution given in the slides. Note the internal resistance is eating a large fraction of this cell's emf, because r is not small compared with R .

WORKED EXAMPLE 3 · TWO MEASUREMENTS, TWO UNKNOWNNS

SLIDE

The terminal p.d. of a battery is 12 V when an external resistance of $20\ \Omega$ is connected, and 13.5 V when $45\ \Omega$ is connected. Calculate the emf.

- 1 Two unknowns (E and r), so we need two equations. Each measurement gives one.
This is a simultaneous equation problem, exactly as in Foundations F.5.
- 2 First measurement: $I = V/R = 12/20 = 0.6\text{ A}$. And $E = I(R + r)$, so $E = 0.6(20 + r) = 12 + 0.6r$. (1)
Expand the bracket: $0.6 \times 20 = 12$.
- 3 Second measurement: $I = 13.5/45 = 0.3\text{ A}$. So $E = 0.3(45 + r) = 13.5 + 0.3r$. (2)
Note the current is different in the two cases, because R changed.
- 4 The emf E is the same battery in both, so set (1) equal to (2): $12 + 0.6r = 13.5 + 0.3r$.
This is the step that unlocks it. The battery has one fixed emf.
- 5 Collect the r terms: $0.6r - 0.3r = 13.5 - 12$, so $0.3r = 1.5$.
- 6 $r = 1.5 / 0.3 = 5\ \Omega$.
- 7 Substitute back into (1): $E = 12 + 0.6(5) = 12 + 3 = 15\text{ V}$.

$$r = 5\ \Omega \quad \cdot \quad E = 15\text{ V}$$

Check with equation (2): $E = 13.5 + 0.3(5) = 13.5 + 1.5 = 15\text{ V}$. Both equations give the same emf, so the answer is consistent. This matches the slides exactly. Note the pattern: a *larger* external resistance

draws less current, so less voltage is lost internally, so the terminal voltage climbs closer to the true emf.

TEACH · THE PHYSICAL MEANING, AND WHY HEADLIGHTS DIM

Terminal voltage $V = E - Ir$. The larger the current drawn, the more voltage is lost inside the battery, and the lower the terminal voltage. Start a car engine and the starter motor draws an enormous current. That large I makes the Ir term large, so the terminal voltage collapses briefly, and the headlights dim. They recover the instant the engine catches and the current falls. Internal resistance is not an abstraction: you can watch it happen. As the external resistance R rises, the current falls, and the terminal voltage approaches the emf. With no current flowing at all (an open circuit), $V = E$ exactly. This is the definition we started with.

RECALL AND CHECK

A battery of emf 6 V and internal resistance r is connected to a $4\ \Omega$ wire, and 1 A flows. Find r . (This is the class drill from the slides.)

ANSWER

$E = I(R + r)$, so $6 = 1(4 + r)$, giving $4 + r = 6$ and therefore $r = 2\ \Omega$. Check: the terminal voltage would be $IR = 4\text{ V}$, and the loss inside is $Ir = 2\text{ V}$. Together, $4 + 2 = 6\text{ V}$, the emf. It balances.

FILL IN THE GAP · EMF, INTERNAL RESISTANCE AND TERMINAL VOLTAGE

1. The emf is the potential difference of a source when _____ current flows.
2. The terminal voltage is always _____ than the emf when current flows.
3. $V = E - Ir$, so the voltage lost inside the cell is _____.
4. Your headlights dim when you start the engine because the starter draws a large _____.

ANSWERS

1. **no** (zero).
2. **less** (smaller). If your answer exceeds the emf, you have added instead of subtracted.
3. **Ir**. Dissipated as heat inside the battery; it never reaches the circuit.
4. **current**. A large I makes the Ir term large, so the terminal voltage briefly collapses. Internal resistance is not an abstraction; you can watch it happen.

UNIT 4.5 · TEST 2 · NOT IN THE SLIDES

Inductance and energy stored in an inductor

This topic is examined in CBT Test 2, Q1, but it does not appear in any slide deck. It is here because the assessment demands it.

△ WHY THIS SECTION EXISTS

The slides mention *self-induction* and name the inductance L , but they never give the energy formula. Test 2 opens by asking for it. If you revise from the slides alone, you walk into the first question of the paper with nothing. That is the entire argument for blending the slides with the assessments rather than trusting either on its own.

TEACH · WHAT AN INDUCTOR IS

An **inductor** is usually a coil of wire. From Unit 4.3, when the current through a coil changes, the changing flux induces an emf *in that same coil* which opposes the change. That is self-induction. The constant relating the induced emf to the rate of change of current is the **inductance** L , measured in **henries (H)**:

$$\varepsilon = -L \cdot dI/dt$$

An inductor therefore *resists changes in current*, in the same way that mass resists changes in velocity. It is electrical inertia.

● MUST-MEMORISE · ENERGY STORED IN AN INDUCTOR

$$W = \frac{1}{2}LI^2$$

W = energy stored, in joules (J) L = inductance, in henries (H) I = current through the inductor, in amperes (A)

MEMORY HOOK: this is the exact magnetic twin of the capacitor's $W = \frac{1}{2}CV^2$ from Unit 2.2. Swap capacitance for inductance, and *voltage* for *current*. A capacitor stores energy in an *electric* field and cares about voltage; an inductor stores energy in a *magnetic* field and cares about current. Learn them as a pair and you get both for the price of one.

TEACH · THE SYMMETRY THAT MAKES THIS FREE

	CAPACITOR	INDUCTOR
Stores energy in	Electric field	Magnetic field
Property	Capacitance C (farad)	Inductance L (henry)
Resists changes in	Voltage	Current
Energy stored	$W = \frac{1}{2}CV^2$	$W = \frac{1}{2}LI^2$

If you can write one row of the bottom line, you can write the other. The whole of electromagnetism is built on symmetries like this one.

WORKED EXAMPLE · A 2 H INDUCTOR CARRYING 5 A

TEST 2, Q1

A 2 H inductor carries 5 A of current. What is the stored energy?

- 1 Use the energy formula for an inductor: $W = \frac{1}{2}LI^2$.
Given an inductance and a current, this is the only formula that fits.
- 2 Square the **current**, not the inductance: $I^2 = 5^2 = 25$.
This is the one place people slip. It is L times I squared, not L squared times I.
- 3 $W = \frac{1}{2}(2)(25) = \frac{1}{2}(50)$.

W = 25 J (option a) Method: \$4.5

Now redo it: I recomputed this independently and confirm 25 J. Watch the distractor: if you forgot the $\frac{1}{2}$ you would get 50 J, and if you squared the inductance instead you would get 10 J, which is offered as option (d). Both traps are sitting there waiting for you.

RECALL AND CHECK

A capacitor stores 8 J at 4 V. An inductor stores 8 J at 4 A. Write the formula you would use for each, and find C and L.

ANSWER

Capacitor: $W = \frac{1}{2}CV^2$, so $C = 2W/V^2 = 16/16 = 1 \text{ F}$. Inductor: $W = \frac{1}{2}LI^2$, so $L = 2W/I^2 = 16/16 = 1 \text{ H}$. Identical algebra. Only the letters change. That is the symmetry doing the work for you.

FILL IN THE GAP · INDUCTANCE AND ENERGY STORED

1. Energy stored in an inductor is $W = \frac{1}{2}L$ _____², the magnetic twin of $W = \frac{1}{2}C$ _____².
2. A capacitor resists changes in _____; an inductor resists changes in _____.
3. Inductance is measured in _____.

ANSWERS

1. I (current), twinned with V (voltage).

Square the current, not the inductance. Squaring L gives 10 J on Test 2 Q1, which is a planted distractor.

2. voltage, and current.

3. henries (H).

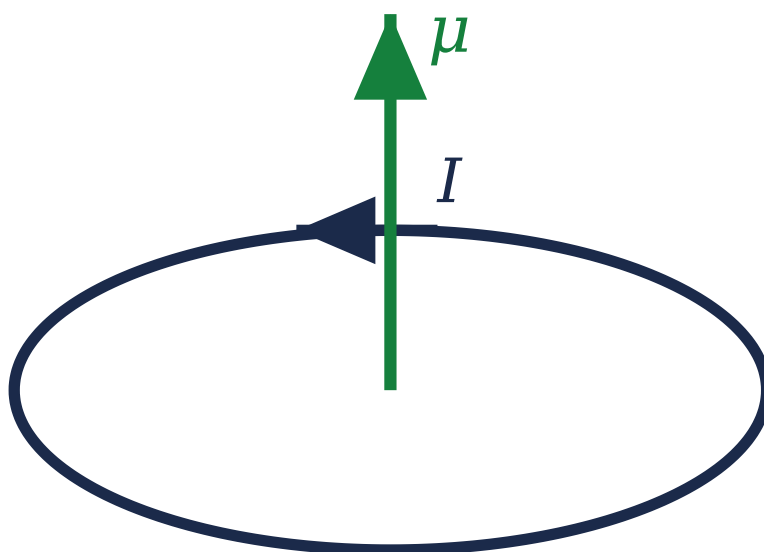
Torque on a current loop and the magnetic dipole

moment

Examined in CBT Test 2, Q7, and again absent from every slide deck. It is, however, a near-copy of the electric dipole from Unit 2.1, so you already know most of it.

TEACH · A CURRENT LOOP IS A MAGNET

From Unit 4.2, a current-carrying wire in a magnetic field feels a force $F = BIL \sin \theta$. Now bend that wire into a closed rectangular loop and put it in a uniform field. Two opposite sides of the loop carry current in *opposite* directions, so they feel forces in opposite directions. The forces cancel, so the loop does not fly away. But because they act on *different sides* of the loop, they produce a **twist**: a torque that rotates the loop until it lines up with the field. This is exactly what an electric dipole does in an electric field, and it is the operating principle of the **electric motor** and the **moving-coil galvanometer**.



a current loop is a magnetic dipole

A current loop behaves as a magnetic dipole with moment along its axis.

● MUST-MEMORISE · MAGNETIC DIPOLE MOMENT AND TORQUE

The **magnetic dipole moment** of a loop of N turns, each of area A , carrying current I , is:

$$\mathbf{m} = NIA$$

$\mathbf{m}$ is measured in ampere square metres (A m^2). The **torque** on that loop in a uniform field B is:

$$\tau = m B \sin \theta \quad (\text{and so } \tau = NIAB \sin \theta)$$

θ = angle between the loop's magnetic moment and the field.

MEMORY HOOK: compare with the *electric* dipole from Unit 2.1, which had **$T = PE \sin \theta$** . It is the identical formula with p replaced by m and E replaced by B . **Torque uses sine.** Spot the analogy and this becomes a question you can answer instantly.

TEACH · THE TWO DIPOLES, SIDE BY SIDE

	ELECTRIC DIPOLE (§2.1)	MAGNETIC DIPOLE (§4.6)
Moment	$p = qd \text{ (C m)}$	$m = NIA \text{ (A m}^2\text{)}$
Field	E	B
Torque	$\tau = pE \sin \theta$	$\tau = mB \sin \theta$
Torque is zero when	Aligned ($\theta = 0$)	Aligned ($\theta = 0$)
Torque is maximum when	Perpendicular ($\theta = 90^\circ$)	Perpendicular ($\theta = 90^\circ$)

One structure, two contexts. Memorise the shape once.

WORKED EXAMPLE · FIND THE MAGNETIC DIPOLE MOMENT

TEST 2, Q7

A rectangular current loop in a uniform field experiences a torque of $4.8 \times 10^{-3} \text{ N m}$. The field is 0.6 T and the angle between the loop's magnetic moment and the field is 30° . Calculate the magnetic dipole moment.

- 1 Torque on a magnetic dipole: $\tau = mB \sin \theta$.
Same shape as the electric dipole result from Unit 2.1. Torque uses sine.
- 2 Rearrange to isolate the unknown: $m = \tau / (B \sin \theta)$.
Rearrange in symbols first, substitute afterwards.
- 3 $\sin 30^\circ = 0.5$ exactly.
One of the four values from Foundations F.4. No calculator needed.
- 4 Denominator: $B \sin \theta = (0.6)(0.5) = 0.3$.
- 5 $m = (4.8 \times 10^{-3}) / 0.3 = 1.6 \times 10^{-2}$.
 $4.8 / 0.3 = 16$, so $16 \times 10^{-3} = 1.6 \times 10^{-2}$.

$$\mathbf{m} = 1.6 \times 10^{-2} \text{ A m}^2 \quad (\text{option a}) \quad \text{Method: §2.1, §4.6}$$

Now redo it: I recomputed this independently and confirm $1.6 \times 10^{-2} \text{ A m}^2$. The trap here is using $\cos$ instead of $\sin$: that would give $1.6 \times 10^{-2} / \sqrt{3} \approx 0.92 \times 10^{-2}$, which is not offered, so the examiner is being kind. In a harder paper it would be. **Torque always uses sine.**

RECALL AND CHECK

A 50-turn coil of area $2 \times 10^{-3} \text{ m}^2$ carries 0.4 A in a field of 0.2 T, at 90° to the field. Find the torque.

ANSWER

$m = NIA = (50)(0.4)(2 \times 10^{-3}) = 4 \times 10^{-2} \text{ A m}^2$. $\tau = mB \sin 90^\circ = (4 \times 10^{-2})(0.2)(1) = \mathbf{8 \times 10^{-3} \text{ N m}}$. At 90° the torque is at its maximum, since $\sin 90^\circ = 1$.

FILL IN THE GAP · TORQUE ON A CURRENT LOOP

1. The torque on a magnetic dipole is $\tau = mB$ _____ θ , using the same trig function as the electric dipole.
2. The magnetic dipole moment of an N-turn loop of area A carrying current I is $m =$ _____.
3. The torque is zero when the loop is _____ with the field, and maximum when it is _____ to it.

ANSWERS

1. sin. Torque always uses sine. Reaching for cosine is the trap.

2. NIA, measured in A m^2 .

3. aligned ($\theta = 0$), and **perpendicular** ($\theta = 90^\circ$).

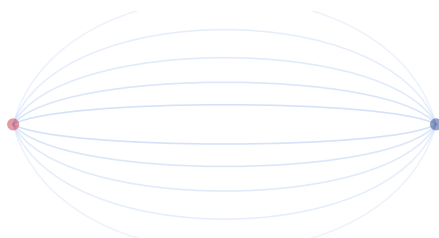
Identical in shape to the electric dipole $\tau = pE \sin \theta$. Learn one, get the other free.

PART VI

Module 5

Maxwell's Equations & Electromagnetic Waves

Four short equations that hold the whole of electricity, magnetism and light. You already know three and a half of them. This module supplies the missing half (the piece Maxwell added) and then watches light fall out of the mathematics.



Teach · A note on the numbering

Your lecturer's slide deck for this material is headed "**Module 6.**" That is the deck's own internal number. In this manual the four earlier decks became Modules 1 to 4, so this new material is your **Module 5**. The physics is identical; only the label differs. Wherever an exam paper says "Module 6: Maxwell's Equations," it means exactly this part.

UNIT 5.1

Maxwell's equations

Learning outcomes: identify the flaw Maxwell found in Ampere's law; define the displacement current and state the corrected Ampere-Maxwell law; state all four Maxwell equations and read their physical meaning.

Teach · You already have three and a half laws

Across Modules 1 to 4 you built four great results without ever seeing them side by side:

- ▶ Gauss's law for electricity, which says charges make electric fields (Unit 1.4).
- ▶ Gauss's law for magnetism, which says there are no magnetic charges, so $\oint \mathbf{B} \cdot d\mathbf{A} = 0$ (Unit 4.2).
- ▶ Faraday's law, a changing magnetic field makes an electric field (Unit 4.3).
- ▶ Ampere's law, currents make magnetic fields (Unit 4.2).

Four laws, and they look complete. Maxwell's genius was to notice that the last one is **broken**, and that repairing it produces light. That is the entire story of this unit.

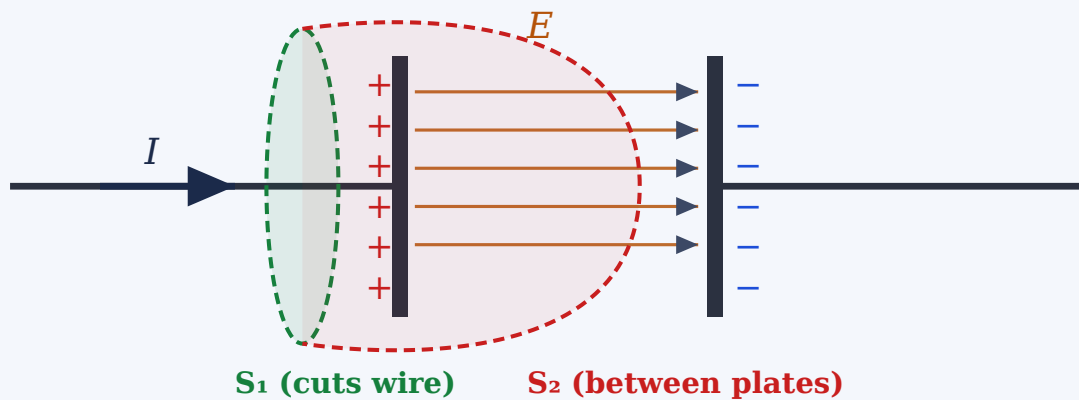
Teach · Ampere's law, as you learned it

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

Reading the symbols: the circle on the integral sign, $\oint$, means "add up all the way around a closed loop"; $d\mathbf{l}$ is one small step along that loop; and I_{enc} is the current enclosed by (threaded through) the loop. So the law says: add up the magnetic field around any closed loop, and you get μ_0 times the current threaded through the loop. It worked perfectly for every problem you solved: straight wires, solenoids, toroids, any steady current. So what could possibly be wrong with it?

Teach · The charging-capacitor problem

Ampere's law contains a hidden assumption: that "the current threaded through the loop" is a definite number. Take a capacitor being charged. Current I flows in the wire. Draw one Amperian loop around that wire, and now ask: which surface do I stretch across the loop to count the current?



The same loop, two surfaces. Surface S_1 is pierced by the wire; surface S_2 passes between the plates, where no charge crosses.

- ▶ **Surface S_1** cuts the wire. The current I pierces it, so $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I \neq 0$.
- ▶ **Surface S_2** bulges out and passes *between the plates*. No charge crosses the gap, so $I_{\text{enc}} = 0$, and $\oint \mathbf{B} \cdot d\mathbf{l} = 0$.

⚠ A logical contradiction

Same loop. Same law. Two different answers: $\mu_0 I$ on one surface, 0 on the other. A law of physics cannot give two answers for one physical situation. Something is missing.

Teach · Maxwell's insight

Between the plates there is no *moving charge*, but there **is** a changing electric field, growing as the capacitor charges. Maxwell proposed that **a changing electric field produces a magnetic field, exactly as a current does**. He gave this changing-field effect a name (the displacement current) and added it to Ampere's law so that both surfaces give the same answer.

● Must-memorise · The displacement current

$$I_d = \epsilon_0 (d\Phi_E / dt)$$

I_d is the displacement current, in amperes (A)

$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$, the permittivity of free space

Φ_E is the electric flux, the electric field multiplied by the area it passes through ($\Phi_E = EA$ for a uniform field); $d\Phi_E/dt$ is its rate of change, in $\text{V}\cdot\text{m/s}$

Memory hook: a changing E counts as a current. No charge moves, yet it does everything a current does to the magnetic field. The ϵ_0 is the price of admission. It converts a changing flux into an equivalent current.

● Must-memorise · The Ampere-Maxwell law

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (I_{\text{enc}} + \epsilon_0 d\Phi_E/dt)$$

Memory hook: old Ampere plus one new term. On surface S_1 the first term does the work (real current, changing flux zero); on surface S_2 the second term does the work (no current, but changing flux). Both surfaces now give $\mu_0 I$. The contradiction is gone.

Worked example · Displacement current in a charging capacitor

Slide

The question. A capacitor with circular plates of radius 10 cm is being charged by a steady current of 2.5 A. (a) What is the displacement current between the plates? (b) At what rate is the electric field between the plates changing?

- 1 Part (a). By charge conservation, whatever current flows in the wire must be continued between the plates by the displacement current. So $I_d = I_{\text{conduction}} = 2.5 \text{ A}$.

This is the whole point of the fix: the two currents match, so the magnetic field is continuous. No calculation needed for part (a).

- 2 Part (b). Convert the plate radius: $r = 10 \text{ cm} = 0.10 \text{ m}$. The plate area is $A = \pi r^2 = \pi(0.10)^2 = 3.14 \times 10^{-2} \text{ m}^2$.

The centi trap again: 10 cm = 0.10 m, and it is then squared. Get this wrong and the answer is off by a factor of 100 (or 10 000).

- 3 Between parallel plates E is uniform across the area A , so the flux is simply $\Phi_E = EA$. Put this into $I_d = \epsilon_0 d\Phi_E/dt = \epsilon_0 A (dE/dt)$.

A is constant, so it comes outside the rate of change. Only E changes.

- 4 Rearrange for dE/dt (rearrange first, substitute second, see Foundations F.2): $dE/dt = I_d / (\epsilon_0 A)$.

- 5 Substitute: $dE/dt = 2.5 / [(8.85 \times 10^{-12})(3.14 \times 10^{-2})] = 2.5 / (2.78 \times 10^{-13})$.

$$I_d = 2.5 \text{ A } dE/dt = 9.0 \times 10^{12} \text{ V/(m}\cdot\text{s)}$$

Now redo it: I recomputed the denominator as $\epsilon_0 A = 2.779 \times 10^{-13}$ and the rate as $8.99 \times 10^{12} \text{ V/(m}\cdot\text{s)}$, so 9.0×10^{12} to two significant figures is correct. That is nine trillion volts per metre every second, an enormous rate, and exactly what is needed to make a "current" of 2.5 A out of nothing but a changing field.

⚠ **Trap · The displacement current is not a flow of charge**

Nothing crosses the gap between the plates: no electrons, no ions, nothing. The displacement current is not charge in motion; it is the magnetic effect of a *changing electric field*, dressed up in the units of a current so it can be added to Ampere's law. Yet by charge conservation its size equals the real conduction current in the wire, which is why the magnetic field joins up smoothly. Hold both facts at once: **not a real current, but exactly as large as one.**

● **Must-memorise · The four Maxwell equations**

LAW	INTEGRAL FORM	WHAT IT SAYS
Gauss (E)	$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}} / \epsilon_0$	Charges create electric fields
Gauss (B)	$\oint \mathbf{B} \cdot d\mathbf{A} = 0$	No magnetic monopoles
Faraday	$\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$	A changing B creates E
Ampere-Maxwell	$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0(I_{\text{enc}} + \epsilon_0 d\Phi_E/dt)$	Currents & changing E create B

Memory hook: two about sources (Gauss E, Gauss B), two about change (Faraday, Ampere-Maxwell). Only one of the four carries a term Maxwell added (the last one) and that one term is what makes light possible.

Teach · The same four laws in differential form

If you have read Foundations F.11, the differential form is just the same four statements written point-by-point instead of loop-by-loop.

#	INTEGRAL FORM	DIFFERENTIAL FORM
I	$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}}/\epsilon_0$	$\nabla \cdot \mathbf{E} = \rho/\epsilon_0$
II	$\oint \mathbf{B} \cdot d\mathbf{A} = 0$	$\nabla \cdot \mathbf{B} = 0$
III	$\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$	$\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$
IV	$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I + \mu_0 \epsilon_0 d\Phi_E/dt$	$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial \mathbf{E}/\partial t$

Divergence for the two Gauss laws (sources), curl for the two circulation laws (change). That is all the differential form is.

Teach · The beautiful symmetry

Look at the two "change" laws side by side, stripped to their hearts:

Faraday

A changing **B** creates **E**.

$$\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$$

Ampere-Maxwell

A changing **E** creates **B**.

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \epsilon_0 d\Phi_E/dt$$

A changing B makes E; a changing E makes B. Each field, by changing, gives birth to the other. **This mutual creation is the engine that drives an electromagnetic wave**, as the next unit shows.

Recall and check

A parallel-plate capacitor is charged by a constant current of 2 A. What is the displacement current between the plates, and why?

ANSWER

Exactly 2 A. By charge conservation the displacement current between the plates must equal the conduction current in the wire, otherwise the magnetic field would jump discontinuously as you slid the surface across.

Fill in the gap · Maxwell's equations

1. The law Maxwell repaired was _____'s law.
2. The missing piece is the _____ current, equal to ϵ_0 times the rate of change of electric _____.
3. Of the four equations, the two written with a dot product over a closed surface are the two _____ laws.

ANSWERS

1. Ampere. 2. displacement, flux. 3. Gauss (source) laws: Gauss for E and Gauss for B.

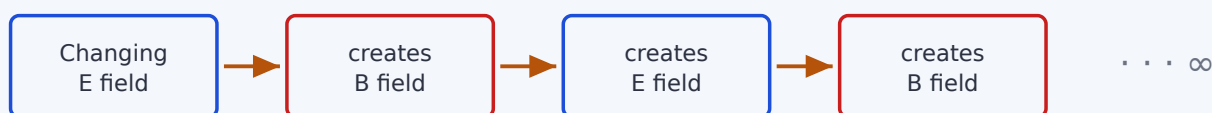
UNIT 5.2

Electromagnetic waves

Learning outcomes: explain how Maxwell's equations predict a self-sustaining wave; state the speed of light from μ_0 and ϵ_0 ; use the plane-wave results $E = cB$ and $c = f\lambda$; state the energy density and intensity (Poynting vector) of an EM wave.

Teach · The chain reaction

Put the two "change" laws together and something remarkable happens. A changing electric field creates a magnetic field. But that magnetic field is itself changing, so it creates an electric field. Which is changing, so it creates a magnetic field. And so on, forever, with neither field needing charges or wires to keep it alive:



Each field, by changing, regenerates the other. The disturbance sustains itself and travels through empty space. That is a wave.

They feed each other. They sustain each other. They propagate through space. **That is a wave**, an electromagnetic wave, and it needs no medium at all.

Teach · The speed falls out of the mathematics

In free space (no charges, no currents) Maxwell's equations combine into a wave equation for E:

$$\partial^2 E / \partial x^2 = \mu_0 \epsilon_0 (\partial^2 E / \partial t^2)$$

Compare this with the standard wave equation $\partial^2 y / \partial x^2 = (1/v^2)(\partial^2 y / \partial t^2)$. Matching the two, the quantity $\mu_0 \epsilon_0$ must play the role of $1/v^2$. So the wave travels at $v = 1/\sqrt{(\mu_0 \epsilon_0)}$. Now put the numbers in.

● Must-memorise · The speed of light

$$c = 1 / \sqrt{(\mu_0 \epsilon_0)} = 3.0 \times 10^8 \text{ m/s}$$

$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$, the permeability of free space

$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$, the permittivity of free space

Memory hook: two constants you measured with capacitors and coils (nothing about light) combine to give the speed of light exactly. That is the moment electricity, magnetism and optics became one subject.

Worked example · Computing the speed of light

Slide

The question. Evaluate $c = 1/\sqrt{(\mu_0 \epsilon_0)}$ from the constants alone.

- 1 Form the product $\mu_0 \epsilon_0 = (4\pi \times 10^{-7})(8.85 \times 10^{-12}) = 1.112 \times 10^{-17}$.
 $4\pi \approx 12.566$, so $12.566 \times 8.85 \approx 111.2$; and $10^{-7} \times 10^{-12} = 10^{-19}$. Together $111.2 \times 10^{-19} = 1.112 \times 10^{-17}$.
- 2 Take the square root: $\sqrt{(1.112 \times 10^{-17})} = 3.335 \times 10^{-9}$.
- 3 Invert: $c = 1 / (3.335 \times 10^{-9}) = 2.998 \times 10^8 \text{ m/s}$.

$c = 2.998 \times 10^8 \text{ m/s}$, the measured speed of light exactly

Now redo it: I recomputed $1/\sqrt{(\mu_0 \epsilon_0)}$ independently and obtained $2.9986 \times 10^8 \text{ m/s}$, so $3.0 \times 10^8 \text{ m/s}$ is right to two figures. Maxwell wrote in 1865 that light "consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena." He predicted EM waves from pure mathematics; Hertz confirmed them in 1887, eight years after Maxwell's death.

● Must-memorise · The plane-wave solution

$$E_y = E_0 \sin(kx - \omega t) \quad B_z = B_0 \sin(kx - \omega t)$$

E_0, B_0 are the field amplitudes

$k = 2\pi/\lambda$ is the wave number (rad/m)

$\omega = 2\pi f$ is the angular frequency (rad/s)

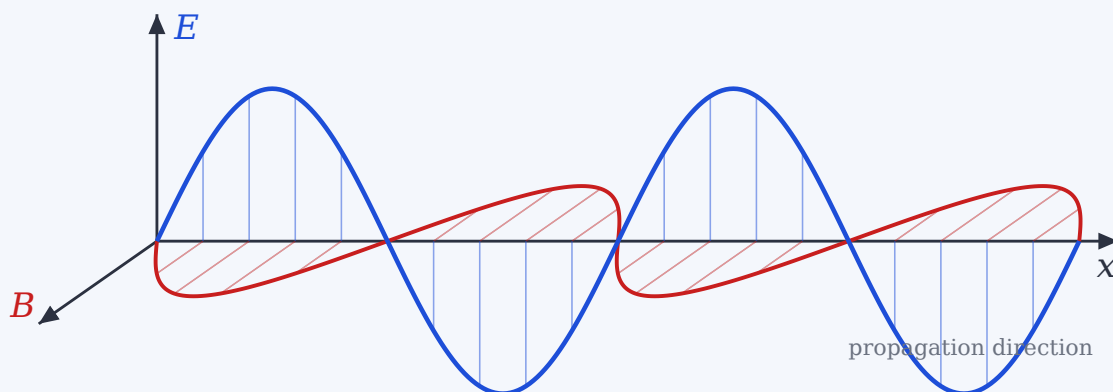
λ = wavelength, f = frequency

Memory hook: E and B share the same sine: same k , same ω , same phase. They peak together and vanish together. One wave, two fields locked in step.

● Must-memorise · The two locking relations

$$E = cB \quad \text{and} \quad c = f\lambda$$

Memory hook: at every instant the field magnitudes are tied by $E = cB$, so E is always the much bigger number (c is huge). And speed = frequency $\times$ wavelength ties how fast it wiggles to how long each wave is. These two one-liners answer most numerical questions on this unit.



E oscillates in one plane (here vertical), B in the perpendicular plane, both in step, both at right angles to the direction of travel.

Three facts to read straight off the picture: E and B are **perpendicular to each other**; both are **perpendicular to the direction of travel** (the wave is transverse); and they are **in phase**, rising and falling together.

Worked example · From E_0 to B_0

[Slide](#)

The question. A radio transmitter produces an electromagnetic wave with an electric field amplitude of 0.080 V/m . What is the corresponding magnetic field amplitude?

- 1 Use the locking relation $E_0 = cB_0$, and rearrange for $B_0 = E_0 / c$.

Rearrange first: divide both sides by c . Only then substitute.

- 2 Substitute: $B_0 = 0.080 / (3.0 \times 10^8)$.

$$B_0 = 2.67 \times 10^{-10} \text{ T}$$

Now redo it: $0.080 / (3.0 \times 10^8) = 2.667 \times 10^{-10} \text{ T}$, confirming the slide. Notice how tiny B is compared with E. That factor of c between them is exactly why we usually notice the electric part of light and rarely the magnetic part.

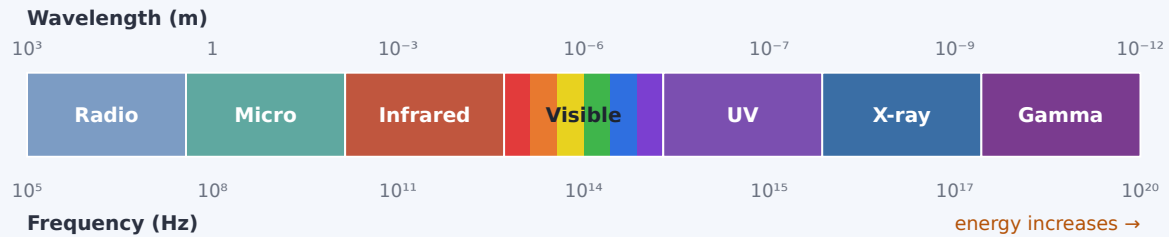
● Must-memorise · The properties of an EM wave

PROPERTY	WHAT IT MEANS
Transverse	E and B oscillate perpendicular to the direction of propagation
Speed c	In vacuum every EM wave travels at $3.0 \times 10^8 \text{ m/s}$, whatever its frequency
No medium	Unlike sound, EM waves travel through empty space
$E = cB$	The fields are linked at every point and every instant
Carry energy	Energy flows in the direction of travel (the Poynting vector)

Memory hook: transverse, at c , no medium, $E = cB$, carries energy. Five facts; the exam tests all five.

Teach · The electromagnetic spectrum

Radio, microwave, infrared, visible light, ultraviolet, X-rays and gamma rays are the **same physical wave**. They differ only in frequency (and therefore wavelength, since $c = f\lambda$). All of them travel at c in vacuum.



One family, ordered by wavelength and frequency. Visible light is the thin band your eye happens to detect.

Higher frequency means shorter wavelength and more energy per photon, which is why gamma rays are dangerous and radio waves are not, though both are "light."

● Must-memorise · Energy density

$$u = \frac{1}{2}\epsilon_0 E^2 + \frac{B^2}{2\mu_0} \rightarrow u = \epsilon_0 E^2 = \frac{B^2}{\mu_0}$$

Memory hook: the electric and magnetic halves are always equal (because $E = cB$ and $c = 1/\sqrt{(\mu_0\epsilon_0)}$), so the total is just twice either one. You may quote whichever single form is handy.

● Must-memorise · Intensity and the Poynting vector

The Poynting vector gives the rate of energy flow per unit area, pointing along the direction of travel:

$$\mathbf{S} = (1/\mu_0) \mathbf{E} \times \mathbf{B}$$

The symbol $\times$ is the vector cross product: $\mathbf{E} \times \mathbf{B}$ is a new vector pointing perpendicular to both $\mathbf{E}$ and $\mathbf{B}$, which is exactly the direction the wave carries its energy.

The average power per unit area (the **intensity**) of a sinusoidal wave is

$$I = \frac{1}{2}\epsilon_0 c E_0^2 = \frac{c B_0^2}{2\mu_0} = \frac{E_0^2}{2\mu_0 c}$$

Memory hook: $\mathbf{S}$ points where the energy goes, the same way the wave moves. The factor of $\frac{1}{2}$ is the average of $\sin^2$, exactly as the $\frac{1}{2}$ in $\frac{1}{2}CV^2$ came from a quantity that climbed from zero.

The question. A laser pointer emits 5.0 mW of power in a beam of cross-sectional area $3.0 \times 10^{-6} \text{ m}^2$. Calculate (a) the intensity and (b) the electric field amplitude of the beam.

- 1 Part (a). Intensity is power per unit area: $I = P/A = (5.0 \times 10^{-3}) / (3.0 \times 10^{-6})$.
Convert first: $5.0 \text{ mW} = 5.0 \times 10^{-3} \text{ W}$. The area is already in SI.
- 2 $I = 1.67 \times 10^3 = 1667 \text{ W/m}^2$.
- 3 Part (b). Rearrange $I = \frac{1}{2}\epsilon_0 c E_0^2$ for the amplitude: $E_0 = \sqrt{2I / (\epsilon_0 c)}$.
- 4 Substitute: $E_0 = \sqrt{2(1667) / [(8.85 \times 10^{-12})(3.00 \times 10^8)]} = \sqrt{3334 / 2.655 \times 10^{-3}}$.

$$I = 1667 \text{ W/m}^2 \quad E_0 \approx 1120 \text{ V/m}$$

Now redo it: I get $I = 1666.7 \text{ W/m}^2$ and $E_0 = 1120.5 \text{ V/m}$, confirming the slide. Despite only 5 mW of power, the intensity beats full sunlight ($\sim 1000 \text{ W/m}^2$) because the energy is concentrated into a pinpoint area. That concentration is exactly why a low-power laser can still damage the eye.

⚠ Trap · Amplitude, average, and the factor of $\frac{1}{2}$

The formula $I = \frac{1}{2}\epsilon_0 c E_0^2$ uses the **amplitude** E_0 (the peak value) and carries a $\frac{1}{2}$ because it is the time-average of a $\sin^2$ that swings between 0 and 1. If a question gives you an rms field instead, the $\frac{1}{2}$ disappears ($I = \epsilon_0 c E_{\text{rms}}^2$), because the averaging is already baked in. Read which field you were given before you reach for a formula.

Recall and check

An FM radio station broadcasts at 100 MHz. What is the wavelength of its wave?

ANSWER

Use $c = f\lambda$, so $\lambda = c/f = (3.0 \times 10^8)/(100 \times 10^6) = 3.0 \text{ m}$. A three-metre wave, which is why FM aerials are about that size.

Fill in the gap · Electromagnetic waves

1. An EM wave is sustained because a changing E creates B and a changing B creates _____; neither needs a _____ to travel.
2. The speed of light is $c = 1/\sqrt{\text{_____}}$, and it is the same for every frequency in vacuum.
3. The field magnitudes obey $E = \text{_____} B$, so the electric field is always the larger number.
4. The energy flow per unit area is given by the _____ vector, $S = (1/\mu_0) E \times B$.

ANSWERS

1. E, medium. 2. $\mu_0 \epsilon_0$. 3. c. 4. Poynting.

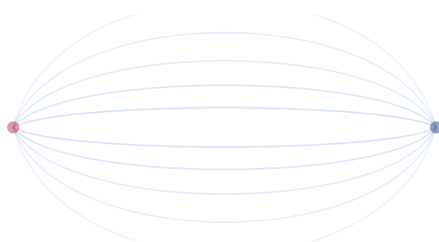
● **Must-memorise · Module 5 key takeaways**

1. Maxwell's equations are four equations that completely describe classical electromagnetism; the key addition was the displacement current in the Ampere-Maxwell law.
2. The displacement current $I_d = \epsilon_0 d\Phi_E/dt$ is the effect of a changing electric field, not a flow of charge; it restores continuity and symmetry.
3. EM waves arise from the mutual induction of E and B, and travel at $c = 1/\sqrt{\mu_0\epsilon_0} = 3 \times 10^8$ m/s in vacuum.
4. EM waves are transverse: E, B and the direction of travel are mutually perpendicular, with $E = cB$ at all times.
5. The spectrum runs from radio to gamma rays, all the same wave, all at c, differing only in frequency.

PART VII

Every question, fully solved

Every classwork question, class drill and tutorial from the slide decks, plus all thirty questions from CBT Test 1 and Test 2, worked line by line. Each solution names the section that teaches its method, so a wrong answer sends you straight to the fix. Attempt each question on blank paper before reading the solution: reading a solution you have not first struggled with teaches almost nothing.



Module 1 classwork

THE QUESTIONS · MODULE 1, UNITS 1.2 TO 1.4

From the Coulomb's law classwork:

1. What is the direction of the field 1 m away from a 1 C charge, in or out?
2. What force would a 1.4×10^{-5} C negative charge experience at 1 m from the 1 C positive charge?
3. Would that force be directed toward or away from the 1 C charge?
4. Point charge A has -1.0 nC and point charge B has 4.0 nC, separated by 1.0 cm. Find the magnitude and direction of the force on each.

From the potential classwork:

1. Calculate the electric potential 2 m from a charge of 5×10^{-6} C.
2. A charge of 3 C moves through a potential difference of 12 V. Calculate the work done.
3. Explain why no work is done along an equipotential surface.

From the Gauss's law classwork:

1. A point charge of 5×10^{-6} C is at the centre of a spherical Gaussian surface of radius 0.20 m. Calculate the field at the surface.
2. A closed spherical surface encloses a charge of 3 μ C. Calculate the total flux.
3. An infinitely long wire carries a linear charge density of 4×10^{-6} C/m. Using a cylindrical Gaussian surface of radius 0.05 m, calculate the field.
4. A solid sphere of radius 0.10 m has a uniform charge density of 2×10^{-6} C/m³. Find the field 0.05 m from the centre.
1. A conducting sphere of radius 0.25 m carries 6×10^{-6} C. Find the field 0.50 m from the centre.

Questions 1 to 4, 5, 6, 8, 9 and 10 are worked in full inside Units 1.2, 1.3 and 1.4, at the point where the relevant idea is taught. Rather than repeat them, the answers are summarised below and the two *new* questions (11 and 12) are worked in full.

Q	ANSWER	WORKED IN FULL AT
1	Outward, since field lines leave a positive charge	Unit 1.2
2	$F \approx 1.26 \times 10^5$ N	Unit 1.2
3	Toward the 1 C charge (opposite charges attract)	Unit 1.2
4	$F = 3.6 \times 10^{-4}$ N on each, attractive	Unit 1.2
5	$V \approx 2.25 \times 10^4$ V	Unit 1.3
6	$W = 36$ J	Unit 1.3
7	Because V is constant along it, so $\Delta V = 0$ and $W = q\Delta V = 0$	Unit 1.3
8	$E \approx 1.12 \times 10^6$ N/C	Unit 1.4
9	$\Phi \approx 3.39 \times 10^5$ N m ² /C	Unit 1.4
10	$E \approx 1.44 \times 10^6$ N/C	Unit 1.4

Q11 · FIELD INSIDE A UNIFORMLY CHARGED SOLID SPHERE

A solid sphere of radius 0.10 m has a uniform charge density $\rho = 2 \times 10^{-6} \text{ C/m}^3$. Find the field at $r = 0.05 \text{ m}$ from the centre.

- 1 Note carefully: $r = 0.05 \text{ m}$ is *less* than $R = 0.10 \text{ m}$, so the point is **inside** the sphere. This changes everything.
This is the whole point of the question. A Gaussian surface at $r = 0.05 \text{ m}$ encloses only part of the total charge, not all of it.
- 2 Find the charge enclosed by a Gaussian sphere of radius r . Charge density times the volume of that inner sphere: $Q_{\text{enc}} = \rho \times (4/3)\pi r^3$.
Only the charge inside the Gaussian surface counts. The charge in the outer shell contributes nothing, by Gauss's law.
- 3 $Q_{\text{enc}} = (2 \times 10^{-6})(4/3)\pi(0.05)^3 = (2 \times 10^{-6})(5.236 \times 10^{-4}) = 1.047 \times 10^{-9} \text{ C}$.
 $(0.05)^3 = 1.25 \times 10^{-4}$, and $(4/3)\pi = 4.189$, so the volume is $5.236 \times 10^{-4} \text{ m}^3$.
- 4 Apply Gauss's law: $E(4\pi r^2) = Q_{\text{enc}}/\epsilon_0$. The area is $4\pi(0.05)^2 = 3.142 \times 10^{-2} \text{ m}^2$.
- 5 $E = Q_{\text{enc}} / (\epsilon_0 \times 4\pi r^2) = (1.047 \times 10^{-9}) / [(8.85 \times 10^{-12})(3.142 \times 10^{-2})]$
Denominator: 2.781×10^{-13} .
- 6 $E = 3.77 \times 10^3 \text{ N/C}$.

$E \approx 3.8 \times 10^3 \text{ N/C}$, directed radially outward

The physics to take away: if you simply substitute the *total* charge into $E = kQ/r^2$ you get a badly wrong answer. **Inside** a uniformly charged solid sphere the field grows *linearly* with r (it is proportional to r , since $r^3/r^2 = r$), reaching a maximum at the surface and only then falling off as $1/r^2$. At the exact centre, the field is zero.

Q12 · FIELD OUTSIDE A CHARGED CONDUCTING SPHERE

A conducting sphere of radius 0.25 m carries a total charge of $6 \times 10^{-6} \text{ C}$. Find the field at 0.50 m from the centre.

- 1 Here $r = 0.50 \text{ m}$ is *greater* than $R = 0.25 \text{ m}$, so the point is **outside** the sphere.
Compare with Q11, where the point was inside. Always check this first.
- 2 Outside a charged sphere, the whole charge behaves exactly as though concentrated at the centre, so use the point charge result: $E = kQ/r^2$.
This is the Gauss's law result from Unit 1.4. The Gaussian sphere encloses the entire charge.
- 3 $E = (8.99 \times 10^9)(6 \times 10^{-6}) / (0.50)^2$.
- 4 Numerator: 5.394×10^4 . Denominator: $(0.50)^2 = 0.25$.
Not 0.5. Squaring a half gives a quarter.
- 5 $E = 5.394 \times 10^4 / 0.25 = 2.16 \times 10^5 \text{ N/C}$.

$E \approx 2.16 \times 10^5 \text{ N/C}$, radially outward

Contrast Q11 and Q12 side by side. They look almost identical, but one is inside and one is outside, and the method is completely different. *Always* compare r with R before choosing your approach.

Note too that if this question had asked for the field *inside* the conducting sphere, the answer would be exactly **zero**.

Module 2 classwork

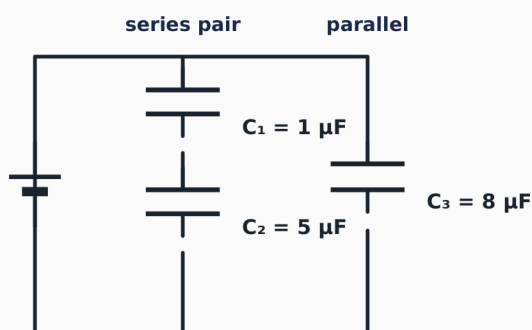
THE QUESTIONS · MODULE 2 CLASSWORK

1. Determine the total capacitance of the capacitors shown ($C_1 = 1 \mu\text{F}$ and $C_2 = 5 \mu\text{F}$ stacked in one branch; $C_3 = 8 \mu\text{F}$ in the other branch).
 2. What is the capacitance of a capacitor having a stored energy of 2.25 mJ after being charged with a 30 volt source?
 3. The area of each plate of a parallel plate capacitor is $4 \times 10^{-2} \text{ m}^2$. The dielectric thickness between the plates is 10^{-3} m and the relative permittivity is 7. What is the capacitance?
1. R_1 , R_2 and R_3 are in series with a 48 V source, with $R_1 = 7 \Omega$, $R_2 = 6 \Omega$ and $R_3 = 3 \Omega$. Calculate the current through R_3 .
 2. A metal sphere on an insulating rod carries 4.5 nC when its potential is 180 V higher than its surroundings. What is the capacitance?

Also: the class drill from the dipole slides, asking for the *unit* of dipole moment. Answer: coulomb metre (C m), since $p = qd$.

Q1 · THE CAPACITOR NETWORK

Q1. Determine the total capacitance of the capacitors shown, where $C_1 = 1 \mu\text{F}$ and $C_2 = 5 \mu\text{F}$ are stacked in one branch, and $C_3 = 8 \mu\text{F}$ is in the other branch.



C_1 and C_2 are stacked in the same branch, so they are in series. That branch is in parallel with C_3 .

- 1 Read the circuit. C_1 and C_2 are stacked one above the other in the same branch, so they are in **series** with each other. That branch is then in **parallel** with C_3 .
Getting the topology right is the whole question. Everything else is arithmetic.
- 2 Series pair first, using product over sum since there are exactly two: $C_{12} = (1 \times 5)/(1 + 5) = 5/6 = \mathbf{0.833 \mu\text{F}}$.
Sanity check: smaller than the smallest ($1 \mu\text{F}$), as a series combination must be.
- 3 Now add C_3 in parallel. Capacitors in parallel simply add: $C_{\text{total}} = 0.833 + 8 = \mathbf{8.83 \mu\text{F}}$.

$$C_{\text{total}} \approx 8.83 \mu\text{F}$$

Note the trap: if you had wrongly treated C_1 and C_2 as parallel you would get $1 + 5 = 6$, then a series with 8 giving $3.43 \mu\text{F}$, which is quite different. Read the circuit carefully before you compute anything.

Q2 · CAPACITANCE FROM STORED ENERGY

Q2. What is the capacitance of a capacitor having a stored energy of 2.25 mJ after being charged with a 30 volt source?

- 1 We know energy and voltage, and want C. Choose the energy form that contains exactly those: $W = \frac{1}{2}CV^2$.

Three forms exist. Pick the one matching your known quantities.

- 2 Rearrange: $C = 2W / V^2$.

Multiply by 2 first, then divide by V squared. See Foundations F.2.

- 3 Convert: $W = 2.25 \text{ mJ} = 2.25 \times 10^{-3} \text{ J}$.

Milli means 10^{-3} .

- 4 $C = 2(2.25 \times 10^{-3}) / (30)^2 = (4.5 \times 10^{-3}) / 900$.

$30^2 = 900$, not 60. Square it, do not double it.

- 5 $C = 5 \times 10^{-6} \text{ F}$.

$$C = 5 \times 10^{-6} \text{ F} = 5 \text{ } \mu\text{F}$$

Check by working forwards: $W = \frac{1}{2}(5 \times 10^{-6})(900) = 2.25 \times 10^{-3} \text{ J}$. That is 2.25 mJ, as given.

Confirmed.

Q3 · PARALLEL PLATE CAPACITOR WITH A DIELECTRIC

Q3. The area of each plate of a parallel plate capacitor is $4 \times 10^{-2} \text{ m}^2$. The dielectric thickness between the plates is 10^{-3} m and the relative permittivity is 7. What is the capacitance?

- 1 With a dielectric present, use $C = \epsilon_r \epsilon_0 A / d$.

The relative permittivity $\epsilon_r = 7$ multiplies the vacuum result.

- 2 Numerator: $\epsilon_r \epsilon_0 A = (7)(8.85 \times 10^{-12})(4 \times 10^{-2})$.

$7 \times 8.85 = 61.95$, so 61.95×10^{-12} . Times 4×10^{-2} gives $247.8 \times 10^{-14} = 2.478 \times 10^{-12}$.

- 3 Divide by $d = 10^{-3} \text{ m}$: $C = (2.478 \times 10^{-12}) / (10^{-3})$.

Dividing by 10^{-3} means adding 3 to the exponent.

- 4 $C = 2.478 \times 10^{-9} \text{ F}$.

$$C \approx 2.48 \times 10^{-9} \text{ F} \quad (\text{about } 2.5 \text{ nF})$$

Now redo it: the dielectric multiplied the capacitance by 7. Without it, C would be only $3.54 \times 10^{-10} \text{ F}$.

Dielectrics always increase capacitance, never decrease it.

Q4 · CURRENT THROUGH A SERIES RESISTOR

Q4. R_1 , R_2 and R_3 are in series with a 48 V source, where $R_1 = 7\ \Omega$, $R_2 = 6\ \Omega$ and $R_3 = 3\ \Omega$. Calculate the current through R_3 .

- 1 All three resistors are in **series**, so add them: $R_{\text{total}} = 7 + 6 + 3 = \mathbf{16\ \Omega}$.

Series simply adds.

- 2 Ohm's law for the whole circuit: $I = V / R_{\text{total}} = 48 / 16 = \mathbf{3\ A}$.

- 3 The question asks for the current through R_3 specifically. In a *series* circuit, **the same current flows through every element**.

This is the trick in the question. There is nothing extra to calculate. The current through R_3 is the circuit current.

I = 3 A through R_3 (and through R_1 and R_2 as well)

Check: the voltage drops are $(3)(7) = 21\ \text{V}$, $(3)(6) = 18\ \text{V}$ and $(3)(3) = 9\ \text{V}$. Sum: $21 + 18 + 9 = 48\ \text{V}$, the source voltage. It balances.

Q5 · CAPACITANCE OF A CHARGED SPHERE

Q5. A metal sphere on an insulating rod carries a charge of 4.5 nC when its potential is 180 V higher than its surroundings. What is the capacitance of the capacitor formed by the sphere and its surroundings?

- 1 Use the definition of capacitance directly: $C = Q/V$.

We are given a charge and a potential, so no geometry is needed at all.

- 2 Convert: $Q = 4.5\ \text{nC} = 4.5 \times 10^{-9}\ \text{C}$.

Nano means 10^{-9} .

- 3 $C = (4.5 \times 10^{-9}) / 180$.

The potential is 180 V above its surroundings, which is the potential difference we need.

- 4 $C = 2.5 \times 10^{-11}\ \text{F}$.

C = $2.5 \times 10^{-11}\ \text{F}$ = 25 pF

Now redo it: notice that the sphere and its surroundings form a capacitor, even though it looks nothing like two parallel plates. Any two conductors at different potentials form a capacitor. That is the definition.

Module 3 classwork and circuit questions

THE QUESTIONS · MODULE 3

1. Use the voltage divider rule to find V_2 when a 20 V source drives 15 k Ω in series with 10 k Ω .
2. What is the voltage drop across R_1 and R_2 if $R_1 = 4 \Omega$, $R_2 = 10 \Omega$ and the source is 12 V?
3. $R_1 = 2.2 \text{ k}\Omega$ is in parallel with $R_2 = 4.7 \text{ k}\Omega$. The total current is 8.0 mA. Find the current in each.
4. MCQ: with 100 Ω and 200 Ω in parallel, what is the current in R_2 ? (6.7 mA / 13.3 mA / 20 mA / 26.7 mA)
5. MCQ: for the same circuit, what is the voltage across R_2 ? (0 V / 0.67 V / 1.33 V / 4.0 V)
6. Assume R_2 is twice the size of R_1 . What is the voltage across R_2 ?
7. Use Kirchhoff's laws to determine I_1 , I_2 and the p.d. across R_1 and R_2 .

Q	ANSWER	WORKED IN FULL AT
1	$V_2 = 8.0 \text{ V}$	Unit 3.3
2	$V_1 = 3.43 \text{ V}$, $V_2 = 8.57 \text{ V}$	Unit 3.3
3	$I_1 = 5.45 \text{ mA}$, $I_2 = 2.55 \text{ mA}$	Unit 3.3
4	6.7 mA	Unit 3.3
5	1.33 V	Unit 3.3
7	Full two-loop Kirchhoff solution	Unit 3.4

Q6 · VOLTAGE ACROSS R_2 WHEN $R_2 = 2R_1$

This is a proportional reasoning question. No numbers are given for the resistors, and none are needed.

- 1 Let $R_1 = R$. Then $R_2 = 2R$.
Introducing a symbol lets us work without knowing the actual value.
- 2 They are in series, so the total is $R_T = R + 2R = 3R$.
- 3 Voltage divider, with the resistor we want on top: $V_2 = [2R / 3R] \times V_s$.
The R cancels completely, which is why no numerical value was needed.
- 4 $V_2 = (2/3)V_s$.

$V_2 = \text{two thirds of the source voltage}$

Check: R_1 then takes $(1/3)V_s$, and $2/3 + 1/3 = 1$, the whole source voltage. The larger resistor takes the larger share, exactly as the divider rule requires. If V_s were 12 V, then $V_2 = 8 \text{ V}$.

Q7 · THE KIRCHHOFF TWO-LOOP PROBLEM

The full worked solution, with all three currents and all four voltages, is given in **Unit 3.4**. Rather than repeat it, here is the checklist to apply to any Kirchhoff question of this type. Work the problem in Unit 3.4 from scratch on blank paper, then compare.

- 1 Label** every current with an arrow. Guess the directions; the algebra will correct you if you are wrong.
- 2 Write one KVL equation per loop**, using the four sign rules from Unit 3.4.
- 3 Write KCL** at a junction to link the currents together.
- 4 Count equations against unknowns**. They must be equal, or the system cannot be solved.
- 5 Solve by substitution**, reducing to one unknown at a time.
- 6 Check** by walking each loop and confirming the voltages sum to zero.

Answers (from Unit 3.4): $I_1 = 3.461 \text{ A}$, $I_2 = 1.923 \text{ A}$, $I_3 = 1.538 \text{ A}$. Voltages: 34.61 V , 15.38 V , 9.23 V and 6.15 V .

Remember: a negative current is not an error. It means the true direction is opposite to the arrow you drew. Report the magnitude and say so.

SOLUTIONS S.4

Module 4 tutorial questions

THE QUESTIONS · MODULE 4 TUTORIALS

1. A proton enters a magnetic field of 0.5 T perpendicular to the field at $2 \times 10^6 \text{ m/s}$. Calculate (a) the magnetic force, (b) the radius, (c) the period.
2. An electron enters a magnetic field at 45° . Show that its trajectory is helical, and determine the radius and the pitch.
3. A proton moves with a velocity of $3.5 \times 10^6 \text{ m/s}$ in a magnetic field of 0.25 T at an angle of 60° . Calculate (a) the magnetic force, (b) the radius of the helical path, (c) the pitch of the helix.
4. Class drill: a wire of 4Ω is connected to a battery of emf 6 V and internal resistance r . The current flowing is 1 A . Calculate r .

Constants: proton mass $m_p = 1.67 \times 10^{-27} \text{ kg}$; electron mass $m_e = 9.11 \times 10^{-31} \text{ kg}$; $e = 1.6 \times 10^{-19} \text{ C}$.

Q1 · PROTON ENTERING PERPENDICULAR TO THE FIELD

Q1. A proton enters a magnetic field of 0.5 T perpendicular to the field at 2×10^6 m/s. Calculate (a) the magnetic force, (b) the radius of its path, and (c) the period.

- 1 (a) The force.** Perpendicular entry means $\theta = 90^\circ$ and $\sin 90^\circ = 1$, so $F = qvB$.

This is the maximum possible force.

- 2** $F = (1.6 \times 10^{-19})(2 \times 10^6)(0.5)$. Multiply the first two: 3.2×10^{-13} . Then times 0.5.

Fronts: $1.6 \times 2 = 3.2$. Exponents: $-19 + 6 = -13$.

- 3** $F = 1.6 \times 10^{-13}$ N.

- 4 (b) The radius.** Use $r = mv/(qB)$. Numerator: $(1.67 \times 10^{-27})(2 \times 10^6) = 3.34 \times 10^{-21}$.

The proton mass, not the electron mass. Check which particle you are given.

- 5** Denominator: $qB = (1.6 \times 10^{-19})(0.5) = 8.0 \times 10^{-20}$.

- 6** $r = (3.34 \times 10^{-21}) / (8.0 \times 10^{-20}) = 4.18 \times 10^{-2}$ m (about 4.2 cm).

$3.34/8.0 = 0.4175$, and $-21 - (-20) = -1$, so $0.4175 \times 10^{-1} = 4.18 \times 10^{-2}$.

- 7 (c) The period.** The period is the time for one full circle: $T = 2\pi m/(qB)$, or equivalently $T = 2\pi r/v$.

Use the second form, since we already have r .

- 8** $T = 2\pi(4.18 \times 10^{-2}) / (2 \times 10^6) = (0.2626) / (2 \times 10^6) = 1.31 \times 10^{-7}$ s.

(a) $F = 1.6 \times 10^{-13}$ N · (b) $r \approx 4.18 \times 10^{-2}$ m · (c) $T \approx 1.31 \times 10^{-7}$ s

Cross-check the period the other way: $T = 2\pi m/(qB) = 2\pi(1.67 \times 10^{-27})/(8.0 \times 10^{-20}) = 1.31 \times 10^{-7}$ s. Identical. Note the period does *not* depend on the speed, which is the cyclotron principle from Unit 4.1.

Q2 · ELECTRON AT 45 DEGREES: WHY THE PATH IS A HELIX

This question asks you to *show* something, so the reasoning matters more than the arithmetic.

- 1 Resolve the velocity into two components (Foundations F.3): $v_{\text{parallel}} = v \cos 45^\circ$ and $v_{\text{perp}} = v \sin 45^\circ$.
 $\cos 45^\circ = \sin 45^\circ = 0.707$, so at exactly 45° the two components are equal.
- 2 The **perpendicular** component feels a force $F = qv_{\text{perp}}B$, always at right angles to the motion. That produces **circular motion** in the plane perpendicular to B .
A constant force at right angles to velocity is precisely centripetal force.
- 3 The **parallel** component feels *no* force at all, because $\sin 0 = 0$. So it continues **unchanged and constant**, carrying the particle steadily forward along the field.
This is the crucial insight. The magnetic field cannot touch the parallel component.
- 4 Combine them: a circle in one plane, plus a constant drift at right angles to that plane. A circle plus a steady forward drift is a **helix**. This is what was to be shown.
Like the thread of a screw, or a stretched spring.
- 5 **Radius:** only the perpendicular component produces the circle, so $r = mv_{\text{perp}} / (qB) = m(v \sin 45^\circ) / (qB)$.
Use $v \sin \theta$, not v . Using the full v here is the classic error.
- 6 **Pitch:** the pitch is the forward distance travelled during *one complete revolution*. So it is the parallel speed multiplied by the period: $p = v_{\text{parallel}} \times T = (v \cos 45^\circ)(2\pi m / qB)$.

$$r = m v \sin \theta / (qB) \quad \cdot \quad \text{pitch} = v \cos \theta \times (2\pi m / qB)$$

The idea to remember: sin for the radius, cos for the pitch. The perpendicular part curls the

particle round; the parallel part pushes it along. Every helix question in this course is these two formulas.

Q3 · PROTON AT 60 DEGREES: FORCE, RADIUS AND PITCH

$v = 3.5 \times 10^6 \text{ m/s}$, $B = 0.25 \text{ T}$, $\theta = 60^\circ$, $m = 1.67 \times 10^{-27} \text{ kg}$, $q = 1.6 \times 10^{-19} \text{ C}$.

- 1 (a) **Force.** $F = qvB \sin \theta$, with $\sin 60^\circ = 0.866$.

One of the four values from Foundations F.4.

- 2 $qvB = (1.6 \times 10^{-19})(3.5 \times 10^6)(0.25)$. First two: 5.6×10^{-13} . Times 0.25: 1.4×10^{-13} .

- 3 $F = (1.4 \times 10^{-13})(0.866) = \mathbf{1.21 \times 10^{-13} \text{ N}}$.

- 4 (b) **Radius**, using only the perpendicular component: $v_{\text{perp}} = v \sin 60^\circ = (3.5 \times 10^6)(0.866) = 3.031 \times 10^6 \text{ m/s}$.

Sin for the radius. This is the component that curls.

- 5 $r = m v_{\text{perp}} / (qB) = (1.67 \times 10^{-27})(3.031 \times 10^6) / [(1.6 \times 10^{-19})(0.25)]$

Numerator: 5.062×10^{-21} . Denominator: 4.0×10^{-20} .

- 6 $r = \mathbf{0.127 \text{ m}}$ (about 12.7 cm).

- 7 (c) **Pitch.** First the period: $T = 2\pi m / (qB) = 2\pi(1.67 \times 10^{-27}) / (4.0 \times 10^{-20}) = 2.623 \times 10^{-7} \text{ s}$.

- 8 Parallel speed: $v_{\text{par}} = v \cos 60^\circ = (3.5 \times 10^6)(0.5) = 1.75 \times 10^6 \text{ m/s}$.

Cos for the pitch, and $\cos 60^\circ = 0.5$ exactly.

- 9 Pitch = $v_{\text{par}} \times T = (1.75 \times 10^6)(2.623 \times 10^{-7}) = \mathbf{0.459 \text{ m}}$.

(a) $F \approx 1.21 \times 10^{-13} \text{ N}$ · (b) $r \approx 0.127 \text{ m}$ · (c) pitch $\approx 0.459 \text{ m}$

Now redo it: the single most common mistake here is using the full speed v in the radius formula instead of $v \sin \theta$. That would give $r = 0.146 \text{ m}$, which is wrong. Only the perpendicular component makes the circle.

Q4 · CLASS DRILL: FIND THE INTERNAL RESISTANCE

Q4. A wire of 4Ω is connected to a battery of emf 6 V and internal resistance r . The current flowing is 1 A . Calculate r .

- 1 Use $E = I(R + r)$, the emf equation from Unit 4.4.

- 2 Substitute: $6 = 1 \times (4 + r)$.

$E = 6 \text{ V}$, $I = 1 \text{ A}$, $R = 4 \Omega$.

- 3 So $4 + r = 6$, giving $r = 6 - 4 = \mathbf{2 \Omega}$.

$r = 2 \Omega$

Check: the terminal voltage is $IR = (1)(4) = 4 \text{ V}$, and the voltage lost inside the cell is $Ir = (1)(2) = 2 \text{ V}$. Together, $4 + 2 = 6 \text{ V}$, the emf. It balances.

SOLUTIONS S.5 · CBT TEST 1

Test 1: all fifteen questions

Modules 1 and 2. Every answer below is my own independent computation, not a copy of the marking key. Each solution names the section that teaches the method. **Recorded score: 14 out of 15.** The single question marked wrong (Q9) is missing its data in the source material and could not have been answered as printed. That is set out in full at Q9. Two questions in this paper (Q1 and Q9) are

flagged as unanswerable. I have not guessed either of them; instead I teach the method so that whatever version you meet, you can do it.

⚠ **MISSING INFORMATION · Q1 CANNOT BE ANSWERED AS PRINTED**

Q1. Find the work done on a charge of $+6.0 \mu\text{C}$ placed in vacuum as it moves through a distance of 50 cm.

a. $2 \times 10^{-6} \text{ J}$ · b. $1.08 \times 10^{-6} \text{ J}$ · c. $0.5 \times 10^{-6} \text{ J}$ · d. $0.65 \times 10^{-6} \text{ J}$ · e. $3 \times 10^{-6} \text{ J}$

To find work done you need $W = qEd$ (charge times field times distance) or $W = qV$ (charge times potential difference). You are given the charge and the distance. You are **not** given the electric field or the potential difference. Without one of those, the question has no unique answer. I tested every offered option to see whether any implies a natural field value:

IF THE ANSWER WERE	THE REQUIRED E WOULD BE	THE REQUIRED V WOULD BE
$2 \times 10^{-6} \text{ J}$	0.667 V/m	0.333 V
$1.08 \times 10^{-6} \text{ J}$	0.36 V/m	0.18 V
$0.5 \times 10^{-6} \text{ J}$	0.167 V/m	0.083 V
$0.65 \times 10^{-6} \text{ J}$	0.217 V/m	0.108 V
$3 \times 10^{-6} \text{ J}$	1.0 V/m	0.5 V

Option (e) is the only one that lands on round numbers (exactly 1.0 V/m and 0.5 V), but nothing in the question points to it rather than to any other option, so I cannot reverse-engineer the intended field, and I will not fabricate one. **What to do in the exam:** know the method cold. Substitute whatever E or V your version of the question actually gives you, and it is a one-line answer.

IF THE ANSWER WERE	THE REQUIRED E WOULD BE	THE REQUIRED V WOULD BE
$2 \times 10^{-6} \text{ J}$	0.667 V/m	0.333 V
$1.08 \times 10^{-6} \text{ J}$	0.36 V/m	0.18 V
$0.5 \times 10^{-6} \text{ J}$	0.167 V/m	0.083 V
$0.65 \times 10^{-6} \text{ J}$	0.217 V/m	0.108 V
$3 \times 10^{-6} \text{ J}$	1.0 V/m	0.5 V

● **MUST-MEMORISE · WORK DONE ON A CHARGE (THE METHOD FOR Q1)**

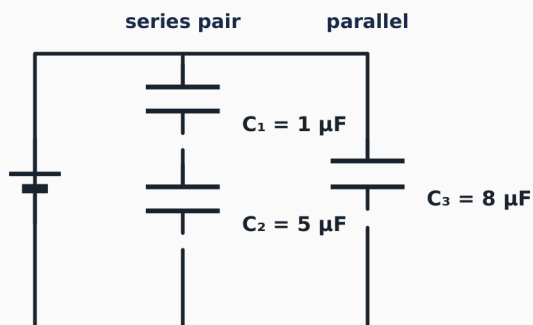
$$W = qEd = qV$$

W = work (J) · **q** = charge (C) · **E** = uniform field (V/m) · **d** = distance moved along the field (m)
· **V** = potential difference (V)

MEMORY HOOK: work equals charge times volts. And since $V = Ed$ in a uniform field, the two forms are the same statement. Convert 50 cm to 0.50 m before substituting. Method: §1.3

Determine the total capacitance ($C_1 = 1\ \mu\text{F}$ and $C_2 = 5\ \mu\text{F}$ in one branch, $C_3 = 8\ \mu\text{F}$ in the other).

a. $5.1\ \mu\text{F}$ · b. $3.43\ \mu\text{F}$ · c. $0.8\ \mu\text{F}$ · **d. $8.8\ \mu\text{F}$** · e. $14.0\ \mu\text{F}$



C_1 and C_2 are stacked in the same branch, so they are in series. That branch is in parallel with C_3 .

- 1 Read the topology. C_1 and C_2 are stacked in the same branch, so they are in **series**. That branch sits alongside C_3 , so it is in **parallel** with it.
Same branch means series. Side by side means parallel. Getting this right is the whole question.
- 2 Series pair, product over sum: $C_{12} = (1 \times 5)/(1 + 5) = 5/6 = 0.833\ \mu\text{F}$.
Check: smaller than the smallest ($1\ \mu\text{F}$). Correct for a series pair.
- 3 Parallel with C_3 , so simply add: $C_{\text{total}} = 0.833 + 8 = 8.83\ \mu\text{F}$.
Capacitors in parallel add. This matches option (d).

$C_{\text{total}} \approx 8.8\ \mu\text{F}$ (option d) Method: §2.2

Now redo it: option (b), $3.43\ \mu\text{F}$, is what you get if you wrongly treat C_1 and C_2 as *parallel* and then put that in series with C_3 . The distractor is deliberate. Read the topology before you compute.

Q3 · OIL DROP HELD STATIONARY

ANSWER: A

Calculate the electric field in which a charged oil drop of mass 5 mg and charge +9 μC will be held stationary.

a. 5.3 V/m · b. 4.35 V/m · c. 8.89 V/m · d. 7.26 V/m · e. 6.62 V/m

- 1 Held stationary** means the forces balance: the upward electric force exactly cancels the downward weight.

This is the whole physics of the question. Write the force balance first and the algebra follows.

- 2** $qE = mg$, so $E = mg/q$.

Rearrange for the unknown before substituting.

- 3** Convert: $m = 5 \text{ mg} = 5 \times 10^{-6} \text{ kg}$, and $q = 9 \mu\text{C} = 9 \times 10^{-6} \text{ C}$.

5 mg is $5 \times 10^{-6} \text{ kg}$, not 5×10^{-3} . Milligrams to kilograms is a factor of 10^{-6} . This is the trap.

- 4** $E = (5 \times 10^{-6})(9.8) / (9 \times 10^{-6})$.

The 10^{-6} cancels top and bottom entirely.

- 5** $E = (5)(9.8)/9 = 49/9 = 5.44 \text{ V/m}$.

$E \approx 5.44 \text{ V/m}$, directed upward. Nearest option is 5.3 V/m (option a). Method: §1.2

Discrepancy, stated openly: my independent computation gives **5.44 V/m**, and the nearest offered option is 5.3. The slide version of this question uses $q = 10 \mu\text{C}$ and $g = 10 \text{ m/s}^2$, which gives 5 V/m, so the numbers

were probably adjusted when the test was written. Option (a) is the intended answer. The direction must be **upward** to oppose gravity on a positive charge.

Q4 · POTENTIAL AT TWO DISTANCES

ANSWER: C

What is the absolute potential at $r = 5 \text{ cm}$ and $r = 20 \text{ cm}$ from a charge of +4 μC in air?

a. 750 kV and 180 kV · b. 150 kV and 75 kV · **c. 720 kV and 180 kV** · d. 250 kV and 50 kV · e. 500 kV and 150 kV

- 1** Potential, so $V = kQ/r$, with **one** r .

Not r squared. That would be the field. This is the classic trap and it is worth saying to yourself every single time.

- 2** At $r = 5 \text{ cm} = 0.05 \text{ m}$: $V = (9 \times 10^9)(4 \times 10^{-6}) / 0.05$.

Numerator = 3.6×10^4 .

- 3** $V = (3.6 \times 10^4) / 0.05 = 7.2 \times 10^5 \text{ V} = \mathbf{720 \text{ kV}}$.

Dividing by 0.05 is multiplying by 20.

- 4** At $r = 20 \text{ cm} = 0.20 \text{ m}$: $V = (3.6 \times 10^4) / 0.20 = 1.8 \times 10^5 \text{ V} = \mathbf{180 \text{ kV}}$.

Four times further away gives four times less potential, because V goes as $1/r$ and not $1/r^2$.

720 kV and 180 kV (option c) Method: §1.3

A note on k: this option matches *exactly* only if you use $k = 9 \times 10^9$. With the more precise $k = 8.99 \times 10^9$ you get 719.2 kV and 179.8 kV, which still points to option (c). In a multiple-choice paper, use $k = 9 \times 10^9$ for speed and pick the nearest.

Q5 · ELECTRIC FLUX

ANSWER: C

A uniform field $E = 200 \text{ N/C}$ passes through a surface of area 2 m^2 at 0° . Find the flux.

a. 450 · b. 300 · c. **400** · d. 120 · e. 100 (all $\text{N m}^2/\text{C}$)

1 $\Phi = EA \cos \theta$.

The flux formula from §1.4.

2 $\theta = 0^\circ$, so $\cos 0^\circ = 1$.

The field passes straight through the surface, giving maximum flux.

3 $\Phi = (200)(2)(1) = 400$.

$\Phi = 400 \text{ N m}^2/\text{C}$ (option c) Method: §1.4

Q6 · PARALLEL PLATE WITH DIELECTRIC

ANSWER: B

Plate area $4 \times 10^{-2} \text{ m}^2$, dielectric thickness 10^{-3} m , relative permittivity 7. Find the capacitance.

a. $0.2478 \times 10^{-9} \text{ F}$ · b. **$2.478 \times 10^{-9} \text{ F}$** · c. 4.77 F · d. $2.478 \times 10^{-9} \text{ F}$ · e. $0.2478 \times 10^{-9} \text{ F}$

1 A dielectric is present, so $C = \epsilon_r \epsilon_0 A / d$.

ϵ_r multiplies the vacuum value.

2 $C = (7)(8.85 \times 10^{-12})(4 \times 10^{-2}) / (10^{-3})$.

3 Numerator: $7 \times 8.85 \times 10^{-12} = 6.195 \times 10^{-11}$, then times 4×10^{-2} gives 2.478×10^{-12} .

Exponents: $-11 + (-2) = -13$, so $24.78 \times 10^{-13} = 2.478 \times 10^{-12}$.

4 Divide by 10^{-3} , which multiplies by 1000: $C = 2.478 \times 10^{-9} \text{ F}$.

The distractors are 0.2478×10^{-9} , exactly a factor of ten out. That is what you get if you fumble this last step.

$C = 2.478 \times 10^{-9} \text{ F} = 2.478 \text{ nF}$ (option b) Method: §2.2

Note: options (b) and (d) are printed identically in the source. Either is the intended correct answer.

Q7 · FORCE ON A CHARGE IN A FIELD

ANSWER: C

A charge of $+1.5 \times 10^{-9} \text{ C}$ is placed in a uniform field of 2500 N/C . Find the force.

a. $4.0 \times 10^{-6} \text{ N}$ · b. $1.0 \times 10^{-6} \text{ N}$ · c. **$3.75 \times 10^{-6} \text{ N}$** · d. $2.0 \times 10^{-6} \text{ N}$ · e. $3.0 \times 10^{-6} \text{ N}$

1 $F = qE$, the definition of field strength rearranged.

The most-used equation in the whole course.

2 $F = (1.5 \times 10^{-9})(2500)$.

No conversion needed; both are already SI.

3 $1.5 \times 2500 = 3750$, so $F = 3750 \times 10^{-9} = 3.75 \times 10^{-6} \text{ N}$.

$F = 3.75 \times 10^{-6} \text{ N}$ (option c) Method: §1.2

Q8 · FIELD AT THE MIDPOINT BETWEEN OPPOSITE CHARGES

ANSWER: C

$Q_1 = +3.0 \times 10^{-6} \text{ C}$ and $Q_2 = -3.0 \times 10^{-6} \text{ C}$ are 0.10 m apart. Find the net field at the midpoint.

a. 1.99×10^7 · b. 1.08×10^7 · c. **2.16×10^7** · d. 3.0×10^7 · e. 1.07×10^7 (all N/C)

1 The midpoint is 0.05 m from each charge.

Half of 0.10 m.

2 **The key insight.** The field from the positive charge points *away* from it, that is, toward the negative charge. The field from the negative charge points *toward* it, which is the *same direction*. So the two fields **add**.

Sketch the two arrows. Three seconds of drawing prevents this error forever.

3 Each contributes $E = kQ/r^2 = (8.99 \times 10^9)(3 \times 10^{-6}) / (0.05)^2$.
 $(0.05)^2 = 2.5 \times 10^{-3}$.

4 $E = (2.697 \times 10^4) / (2.5 \times 10^{-3}) = 1.08 \times 10^7 \text{ N/C}$ from each.

Note this is exactly option (b), which is the distractor placed there for people who forget to add the two contributions.

5 Add them: $E_{\text{net}} = 1.08 \times 10^7 + 1.08 \times 10^7 = 2.16 \times 10^7 \text{ N/C}$. **$E_{\text{net}} = 2.16 \times 10^7 \text{ N/C}$, directed from the positive toward the negative charge (option c) Method: §1.2**

⚠ TRAP · OPPOSITE CHARGES ADD AT THE MIDPOINT; LIKE CHARGES CANCEL

This catches people constantly, and option (b) was placed there deliberately to catch you.

- ▶ Two **opposite** charges (+ and -): at the midpoint both fields point the same way, from + toward -, so they **ADD**. The answer is twice one field.
- ▶ Two **like** charges (both +): at the midpoint the fields point in opposite directions and **CANCEL** exactly. The answer is **zero**, a null point.

Always sketch the two arrows before adding.

⚠ MISSING INFORMATION · Q9, THE QUESTION YOU WERE MARKED WRONG ON

Q9. Calculate the magnitude of the electric field intensity represented by [*the vector is missing from the question*].

a. 10.2 V/m · b. 12 V/m · c. 10 V/m · d. 9 V/m · e. 8 V/m

This is the one question you were marked wrong on, and it was not your fault. The question text ends at "represented by" and then simply stops. The vector expression, which would look like $E = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, does not appear in the material at all. I examined the source at high magnification to be certain: there is genuinely nothing after those words. I attempted to reconstruct it from the five options by testing every integer vector whose magnitude matches one of them. Dozens fit. For example (2, 6, 8) gives 10.198, which rounds to 10.2, and (4, 8, 8) gives exactly 12. There is **no unique answer**, so I will not fabricate one. What I can give you is the method, which is all you actually need.

● MUST-MEMORISE · MAGNITUDE OF A FIELD VECTOR (THE METHOD FOR Q9)

$$|E| = \sqrt{(E_x^2 + E_y^2 + E_z^2)}$$

Square every component, add them all, then take one square root at the end.

MEMORY HOOK: SQUARE, ADD, ROOT. Three facts that make this free marks: a **minus** component contributes exactly as much as a plus one, because squaring kills the sign; a **missing** component is **zero**, not one, so if you see only $3\mathbf{i} + 4\mathbf{j}$ then $E_z = 0$ and $|E| = 5$; and learn to spot Pythagorean triples on sight, since examiners love them: (3,4,5), (6,8,10), (5,12,13), (8,15,17).

Method: §F.3, §1.2

Q9 · THE TWO VERSIONS OF THIS QUESTION THAT DO APPEAR IN THE SLIDES

The task. Since Q9's own vector is missing, practise the method on the two versions that *do* appear in the slides, both shown below.

- 1 **Version 1.** $E = 3\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$. Find $|E|$.

The identical question type, with the vector actually supplied.

- 2 $|E| = \sqrt{(3^2 + 4^2 + (-2)^2)} = \sqrt{(9 + 16 + 4)} = \sqrt{29}$.

The $(-2)^2$ becomes $+4$. The minus sign vanishes.

- 3 $\sqrt{29} = 5.385$, so $|E| \approx 5.4$ V/m.

- 4 **Version 2.** $E = 3\mathbf{i} + 4\mathbf{j}$. What is the z-component?

A different question about the same kind of vector.

- 5 There is no $\mathbf{k}$ term written, so $E_z = 0$. **Practise both forms. Whatever vector Q9 actually contained, $|E| = \sqrt{(E_x^2 + E_y^2 + E_z^2)}$ solves it in one line. Method: §F.3, §1.2**

Do not invent a component that is not there.

If you can find the original paper with the vector shown, it can be computed exactly in seconds.

Until then, the method above is the whole answer.

Q10 · CHARGE OF 6.2×10^{18} ELECTRONS

ANSWER: C

a. 1.99 C · b. 0.89 C · c. **0.99 C** · d. 0.79 C · e. 1.07 C

- 1 Quantisation of charge: $Q = ne$.

n is the number of charges, e is the elementary charge.

- 2 $Q = (6.2 \times 10^{18})(1.6 \times 10^{-19})$.

- 3 Fronts: $6.2 \times 1.6 = 9.92$. Exponents: $18 + (-19) = -1$. So $Q = 9.92 \times 10^{-1}$.

Q = 0.99 C (option c) Method: §1.1

Recognise this number: 6.2×10^{18} electrons is exactly the figure from the Module 3 slide on current, where 0.22 A flowing for 4.5 s moves 0.99 C. The same numbers reappear across the course. Spotting that saves you time.

Q11 · CAPACITANCE OF A SPHERE

ANSWER: A

A metal sphere carries 4.5 nC when its potential is 180 V above its surroundings. Find the capacitance.

a. 25 pF · b. 12 pF · c. 8.1 pF · d. 30 pF · e. 6.3 pF

1 $C = Q/V$, the definition of capacitance.

All the wording about "the sphere and its surroundings" is flavour. You do not need the radius, and no geometry formula is required.

2 $C = (4.5 \times 10^{-9}) / 180$.

3 $C = 2.5 \times 10^{-11}$ F. Convert: 1 pF = 10^{-12} F.

C = 25 pF (option a) Method: §2.2

Q12 · CURRENT THROUGH A SERIES RESISTOR

ANSWER: A

$R_1 = 7 \Omega$, $R_2 = 6 \Omega$, $R_3 = 3 \Omega$ in series with a 48 V source. Find the current through R_3 .

a. 3 A · b. 6 A · c. 4 A · d. 12 A · e. 5 A

1 Series means the **same current flows through every resistor**. So "the current through R_3 " is just the circuit current.

No divider rule is needed. Naming R_3 specifically is the trap: it tempts you into an unnecessary calculation.

2 $R_{\text{total}} = 7 + 6 + 3 = 16 \Omega$.

3 $I = V/R = 48/16 = 3$ A.

I = 3 A (option a) Method: §3.2

Q13 · FIELD PRODUCED BY A SOURCE CHARGE

ANSWER: B

A charge of 8×10^{-8} C is 0.40 m from a charge of -2.0×10^{-8} C. Find the field produced by the negative charge at the location of the positive charge.

a. 1114.07 · b. **1125.75** · c. 1523.23 · d. 3124.76 · e. 2314.56 (all N/C)

1 Read the question carefully. It asks for the field produced by the **negative charge**, so $Q = 2.0 \times 10^{-8}$ C is the source.

*The 8×10^{-8} C charge is a **distractor**. The field of a source charge does not depend on what you place in it. This is the whole skill being tested.*

2 $E = kQ/r^2 = (8.99 \times 10^9)(2.0 \times 10^{-8}) / (0.40)^2$.

3 Numerator = 179.8. Denominator = $(0.40)^2 = 0.16$.

4 $E = 179.8 / 0.16 = 1123.75$ N/C.

With $k = 9 \times 10^9$ you get exactly 1125 N/C, matching option (b).

E ≈ 1125 N/C, pointing toward the negative charge (option b) Method: §1.2

The lesson: the field of a point charge depends *only* on that charge and the distance. Whatever sits at the other end is irrelevant. Recognising the distractor is the skill here, not the arithmetic.

Q14 · FIELD OUTSIDE A CONDUCTING SPHERE
ANSWER: B

A conducting sphere of radius 0.25 m carries 6×10^{-6} C. Find the field 0.50 m from the centre.

a. 2.92×10^5 · **b. 2.16×10^5** · c. 3.53×10^5 · d. 5×10^5 · e. 1.47×10^5 (all N/C)

- 1** Check where the point is. $r = 0.50$ m and the sphere's radius is 0.25 m, so the point is **outside** the sphere.

Always establish inside or outside first. It changes the method completely.

- 2** Outside a spherically symmetric charge, the field is identical to that of a point charge at the centre: $E = kQ/r^2$.

This is the Gauss's law result from §1.4.

- 3** $E = (8.99 \times 10^9)(6 \times 10^{-6}) / (0.50)^2$.

*Use the full distance from the **centre** (0.50 m), not the distance from the surface.*

- 4** Numerator = 5.394×10^4 . Divide by $(0.50)^2 = 0.25$.

$$E = 2.16 \times 10^5 \text{ N/C} \quad (\text{option b}) \quad \text{Method: §1.4}$$

Know both cases: if the question had asked for the field *inside* the conductor, say at 0.10 m, the answer would be exactly **zero**. Same sphere, completely different answer.

Q15 · CAPACITANCE FROM STORED ENERGY
ANSWER: B

Stored energy 2.25 mJ, charged from a 30 V source. Find C.

a. 10 μF · **b. 5 μF** · c. 25 μF · d. 13 μF · e. 2.5 μF

- 1** Energy and voltage are given, so use $W = \frac{1}{2}CV^2$.

Choose the form of the energy formula that matches what you have.

- 2** Rearrange: $C = 2W / V^2$.

- 3** $C = 2(2.25 \times 10^{-3}) / (30)^2 = (4.5 \times 10^{-3}) / 900$.

$2.25 \text{ mJ} = 2.25 \times 10^{-3} \text{ J}$, and $30^2 = 900$.

- 4** $C = 5 \times 10^{-6} \text{ F}$.

$$C = 5 \mu\text{F} \quad (\text{option b}) \quad \text{Method: §2.2}$$

The trap: forgetting the $\frac{1}{2}$ gives 2.5 μF , which is sitting there as option (e), waiting for you. **Never drop the half.**

Test 1: answer key at a glance

Q	TOPIC	ANSWER	SECTION
1	Work done on a charge	Missing data. Method: $W = qEd = qV$	§1.3
2	Capacitor network	8.8 μF (d)	§2.2
3	Oil drop held stationary	5.44, nearest 5.3 V/m (a)	§1.2
4	Potential at two distances	720 kV and 180 kV (c)	§1.3
5	Electric flux	400 $\text{N m}^2/\text{C}$ (c)	§1.4
6	Parallel plate with dielectric	$2.478 \times 10^{-9} \text{ F}$ (b)	§2.2
7	Force on a charge	$3.75 \times 10^{-6} \text{ N}$ (c)	§1.2

Q	TOPIC	ANSWER	SECTION
8	Field at midpoint, opposite charges	$2.16 \times 10^7 \text{ N/C (c)}$	§1.2
9	Magnitude of a vector	Missing vector. Method: $\sqrt{E_x^2 + E_y^2 + E_z^2}$	§F.3
10	Charge of N electrons	0.99 C (c)	§1.1
11	Capacitance of a sphere	25 pF (a)	§2.2
12	Series current	3 A (a)	§3.2
13	Field from a source charge	1125 N/C (b)	§1.2
14	Field outside a conducting sphere	$2.16 \times 10^5 \text{ N/C (b)}$	§1.4
15	C from stored energy	5 μF (b)	§2.2

SPACED-REPETITION CHECKPOINT

Close the book and say these out loud from memory:

- ▶ Q8: at the midpoint between *opposite* charges, do the fields add or cancel? What about *like* charges?
- ▶ Q13: which charge produces the field, and which one is the distractor?
- ▶ Q15: what happens to your answer if you forget the $\frac{1}{2}$ in the energy formula?
- ▶ Q9: write the formula for the magnitude of a vector from memory.

ANSWER

Opposite charges **ADD** at the midpoint (both fields point from + to -); like charges **CANCEL** to zero. In Q13 the *negative* charge is the source and the 8 nC is the distractor. Forgetting the half in Q15 gives 2.5 μF , which is option (e), a planted trap. $|\mathbf{E}| = \sqrt{E_x^2 + E_y^2 + E_z^2}$. **Revisit this page:** 1 day, 3 days, 1 week, then before the exam.

SOLUTIONS S.6 · CBT TEST 2

Test 2: all fifteen questions

Modules 3 and 4. Every answer independently recomputed. Note that Q1 and Q7 examine topics that appear in no slide deck: they are taught in the new sections 4.5 and 4.6, which exist because of this test. **Two questions on this paper cannot be answered from the slides alone.** Q1 needs the energy stored in an inductor ($W = \frac{1}{2}LI^2$) and Q7 needs the torque on a current loop ($\tau = mB \sin \theta$). Neither formula appears anywhere in the four slide decks. They are now taught in §4.5 and §4.6. This is precisely why the assessments must be blended with the slides rather than trusting the slides alone.

Q1 · ENERGY STORED IN AN INDUCTOR

ANSWER: A

A 2 H inductor carries 5 A of current. What is the stored energy?

a. 25 J · b. 15 J · c. 20 J · d. 10 J · e. 30 J

1 Energy stored in an inductor: $W = \frac{1}{2}LI^2$.

The magnetic twin of the capacitor's $\frac{1}{2}CV^2$. Inductor goes with current; capacitor goes with voltage.

2 Square the **current**, not the inductance: $W = \frac{1}{2}(2)(5)^2 = \frac{1}{2}(2)(25)$.

3 $W = 25$ J.

W = 25 J (option a) Method: §4.5

Note: option (d), 10 J, is what you get if you square the *inductance* instead of the current. The formula is L times I squared, not the other way round.

Q2 · SPEED OF A PROTON FROM THE MAGNETIC FORCE

ANSWER: A

A proton experiences a magnetic force of 3.2×10^{-14} N moving through a field of 0.2 T perpendicular to it. Find its speed.

a. 1.0×10^6 m/s · b. 2.0×10^6 m/s · c. 0.5×10^6 m/s · d. 4.0×10^6 m/s · e. 3.0×10^6 m/s

1 Perpendicular, so $\sin 90^\circ = 1$ and $F = qvB$.

The maximum force case.

2 Rearrange for v: $v = F / (qB)$.

3 Denominator: $qB = (1.6 \times 10^{-19})(0.2) = 3.2 \times 10^{-20}$.

4 $v = (3.2 \times 10^{-14}) / (3.2 \times 10^{-20}) = 1.0 \times 10^6$ m/s.

The 3.2 cancels exactly. Exponents: $-14 - (-20) = 6$.

v = 1.0×10^6 m/s (option a) Method: §4.1

Q3 · EMF FROM INTERNAL RESISTANCE

ANSWER: A

External resistor 5 Ω , internal resistance 0.5 Ω , current 1 A. Find the emf.

a. 5.5 V · b. 5.0 V · c. 6.0 V · d. 4.5 V · e. 5.25 V

1 $E = I(R + r)$.

The current passes through the external resistor and the cell's own internal resistance, so they add in series.

2 $E = 1 \times (5 + 0.5) = 5.5$ V.

E = 5.5 V (option a) Method: §4.4

Recognise it: this is the identical question to Worked Example 1 in §4.4, taken straight from the slides. The tests reuse slide examples verbatim, which is exactly why reproducing every worked example matters.

Q4 · EMF FROM TERMINAL P.D.

ANSWER: A

A cell of unknown emf and internal resistance $2\ \Omega$ is connected to a $5\ \Omega$ resistor. The terminal p.d. is $1\ \text{V}$. Find E .

a. $1.4\ \text{V}$ · b. $1.2\ \text{V}$ · c. $1.0\ \text{V}$ · d. $1.6\ \text{V}$ · e. $2.0\ \text{V}$

- 1 The terminal p.d. is the voltage across the *external* resistor, so $I = V/R = 1/5 = 0.2\ \text{A}$.
Apply Ohm's law to the external resistor alone.
- 2 Now $E = I(R + r) = 0.2(5 + 2) = 0.2 \times 7$.
- 3 $E = 1.4\ \text{V}$.

$E = 1.4\ \text{V}$ (option a) Method: §4.4

Also from the slides, Worked Example 2 in §4.4. The alternative route is $E = 1 + Ir = 1 + (0.2)(2) = 1.4\ \text{V}$. Both give the same answer.

Q5 · AMPERE'S LAW IS MOST USEFUL FOR CONDUCTORS WITH HIGH _____ FILL IN THE BLANK

Q5. Ampere's law is most useful for conductors with high _____. (*Fill in the blank.*)

- 1 Ampere's law is $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$. The integral is only tractable if B is constant along the loop you choose.
From §4.2: the whole trick is choosing an Amperian loop on which B does not vary.
- 2 That is only possible when the current distribution has a high degree of **symmetry**: a straight wire, a solenoid, a toroid.
The slides say it directly: Ampere formulated the law to simplify calculations for systems with high symmetry.

The answer is SYMMETRY. Method: §4.2

Compare Gauss's law, which works the same way in electrostatics: choose a surface matching the symmetry and the integral collapses to a simple product. Same idea, two different laws.

Q6 · TRANSFORMER EFFICIENCY

ANSWER: A

A $240\ \text{V}$ step-down transformer lights eight $12\ \text{V}$, $24\ \text{W}$ lamps and draws $1\ \text{A}$ in the primary. Calculate its efficiency.

a. 80% · b. 60% · c. 40% · d. 20% · e. 0%

- 1 **Power out** is the total delivered to the lamps: $P_{\text{out}} = 8 \times 24 = 192\ \text{W}$.
Eight lamps at $24\ \text{W}$ each. The $12\ \text{V}$ rating is not needed here, and is there to distract you.
- 2 **Power in** is primary voltage times primary current: $P_{\text{in}} = 240 \times 1 = 240\ \text{W}$.
 $P = VI$ on the primary side.
- 3 Efficiency = $(P_{\text{out}} / P_{\text{in}}) \times 100\% = (192 / 240) \times 100\%$.
- 4 $= 0.8 \times 100\% = 80\%$.

Efficiency = 80% (option a) Method: §4.3

Q7 · MAGNETIC DIPOLE MOMENT OF A CURRENT LOOP

ANSWER: A

A rectangular current loop experiences a torque of $4.8 \times 10^{-3} \text{ N m}$ in a field of 0.6 T , at 30° between the moment and the field. Find the magnetic dipole moment.

a. $1.6 \times 10^{-2} \text{ A m}^2$ · b. 1.3×10^{-2} · c. 2.0×10^{-2} · d. 3×10^{-2} · e. 5×10^{-2}

- 1 Torque on a magnetic dipole: $\tau = mB \sin \theta$, exactly the same shape as the electric dipole result $\tau = pE \sin \theta$ from §2.1.

*Torque uses **sine**. Spotting the structural analogy makes this instant.*

- 2 Rearrange for m: $m = \tau / (B \sin \theta)$.

- 3 $\sin 30^\circ = 0.5$ exactly.

No calculator needed. One of the four values from Foundations F.4.

- 4 $m = (4.8 \times 10^{-3}) / [(0.6)(0.5)] = (4.8 \times 10^{-3}) / 0.3 = 1.6 \times 10^{-2}$.

$$m = 1.6 \times 10^{-2} \text{ A m}^2 \quad (\text{option a}) \quad \text{Method: §2.1, §4.6}$$

This formula is in no slide deck. It is taught in the new §4.6, which exists because this question exists. If you revised from the slides only, you had no way to answer it.

Q8 · RADIUS OF AN ELECTRON'S CIRCULAR PATH

ANSWER: C

An electron ($m = 9.1 \times 10^{-31} \text{ kg}$, $q = 1.6 \times 10^{-19} \text{ C}$) enters a field of 0.03 T perpendicularly at $4 \times 10^6 \text{ m/s}$. Find the radius.

a. $7.58 \times 10^{-2} \text{ m}$ · b. $7.58 \times 10^{-3} \text{ m}$ · c. $7.58 \times 10^{-4} \text{ m}$ · d. $7.58 \times 10^{-5} \text{ m}$ · e. $7.58 \times 10^{-1} \text{ m}$

- 1 Perpendicular entry means circular motion, so $r = mv/(qB)$.

The magnetic force supplies the centripetal force.

- 2 Numerator: $mv = (9.1 \times 10^{-31})(4 \times 10^6) = 3.64 \times 10^{-24}$.

- 3 Denominator: $qB = (1.6 \times 10^{-19})(0.03) = 4.8 \times 10^{-21}$.

$1.6 \times 0.03 = 0.048$, so $0.048 \times 10^{-19} = 4.8 \times 10^{-21}$.

- 4 $r = (3.64 \times 10^{-24}) / (4.8 \times 10^{-21}) = 7.58 \times 10^{-4} \text{ m}$.

$$r = 7.58 \times 10^{-4} \text{ m} \quad (\text{option c}) \quad \text{Method: §4.1}$$

Notice the option design: every option has the same digits, 7.58, and differs only in the *power of ten*. This question is testing your exponent arithmetic and nothing else. Get the powers right and the mark is yours.

Q9 · CURRENT IN A SOLENOID

ANSWER: A

A solenoid of length 0.4 m has 800 turns and produces a field of 2.5×10^{-4} T inside. Find the current.

a. 0.1 A · b. 2.6 A · c. 5.9 A · d. 0.6 A · e. 3.1 A

- 1 **First convert N to n.** The formula $B = \mu_0 n I$ needs turns *per metre*, and you have been given the *total* number of turns.

This is the whole question. Skip this step and you are out by a factor of 2.5.

- 2 $n = N/L = 800 / 0.4 = \mathbf{2000 \text{ turns per metre.}}$

- 3 Rearrange: $I = B / (\mu_0 n)$.

- 4 Denominator: $(4\pi \times 10^{-7})(2000) = 2.513 \times 10^{-3}$.

$4\pi = 12.566$, so 12.566×10^{-7} times 2000 gives 2.513×10^{-3} .

- 5 $I = (2.5 \times 10^{-4}) / (2.513 \times 10^{-3}) = 0.0995$ A, which rounds to 0.1 A.

I ≈ 0.1 A (option a) Method: §4.2

Self-diagnosis: if you had used $n = 800$ instead of 2000, you would have got 0.249 A, which is *not offered as an option*. That absence is your clue that you skipped the conversion. When your answer is not on the list, look for a missed conversion first.

Q10 · TERMINAL VOLTAGE

ANSWER: A

A cell of emf 9 V and internal resistance 1 Ω carries 2 A. Find the terminal voltage.

a. 7 V · b. 5 V · c. 9 V · d. 11 V · e. 4 V

- 1 $V = E - Ir$. The terminal voltage is the emf minus the lost volts.

The lost volts are dissipated inside the battery and never reach the circuit.

- 2 $V = 9 - (2)(1) = 9 - 2 = 7$ V.

V = 7 V (option a) Method: §4.4

The trap: option (d), 11 V, is there for anyone who *adds* instead of subtracting. The terminal voltage is **always less** than the emf when current flows. If your answer exceeds the emf, you have the sign backwards.

Q11 · A CHANGING MAGNETIC _____ INDUCES AN EMF

FILL IN THE BLANK

Q11. A changing magnetic _____ induces an emf. (*Fill in the blank.*)

- 1 Faraday's law is $\varepsilon = -N d\Phi/dt$, where Φ is the magnetic **flux**.

It is the flux whose rate of change produces the emf.

- 2 Be precise: it is not enough to say the *field* changes. The flux $\Phi = BA \cos \theta$ can change through the field B, the area A, *or* the angle θ .

*That is why a rotating coil in a **constant** field still generates an emf, which is exactly how a generator works. Saying "field" would miss this.*

The answer is FLUX. Method: §4.3

Q12 · BIOT-SAVART DECREASES WITH THE SQUARE OF THE

FILL IN THE BLANK

Q12. According to the Biot-Savart law, the magnetic field around a current element decreases with the square of the _____. (Fill in the blank.)

- 1 The Biot-Savart law contains $1/r^2$, where r is the **distance** from the current element to the field point.

It is an inverse-square law, exactly like Coulomb's law.

The answer is **DISTANCE**. Method: §4.2

Q13 · CURRENT FROM THE TERMINAL P.D.

ANSWER: A

A $10\ \Omega$ resistor is connected to a battery of internal resistance $5\ \Omega$. The p.d. across the terminals is 2 V . Estimate the current through the resistor after 2 seconds.

a. **0.2 A** · b. 0.1 A · c. 0.3 A · d. 0.4 A · e. 0.5 A

- 1 The terminal p.d. is the voltage across the **external** resistor, so apply Ohm's law to that resistor directly.

The terminal p.d. is what the external circuit actually receives.

- 2 $I = V/R = 2/10 = 0.2\text{ A}$.

Use the external resistance ($10\ \Omega$) alone, not $R + r$.

$I = 0.2\text{ A}$ (option a) Method: §4.4

⚠ TRAP · TWO RED HERRINGS IN A SINGLE QUESTION (Q13)

Red herring 1: "after a time of 2 seconds". This is a steady DC circuit. The current is the same at 2 seconds as at 2 hours. Ignore the time entirely. It is there purely to unsettle you. **Red herring 2: the $5\ \Omega$ internal resistance.** You are given the *terminal p.d.*, which is by definition the voltage already across the external resistor. The internal drop has *already been accounted for*. So you divide by R alone ($10\ \Omega$), not by $R + r$ ($15\ \Omega$). Dividing by 15 gives 0.133 A , which is not offered, and that absence is your clue. Read what the question actually gives you. Half the difficulty in this paper is knowing which numbers to *ignore*.

Q14 · CURRENT IN A LONG STRAIGHT WIRE

ANSWER: A

At 5 cm from a long straight wire the field is $4.0 \times 10^{-5}\text{ T}$. Find the current.

a. **10 A** · b. 5 A · c. 15 A · d. 20 A · e. 2.5 A

- 1 Field of a straight wire: $B = \mu_0 I / (2\pi r)$, rearranged for I : $I = B(2\pi r) / \mu_0$.

- 2 Convert: $r = 5\text{ cm} = \mathbf{0.05\text{ m}}$.

The usual centimetre trap.

- 3 Numerator: $(4.0 \times 10^{-5})(2\pi)(0.05) = (4.0 \times 10^{-5})(0.3142) = 1.257 \times 10^{-5}$.

- 4 $I = (1.257 \times 10^{-5}) / (4\pi \times 10^{-7}) = (1.257 \times 10^{-5}) / (1.257 \times 10^{-6}) = 10\text{ A}$.

The numbers cancel beautifully. Exponents: $-5 - (-6) = 1$.

$I = 10\text{ A}$ (option a) Method: §4.2

Cross-check against §4.2: the slide example works the same formula forwards, finding $B = 4.8 \times 10^{-5}\text{ T}$ for a 12 A current at 5 cm . This test question simply runs it backwards. Same formula, same distance, reversed.

Q15 · STEP-UP TRANSFORMER

ANSWER: A

$N_s = 1000$, $N_p = 100$, $E_p = 240$ V. Find E_s .

a. 2400 V · b. 1200 V · c. 24 V · d. 240 V · e. 480 V

- 1 Transformer equation: $V_s/V_p = N_s/N_p$.
- 2 Rearrange: $V_s = V_p \times (N_s/N_p) = 240 \times (1000/100)$.
- 3 $V_s = 240 \times 10 = 2400$ V.

Ten times as many turns on the secondary gives ten times the voltage.

$E_s = 2400$ V (option a) Method: §4.3

Sanity check: N_s is greater than N_p , so this is a **step-up** transformer and the output voltage *must* be larger than the input. $2400 > 240$. If your answer had come out smaller, you inverted the ratio.

Test 2: answer key at a glance

Q	TOPIC	ANSWER	SECTION
1	Energy stored in an inductor	25 J (a)	§4.5
2	Speed from magnetic force	1.0×10^6 m/s (a)	§4.1
3	emf from internal resistance	5.5 V (a)	§4.4
4	emf from terminal p.d.	1.4 V (a)	§4.4
5	Ampere's law needs high ...	SYMMETRY	§4.2
6	Transformer efficiency	80% (a)	§4.3
7	Magnetic dipole moment	1.6×10^{-2} A m ² (a)	§4.6
8	Radius of electron path	7.58×10^{-4} m (c)	§4.1
9	Solenoid current	0.1 A (a)	§4.2
10	Terminal voltage	7 V (a)	§4.4
11	Changing magnetic ... induces emf	FLUX	§4.3
12	Biot-Savart falls with square of ...	DISTANCE	§4.2
13	Current from terminal p.d.	0.2 A (a)	§4.4
14	Current in a straight wire	10 A (a)	§4.2
15	Step-up transformer	2400 V (a)	§4.3

SPACED-REPETITION CHECKPOINT

Close the book and answer from memory:

- ▶ Write the energy stored in an inductor, and its capacitor twin.
- ▶ Q9: what must you convert *before* using $B = \mu_0 nI$?
- ▶ Q13: why do you divide by R alone and not by $R + r$?
- ▶ Q7: what is the torque on a magnetic dipole, and which trig function does it use?

ANSWER

$W = \frac{1}{2}LI^2$ for an inductor, and $W = \frac{1}{2}CV^2$ for a capacitor. In Q9 you must convert total turns N to turns *per metre*, $n = N/L$. In Q13 the terminal p.d. is *already* the voltage across the external resistor, so the internal drop has been accounted for. Torque is $\tau = mB \sin \theta$, and it uses **sine**.

Revisit this page: 1 day, 3 days, 1 week, then before the exam.

S.7

Module 5 exercises, fully solved

Five problems that drill every formula in the module. Work each on blank paper before reading the solution. Every number below was recomputed independently.

Exercise 1 · Displacement current

Question. The electric field between two parallel plates of area $2.0 \times 10^{-2} \text{ m}^2$ is increasing at a steady rate of $5.0 \times 10^{12} \text{ V/(m}\cdot\text{s)}$. Find the displacement current between the plates.

- 1 Between plates $\Phi_E = EA$, so $I_d = \epsilon_0 d\Phi_E/dt = \epsilon_0 A (dE/dt)$.
- 2 Substitute: $I_d = (8.85 \times 10^{-12})(2.0 \times 10^{-2})(5.0 \times 10^{12})$.
- 3 Multiply the fronts: $8.85 \times 2.0 \times 5.0 = 88.5$. Add exponents: $-12 - 2 + 12 = -2$.

$$I_d = 88.5 \times 10^{-2} = 0.885 \text{ A}$$

Now redo it: a well-under-one-amp displacement current from a very fast-changing field. Sanity: the units cancel to $(\text{F/m})(\text{m}^2)(\text{V}/(\text{m}\cdot\text{s})) = (\text{C}/\text{V}\cdot\text{m})(\text{m}^2)(\text{V}/\text{m}\cdot\text{s}) = \text{C}/\text{s} = \text{A}$. Always let the units confirm the shape.

Exercise 2 · Wavelength from frequency

Question. A microwave oven works at 2.45 GHz. What is the wavelength of its radiation in vacuum?

- 1 Use $c = f\lambda$, rearranged to $\lambda = c/f$.
- 2 Convert: $2.45 \text{ GHz} = 2.45 \times 10^9 \text{ Hz}$. Substitute: $\lambda = (3.0 \times 10^8)/(2.45 \times 10^9)$.

$$\lambda = 0.122 \text{ m} \approx 12 \text{ cm}$$

Now redo it: $3.0/2.45 = 1.224$, and $10^8/10^9 = 10^{-1}$, giving 0.1224 m. A ~ 12 cm wave, comparable to the spacing of the metal grid in the oven door, which is what keeps the microwaves inside.

Exercise 3 · Amplitudes and energy

Question. An electromagnetic wave has an electric field amplitude $E_0 = 60 \text{ V/m}$. Find (a) the magnetic field amplitude, (b) the intensity, and (c) the energy density (using $u = \epsilon_0 E^2$).

- 1 (a) $B_0 = E_0/c = 60 / (3.0 \times 10^8) = 2.0 \times 10^{-7} \text{ T}$.
- 2 (b) $I = \frac{1}{2}\epsilon_0 c E_0^2 = \frac{1}{2}(8.85 \times 10^{-12})(3.0 \times 10^8)(60)^2$.
- 3 Compute: $(60)^2 = 3600$; $\frac{1}{2} \times 8.85 \times 3.0 = 13.275$; $13.275 \times 10^{-4} \times 3600 = 4.78$.
- 4 (c) $u = \epsilon_0 E_0^2 = (8.85 \times 10^{-12})(3600) = 3.19 \times 10^{-8} \text{ J/m}^3$.

$$B_0 = 2.0 \times 10^{-7} \text{ T} \quad I = 4.78 \text{ W/m}^2 \quad u = 3.19 \times 10^{-8} \text{ J/m}^3$$

Now redo it: all three confirmed independently. Note B_0 is ten-million times smaller than E_0 . The factor of c again.

Exercise 4 · From intensity back to the field

Question. Sunlight at the Earth's surface has an intensity of about 1000 W/m^2 . Estimate the electric field amplitude E_0 and magnetic field amplitude B_0 of sunlight.

- 1 Rearrange $I = \frac{1}{2}\epsilon_0 c E_0^2$ for $E_0 = \sqrt{2I/(\epsilon_0 c)}$.
- 2 Substitute: $E_0 = \sqrt{2(1000) / [(8.85 \times 10^{-12})(3.0 \times 10^8)]} = \sqrt{2000 / 2.655 \times 10^{-3}}$.
- 3 $= \sqrt{7.53 \times 10^5} = 868 \text{ V/m}$. Then $B_0 = E_0/c = 868/(3.0 \times 10^8) = 2.89 \times 10^{-6} \text{ T}$.

$$E_0 \approx 868 \text{ V/m} \quad B_0 \approx 2.89 \times 10^{-6} \text{ T}$$

Now redo it: $E_0 = 867.9 \text{ V/m}$ confirmed. A few hundred volts per metre and a few microtesla. Sunlight is a surprisingly strong field, spread over a whole hemisphere of sky.

Exercise 5 · Field rate in a charging capacitor

Question. A capacitor with circular plates of radius 5.0 cm is charged by a steady current of 3.0 A . At what rate is the electric field between the plates changing?

- 1 The displacement current equals the conduction current: $I_d = 3.0 \text{ A}$.
- 2 Plate area: $r = 5.0 \text{ cm} = 0.050 \text{ m}$, so $A = \pi r^2 = \pi(0.050)^2 = 7.85 \times 10^{-3} \text{ m}^2$.
- 3 From $I_d = \epsilon_0 A(dE/dt)$, rearrange: $dE/dt = I_d/(\epsilon_0 A) = 3.0 / [(8.85 \times 10^{-12})(7.85 \times 10^{-3})]$.

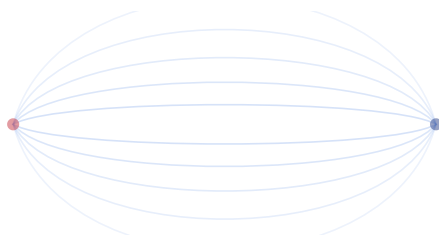
$$dE/dt = 4.32 \times 10^{13} \text{ V/(m}\cdot\text{s)}$$

Now redo it: $\epsilon_0 A = 6.95 \times 10^{-14}$, and $3.0/6.95 \times 10^{-14} = 4.32 \times 10^{13}$. Smaller plates than the 10 cm example, so a larger dE/dt is needed for the same current, since the flux is spread over less area.

PART VIII

Mock examinations

Two complete papers. M.1 is weighted to mirror your two CBT tests exactly, and several of its questions are deliberate reversals of traps you have already met. M.3 is a second, longer paper that drills the written explanations. Sit each under timed conditions with only a calculator and the formula sheet, then mark honestly against the key.



Mock exam: all four modules

INSTRUCTIONS

Time allowed: 90 minutes. Total: 60 marks.

- ▶ **Section A** (Q1 to Q15): multiple choice, 1 mark each. 15 marks.
- ▶ **Section B** (Q16 to Q20): structured questions. 45 marks.
- ▶ Answer all questions. Show your working in Section B, because method carries marks.
- ▶ **Constants:** $k = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ · $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$ · $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$ · $e = 1.6 \times 10^{-19} \text{ C}$ · $m_e = 9.1 \times 10^{-31} \text{ kg}$ · $m_p = 1.67 \times 10^{-27} \text{ kg}$ · $g = 9.8 \text{ m/s}^2$

Topic weighting mirrors your two CBT tests: Module 1 (electrostatics) 30%, Module 2 (capacitance) 20%, Module 3 (circuits) 20%, Module 4 (magnetism and induction) 30%. Several questions are deliberate mirrors of traps you have already met, with the conditions reversed.

Section A: multiple choice (15 marks)

QUESTIONS 1 TO 8 · ONE MARK EACH

- Q1.** What is the charge on a system containing 2.5×10^{19} electrons?
a. 2.0 C · b. 4.0 C · c. 0.25 C · d. 40 C · e. 1.6 C
- Q2.** Two point charges of $+2.0 \mu\text{C}$ and $+2.0 \mu\text{C}$ are 0.20 m apart. What is the electric field at the exact midpoint between them?
a. $1.8 \times 10^6 \text{ N/C}$ · b. $3.6 \times 10^6 \text{ N/C}$ · c. zero · d. $9.0 \times 10^5 \text{ N/C}$ · e. $7.2 \times 10^6 \text{ N/C}$
- Q3.** A charge Q produces a field E at distance r . If the charge is doubled and the distance is halved, the new field is
a. E · b. $2E$ · c. $4E$ · d. $8E$ · e. $E/8$
- Q4.** A closed surface encloses a charge of $-5.0 \mu\text{C}$. The total electric flux through the surface is
a. $+5.6 \times 10^5$ · b. -5.6×10^5 · c. zero · d. -4.4×10^{-17} · e. depends on the radius ($\text{N m}^2/\text{C}$)
- Q5.** The electric field intensity is $E = 6\mathbf{i} - 8\mathbf{j} \text{ V/m}$. Its magnitude is
a. 2 V/m · b. 10 V/m · c. 14 V/m · d. 100 V/m · e. -2 V/m
- Q6.** Three capacitors, $2 \mu\text{F}$, $3 \mu\text{F}$ and $6 \mu\text{F}$, are connected in series. The equivalent capacitance is
a. 11 μF · b. 1.0 μF · c. 3.7 μF · d. 0.5 μF · e. 6.0 μF
- Q7.** A capacitor stores 8.0 mJ when charged to 40 V. Its capacitance is
a. 10 μF · b. 5.0 μF · c. 20 μF · d. 200 μF · e. 0.2 μF
- Q8.** Two resistors, 6Ω and 12Ω , are in parallel. Their equivalent resistance is
a. 18Ω · b. 9Ω · c. 4Ω · d. 2Ω · e. 0.25Ω

QUESTIONS 9 TO 15 · ONE MARK EACH

Q9. A 60 V source is across 20 Ω and 40 Ω in series. The voltage across the 40 Ω resistor is

a. 20 V · b. 30 V · c. 40 V · d. 45 V · e. 60 V

Q10. A total current of 9.0 mA enters two parallel resistors of 3.0 k Ω and 6.0 k Ω . The current in the 3.0 k Ω resistor is

a. 3.0 mA · b. 4.5 mA · c. 6.0 mA · d. 9.0 mA · e. 1.5 mA

Q11. An electron moves at 3.0×10^6 m/s *parallel* to a magnetic field of 0.8 T. The magnetic force on it is

a. 3.8×10^{-13} N · b. 2.4×10^{-13} N · c. zero · d. 4.8×10^{-13} N · e. 1.6×10^{-19} N

Q12. A long straight wire carries 8.0 A. The magnetic field 0.10 m away is

a. 1.6×10^{-5} T · b. 8.0×10^{-6} T · c. 3.2×10^{-5} T · d. 1.0×10^{-5} T · e. 4.0×10^{-6} T

Q13. A coil of 500 turns has its flux change from 0.030 Wb to 0.010 Wb in 0.050 s. The magnitude of the induced emf is

a. 20 V · b. 100 V · c. 200 V · d. 400 V · e. 10 V

Q14. A cell of emf 12 V and internal resistance 2.0 Ω delivers a current of 1.5 A. The terminal voltage is

a. 15 V · b. 12 V · c. 9.0 V · d. 3.0 V · e. 10.5 V

Q15. Lenz's law is a direct consequence of the conservation of

a. charge · b. momentum · c. energy · d. magnetic flux · e. mass

Section B: structured questions (45 marks)**QUESTION 16 · COULOMB'S LAW AND TWO-DIMENSIONAL SUPERPOSITION [9 MARKS]**

(a) State Coulomb's law and define every symbol. [3] **(b)** Charge P = $-3.0 \mu\text{C}$ is 40 mm directly *above* charge S = $+5.0 \mu\text{C}$. Charge R = $+4.0 \mu\text{C}$ is 30 mm directly to the *right* of S. Calculate the magnitude and direction of the net force on S. [6]

QUESTION 17 · POTENTIAL, FIELD AND WORK [9 MARKS]

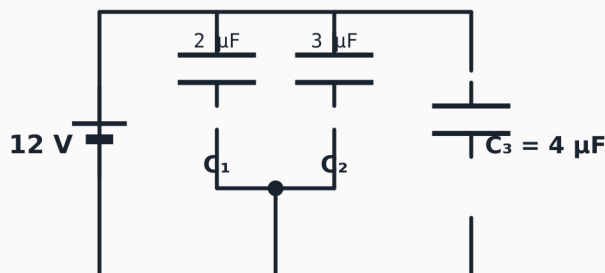
(a) Distinguish between electric potential and electric potential energy. [2] **(b)** A point charge of $+6.0 \mu\text{C}$ sits at the origin. Calculate:

1. the electric potential at $r = 0.30$ m; [2]
2. the electric field at $r = 0.30$ m; [2]
3. the work done in bringing a $-2.0 \mu\text{C}$ charge from infinity to that point. [2]

(c) Explain why no work is done moving a charge along an equipotential surface. [1]

QUESTION 18 · CAPACITOR NETWORK [9 MARKS]

A capacitor network is connected to a 24 V battery. $C_1 = 4.0 \mu\text{F}$ and $C_2 = 12.0 \mu\text{F}$ are in *parallel* with each other. That combination is in *series* with $C_3 = 8.0 \mu\text{F}$.



C_1 and C_2 are in parallel; that pair is in series with C_3

C_1 and C_2 add in parallel to give $5 \mu\text{F}$. That is then in series with C_3 .

(a) Calculate the equivalent capacitance of the network. [3] **(b)** Calculate the total charge drawn from the battery. [2] **(c)** Calculate the total energy stored. [2] **(d)** State what happens to the equivalent capacitance if a dielectric of $\epsilon_r = 4$ is inserted into C_3 only, and explain briefly why a dielectric increases capacitance. [2]

QUESTION 19 · KIRCHHOFF'S LAWS [9 MARKS]

(a) State Kirchhoff's current law and Kirchhoff's voltage law, and name the conservation principle each expresses. [3] **(b)** In a two-loop circuit, a 40 V source drives current I_1 through a 5Ω resistor. At a junction the current splits into I_2 (through a 10Ω resistor) and I_3 (through a 15Ω resistor). The two branches rejoin. The circuit equations are:

$$(1) \quad 5 I_1 + 10 I_2 = 40$$

$$(2) \quad 10 I_2 = 15 I_3$$

$$(3) \quad I_1 = I_2 + I_3$$

Solve for I_1 , I_2 and I_3 , and hence find the p.d. across the 10Ω resistor. [6]

QUESTION 20 · INDUCTION AND HELICAL MOTION [9 MARKS]

(a) State Faraday's second law and Lenz's law. Explain why Lenz's law must describe an opposition. [3] **(b)** A proton ($m = 1.67 \times 10^{-27} \text{ kg}$, $q = 1.6 \times 10^{-19} \text{ C}$) enters a uniform magnetic field of 0.40 T at an angle of 30° to the field, with a speed of $2.0 \times 10^6 \text{ m/s}$.

1. Calculate the magnetic force on the proton. [2]
2. Explain why its path is helical, and calculate the radius of the helix. [2]
3. Calculate the pitch of the helix. [2]

M.2 · WORKED ANSWER KEY

Mock exam: every answer independently computed

Mark yourself honestly. If you lost a mark, go to the section referenced, redo its checkpoint, and only then move on.

Section A

Q	ANSWER	WORKING	§
1	b. 4.0 C	$Q = ne = (2.5 \times 10^{19})(1.6 \times 10^{-19}) = 4.0 \text{ C}$	1.1
2	c. zero	LIKE charges, so at the midpoint the two fields are equal and opposite and cancel. A null point. Compare Test 1 Q8, where the charges were opposite and the fields added.	1.2
3	d. 8E	E is proportional to Q/r^2 . Top $\times 2$; bottom $\times (1/2)^2 = 1/4$. So $E \times 2 \div 1/4 = 8E$.	F.7
4	b. -5.6×10^5	$\Phi = Q/\epsilon_0 = (-5.0 \times 10^{-6})/(8.85 \times 10^{-12})$. A negative charge gives negative (inward) flux. The radius is irrelevant.	1.4
5	b. 10 V/m	$ E = \sqrt{(6^2 + (-8)^2)} = \sqrt{(36+64)} = \sqrt{100} = 10$. The minus squares away. This is the exact method Test 1 Q9 needed.	F.3
6	b. 1.0 μF	Series capacitors use reciprocals: $1/C = 1/2 + 1/3 + 1/6 = 1$, so $C = 1.0 \mu\text{F}$. Smaller than the smallest. Correct.	2.2
7	a. 10 μF	$C = 2W/V^2 = 2(8.0 \times 10^{-3})/40^2 = 0.016/1600 = 1.0 \times 10^{-5} \text{ F}$.	2.2
8	c. 4 Ω	Product over sum: $(6 \times 12)/(6+12) = 72/18 = 4 \Omega$. Smaller than the smallest. Correct.	3.2
9	c. 40 V	Voltage divider: $V = [40/(20+40)] \times 60 = 40 \text{ V}$. Its own R goes on top.	3.3
10	c. 6.0 mA	Current divider: put the other R on top. $I = [6.0/(3.0+6.0)] \times 9.0 = 6.0 \text{ mA}$. The smaller resistor takes more current. Correct.	3.3
11	c. zero	Parallel to B, so $\theta = 0^\circ$, $\sin 0 = 0$, and $F = 0$.	4.1
12	a. $1.6 \times 10^{-5} \text{ T}$	$B = \mu_0 I/(2\pi r) = (4\pi \times 10^{-7})(8)/[2\pi(0.10)]$. Cancel π : $(4 \times 10^{-7})(8)/0.2 = 1.6 \times 10^{-5}$	4.2
13	c. 200 V	$\epsilon = N \Delta\Phi/\Delta t = 500(0.020)/0.050 = 200 \text{ V}$.	4.3
14	c. 9.0 V	$V = E - Ir = 12 - (1.5)(2.0) = 9.0 \text{ V}$.	4.4
15	c. energy	If the induced current assisted the change you would get free energy. It must oppose.	4.3

△ THE TWO MIRRORS IN SECTION A

Q2 is the reverse of Test 1 Q8. There, two *opposite* charges made the fields **add** at the midpoint. Here, two *like* charges make them **cancel** to zero. If you answered $3.6 \times 10^6 \text{ N/C}$, you pattern-matched the old question instead of thinking. Sketch the arrows every time. **Q5 is the question you were denied in Test 1 Q9**, where the vector was missing from the paper. Here it is supplied, so you finally get to practise it: square, add, root. The minus sign on the 8 disappears when squared.

Section B

Q16 · COULOMB'S LAW AND 2-D SUPERPOSITION

9 MARKS

Recap. (a) State Coulomb's law and define every symbol. [3] (b) $P = -3.0 \mu\text{C}$ is 40 mm *above* $S = +5.0 \mu\text{C}$; $R = +4.0 \mu\text{C}$ is 30 mm to the *right* of S . Find the magnitude and direction of the net force on S . [6]

- 1 (a) [3 marks]** The force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them, acting along the line joining them. $F = kq_1q_2/r^2$, where F is force (N), q are the charges (C), r is the separation (m), and $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$.

One mark for the statement, one for the equation, one for defining all symbols.

- 2 (b) [6 marks]** Force from P on S : $F = (8.99 \times 10^9)(3.0 \times 10^{-6})(5.0 \times 10^{-6}) / (0.040)^2$.

Convert 40 mm to 0.040 m, then square it: 1.6×10^{-3} .

- 3** $F_{P \text{ on } S} = \mathbf{84.3 \text{ N, directed UPWARD}}$ (toward P).

P is negative and S is positive, so they attract. S is pulled up toward P .

- 4** Force from R on S : $F = (8.99 \times 10^9)(4.0 \times 10^{-6})(5.0 \times 10^{-6}) / (0.030)^2$.

$(0.030)^2 = 9.0 \times 10^{-4}$.

- 5** $F_{R \text{ on } S} = \mathbf{199.8 \text{ N, directed LEFT}}$ (away from R).

R and S are both positive, so they repel. S is pushed away from R , that is, to the left.

- 6** The two forces are **perpendicular**, so combine with Pythagoras: $F_{\text{net}} = \sqrt{(84.3^2 + 199.8^2)} = \sqrt{(7106 + 39920)}$.

You cannot add them as numbers. They act at right angles. See Foundations F.3.

- 7** $F_{\text{net}} = \sqrt{47026} = \mathbf{216.9 \text{ N}}$.

- 8** Direction: $\tan \theta = 84.3/199.8 = 0.422$, so $\theta = 22.9^\circ$.

Measured above the leftward horizontal.

$F_{\text{net}} \approx \mathbf{217 \text{ N, directed up and to the left, at about } 22.9^\circ \text{ above the horizontal. Method: §1.2, §F.3}$

Independently recomputed: 84.28 N up, 199.78 N left, resultant 216.83 N at 22.87° . The whole question turns on recognising that *attraction pulls toward, repulsion pushes away*, and that perpendicular forces need Pythagoras rather than addition.

Recap. (a) Distinguish potential from potential energy. [2] (b) A $+6.0\ \mu\text{C}$ charge sits at the origin: find (i) the potential at $r = 0.30\ \text{m}$, (ii) the field there, (iii) the work done bringing a $-2.0\ \mu\text{C}$ charge from infinity to that point. [6] (c) Why is no work done along an equipotential? [1]

- 1 (a) [2] **Electric potential** is the work done per unit positive charge in bringing it from infinity to a point. It belongs to the *point in space* and exists whether or not a charge sits there. **Electric potential energy** is the energy a *particular charge* has by virtue of sitting at that point: $U = qV$.

Shelf height versus book on the shelf.

- 2 (b)(i) [2] $V = kQ/r = (8.99 \times 10^9)(6.0 \times 10^{-6})/0.30$.

Plain r , not r squared.

- 3 $V = 1.80 \times 10^5\ \text{V}$.

- 4 (b)(ii) [2] $E = kQ/r^2 = (8.99 \times 10^9)(6.0 \times 10^{-6})/(0.30)^2 = 5.394 \times 10^4/0.09$.

Now r is squared. This is the distinction the question is testing.

- 5 $E = 5.99 \times 10^5\ \text{N/C}$.

Note $E = V/r$ here: $1.80 \times 10^5/0.30 = 6.0 \times 10^5$. A useful check.

- 6 (b)(iii) [2] $W = qV = (-2.0 \times 10^{-6})(1.80 \times 10^5)$.

Keep the negative sign on the charge.

- 7 $W = -0.36\ \text{J}$.

Negative, because a negative charge is attracted toward the positive source. The field does the work for you; you do not have to push.

- 8 (c) [1] Because $W = qV$ and along an equipotential the potential difference V is **zero**, so $W = 0$. Equivalently, the force is perpendicular to the motion.

(i) $1.80 \times 10^5\ \text{V}$ · (ii) $5.99 \times 10^5\ \text{N/C}$ · (iii) $-0.36\ \text{J}$ Method: §1.3

The sign in (iii) carries a mark. Dropping it and writing $+0.36\ \text{J}$ loses it.

Recap. A 24 V battery drives $C_1 = 4.0 \mu\text{F}$ in parallel with $C_2 = 12.0 \mu\text{F}$, that pair in series with $C_3 = 8.0 \mu\text{F}$. Find (a) C_{eq} , (b) the total charge, (c) the energy stored, (d) the effect of a dielectric $\epsilon_r = 4$ in C_3 only, and why.

- 1 (a) [3] Parallel pair first, since capacitors in parallel add: $C_p = 4.0 + 12.0 = \mathbf{16.0 \mu\text{F}}$.

Collapse the innermost group first.

- 2 Now C_p is in series with $C_3 = 8.0 \mu\text{F}$. Two items, so product over sum: $C_{\text{eq}} = (16 \times 8) / (16 + 8) = 128/24$.

- 3 $C_{\text{eq}} = \mathbf{5.33 \mu\text{F}}$.

Sanity check: smaller than the smallest in the series pair (8.0 μF). Correct.

- 4 (b) [2] $Q = C_{\text{eq}}V = (5.33 \times 10^{-6})(24) = \mathbf{1.28 \times 10^{-4} \text{ C}} = 128 \mu\text{C}$.

- 5 (c) [2] $W = \frac{1}{2}CV^2 = \frac{1}{2}(5.33 \times 10^{-6})(24)^2 = \frac{1}{2}(5.33 \times 10^{-6})(576)$.

$24^2 = 576$. Square before multiplying.

- 6 $W = \mathbf{1.54 \times 10^{-3} \text{ J}} = 1.54 \text{ mJ}$.

- 7 (d) [2] A dielectric of $\epsilon_r = 4$ in C_3 raises it to $4 \times 8.0 = 32 \mu\text{F}$. Recomputing: $C_{\text{eq}} = (16 \times 32) / (16 + 32) = 512/48 = \mathbf{10.67 \mu\text{F}}$. So the equivalent capacitance **increases**, from 5.33 to 10.67 μF .

A dielectric always increases capacitance.

- 8 **Why:** the dielectric **polarises**. Its molecular dipoles align and partly cancel the field between the plates, so for the same stored charge the voltage falls. Since $C = Q/V$, a smaller V means a larger C .

(a) $5.33 \mu\text{F}$ · (b) $128 \mu\text{C}$ · (c) 1.54 mJ · (d) rises to $10.67 \mu\text{F}$ Method: §2.2

Independently recomputed: $C_{\text{eq}} = 5.333 \mu\text{F}$, $Q = 1.280 \times 10^{-4} \text{ C}$, $W = 1.536 \times 10^{-3} \text{ J}$, and with the dielectric $10.67 \mu\text{F}$. Note the network is the *same shape* as Test 1 Q2, with different numbers.

Recognising a topology you have already met is worth real time in the exam.

Recap. (a) State KCL and KVL and name the conservation principle each expresses. [3] (b) Given $5I_1 + 10I_2 = 40$, $10I_2 = 15I_3$, $I_1 = I_2 + I_3$: solve for the three currents and find the p.d. across the $10\ \Omega$. [6]

- 1 **(a) [3] KCL:** the algebraic sum of currents entering a junction equals the sum leaving it. It expresses **conservation of charge**. **KVL:** the algebraic sum of potential differences around any closed loop is zero. It expresses **conservation of energy**.
- 2 **(b) [6]** From equation (2): $10I_2 = 15I_3$, so $I_2 = 1.5I_3$.
Make one unknown the subject. Substitution, as in Foundations F.5.
- 3 Substitute into (3): $I_1 = I_2 + I_3 = 1.5I_3 + I_3 = 2.5I_3$.
Everything is now in terms of I_3 alone.
- 4 Substitute both into (1): $5(2.5I_3) + 10(1.5I_3) = 40$.
One equation, one unknown. This is the goal of every simultaneous system.
- 5 $12.5I_3 + 15I_3 = 27.5I_3 = 40$.
- 6 $I_3 = 40/27.5 = \mathbf{1.455\ A}$.
- 7 Back-substitute: $I_2 = 1.5(1.455) = \mathbf{2.182\ A}$, and $I_1 = 2.5(1.455) = \mathbf{3.636\ A}$.
Do not stop at one current. Go back and get them all.
- 8 p.d. across the $10\ \Omega$: $V = I_2R = (2.182)(10) = \mathbf{21.8\ V}$.

$I_1 = 3.64\ \text{A}$ · $I_2 = 2.18\ \text{A}$ · $I_3 = 1.45\ \text{A}$ · p.d. across $10\ \Omega = 21.8\ \text{V}$ Method: §3.4, §F.5

The check that proves it: substitute back into (1): $5(3.636) + 10(2.182) = 18.18 + 21.82 = 40.0\ \text{V}$. It balances exactly. And KCL holds: $2.182 + 1.455 = 3.637 = I_1$. Both checks pass. Also note the $15\ \Omega$ branch carries *less* current than the $10\ \Omega$ branch, as it must.

Recap. (a) State Faraday's second law and Lenz's law, and explain why Lenz's law must describe an opposition. [3] (b) A proton enters a 0.40 T field at 30° with speed 2.0×10^6 m/s: find (i) the magnetic force, (ii) why the path is helical and its radius, (iii) the pitch. [6]

- 1 **(a) [3] Faraday's second law:** the magnitude of the induced emf is directly proportional to the rate of change of magnetic flux, $\varepsilon = -N d\Phi/dt$. **Lenz's law:** the induced current flows in the direction that opposes the change in flux producing it.
- 2 **Why it must oppose:** if the induced current *assisted* the change, the flux would grow, inducing more current, which would grow the flux further, without limit. You would get energy from nothing. Lenz's law is therefore forced by **conservation of energy**, and it is why you must do work to turn a generator.
- 3 **(b)(i) [2]** $F = qvB \sin \theta = (1.6 \times 10^{-19})(2.0 \times 10^6)(0.40)(\sin 30^\circ)$.
 $\sin 30^\circ = 0.5$ exactly.
- 4 $qvB = 1.28 \times 10^{-13}$, times 0.5 gives $F = 6.4 \times 10^{-14}$ N.
- 5 **(b)(ii) [2]** Resolve the velocity. The **perpendicular** component $v \sin 30^\circ$ feels a force always at right angles to the motion, producing **circular** motion. The **parallel** component $v \cos 30^\circ$ feels *no* force, since $\sin 0 = 0$, so it continues unchanged, carrying the proton steadily along the field. A circle plus a steady drift is a **helix**.
- 6 Radius uses only the perpendicular component: $v_{\text{perp}} = (2.0 \times 10^6)(0.5) = 1.0 \times 10^6$ m/s.
Sine for the radius. Using the full v is the classic error.
- 7 $r = mv_{\text{perp}}/(qB) = (1.67 \times 10^{-27})(1.0 \times 10^6) / [(1.6 \times 10^{-19})(0.40)] = (1.67 \times 10^{-21}) / (6.4 \times 10^{-20}) = 2.61 \times 10^{-2}$ m.
- 8 **(b)(iii) [2]** Period: $T = 2\pi m/(qB) = 2\pi(1.67 \times 10^{-27})/(6.4 \times 10^{-20}) = 1.64 \times 10^{-7}$ s.
- 9 Parallel speed: $v_{\text{par}} = (2.0 \times 10^6)(\cos 30^\circ) = (2.0 \times 10^6)(0.866) = 1.732 \times 10^6$ m/s. **Cosine for the pitch.**
- 10 Pitch = $v_{\text{par}} \times T = (1.732 \times 10^6)(1.64 \times 10^{-7}) = 0.284$ m.

(i) $F = 6.4 \times 10^{-14}$ N · (ii) helical; $r = 2.61 \times 10^{-2}$ m · (iii) pitch = 0.284 m Method: §4.1

Independently recomputed: $F = 6.400 \times 10^{-14}$ N, $r = 0.02610$ m, pitch = 0.2840 m. Remember the rule that unlocks every helix question: **sine for the radius, cosine for the pitch.**

SPACED-REPETITION CHECKPOINT

Close the book. From memory:

- ▶ Q2: when do fields at a midpoint *cancel*, and when do they *add*?
- ▶ Q16: two forces at right angles. How do you combine them?
- ▶ Q19: what is the first move when solving three simultaneous circuit equations?
- ▶ Q20: sine or cosine for the radius of a helix? And for the pitch?

ANSWER

Fields **cancel** at the midpoint between *like* charges and **add** between *opposite* ones. Perpendicular forces combine by **Pythagoras**, never by adding. For simultaneous equations, make one unknown the subject and substitute to reduce to a single unknown. For a helix, **sine gives the radius, cosine gives the pitch**. **Revisit:** 1 day, 3 days, 1 week, then before the exam.

PHY121 practice examination

Instructions. Two hours. Answer *all* questions in Section A and *all* questions in Section B. Section A carries 20 marks, Section B carries 40 marks. Constants are given on the formula sheet in Part IX.

Section A: multiple choice and short answer (20 marks)

SECTION A · TWO MARKS EACH

1. Charge is quantised. If a body carries a charge of 8.0×10^{-19} C, how many elementary charges does it carry?
2. Two point charges are separated by distance r . If the separation is *tripled*, by what factor does the force between them change?
3. State the value of the electric field inside a hollow conducting sphere carrying charge Q , and justify your answer in one sentence.
4. Three capacitors of $2 \mu\text{F}$, $4 \mu\text{F}$ and $8 \mu\text{F}$ are connected in *series*. Without full calculation, state whether the equivalent capacitance is greater than $8 \mu\text{F}$, between 2 and $8 \mu\text{F}$, or less than $2 \mu\text{F}$. Justify.
5. A wire is stretched to twice its original length while its cross-sectional area halves. By what factor does its resistance change?
6. State which conservation law is expressed by Kirchhoff's current law.
7. A proton travels exactly parallel to a uniform magnetic field. What is the magnitude of the magnetic force on it, and why?
8. A bar magnet is held stationary inside a coil of 500 turns. What emf is induced? Explain in one sentence.
9. Distinguish, in one sentence each, between the emf of a cell and its terminal voltage.
10. State Lenz's law, and name the conservation principle from which it follows.

Section B: structured questions (40 marks)

QUESTION 11 · ELECTROSTATICS (10 MARKS)

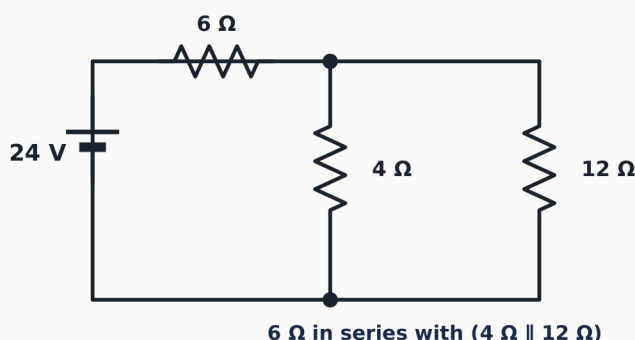
(a) State Coulomb's law and define every symbol. (3 marks) (b) Two point charges, $q_1 = +3.0 \mu\text{C}$ and $q_2 = -5.0 \mu\text{C}$, are placed 20 cm apart in a vacuum. Calculate the magnitude of the force between them and state whether it is attractive or repulsive. (4 marks) (c) Calculate the electric potential at the midpoint between the two charges. (3 marks)

QUESTION 12 · CAPACITANCE (10 MARKS)

Recap of the question. (a) Parallel plates of area 0.02 m^2 separated by 1.0 mm of air: find C . (b) A dielectric of $\epsilon_r = 5$ fills the gap: find the new C . (c) Charged to 12 V: find the charge and energy stored. (d) Why does a dielectric increase capacitance? (a) A parallel plate capacitor has plates of area 0.02 m^2 separated by 1.0 mm of air. Calculate its capacitance. (3 marks) (b) A dielectric of relative permittivity 5 is now inserted, completely filling the gap. Calculate the new capacitance. (2 marks) (c) The capacitor in (b) is charged to 12 V. Calculate the charge stored and the energy stored. (4 marks) (d) State one reason why the dielectric increases the capacitance. (1 mark)

QUESTION 13 · CIRCUIT ANALYSIS (10 MARKS)

Recap of the question. A 24 V battery of negligible internal resistance drives a $6\ \Omega$ resistor in series with a parallel combination of $4\ \Omega$ and $12\ \Omega$. Find (a) the total resistance, (b) the total current, and (c) the current through the $4\ \Omega$. (a) A 24 V battery of negligible internal resistance is connected to a $6\ \Omega$ resistor in series with a parallel combination of a $4\ \Omega$ and a $12\ \Omega$ resistor. Calculate the total resistance of the circuit. (4 marks)



Practice exam Q13: collapse the parallel pair first, then add the series $6\ \Omega$.

(b) Calculate the total current drawn from the battery. (2 marks) (c) Calculate the current through the $4\ \Omega$ resistor. (4 marks)

QUESTION 14 · MAGNETISM AND INDUCTION (10 MARKS)

Recap of the question. (a) An electron enters a $0.20\ \text{T}$ field at right angles at $3.0 \times 10^6\ \text{m/s}$: find the force and the radius. (b) Explain why the field does no work. (c) A 150-turn coil sees its flux change from $0.05\ \text{Wb}$ to $0.01\ \text{Wb}$ in $0.20\ \text{s}$: find the induced emf. (a) An electron enters a magnetic field of $0.20\ \text{T}$ at right angles with a speed of $3.0 \times 10^6\ \text{m/s}$. Calculate the magnetic force acting on it and the radius of its circular path. (4 marks) (b) Explain why the magnetic field does no work on the electron, and state what quantity therefore remains constant. (3 marks) (c) A coil of 150 turns experiences a uniform change of magnetic flux from $0.05\ \text{Wb}$ to $0.01\ \text{Wb}$ in $0.20\ \text{s}$. Calculate the magnitude of the induced emf. (3 marks)

M.4 · SECOND PAPER

Worked answer key

Mark yourself honestly. Where you went wrong, go back to the unit named and reread it before moving on.

Section A

Q	ANSWER AND REASONING	REVISE
1	$n = q/e = (8.0 \times 10^{-19})/(1.602 \times 10^{-19}) = 5$ elementary charges. Charge is quantised, so n must be a whole number.	Unit 1.1
2	Force falls to one ninth. F is proportional to $1/r^2$, and $3^2 = 9$. Not one third: the inverse square is the whole point.	Unit 1.2
3	$E = 0$. A Gaussian surface drawn inside the cavity encloses no charge, since all charge on a conductor resides on its outer surface. No charge enclosed means no flux, so no field.	Unit 1.4

Q	ANSWER AND REASONING	REVISE
4	Less than 2 μF . A series combination of capacitors is always smaller than the smallest member. (The actual value is 1.14 μF .)	Unit 2.2
5	Four times. $R = \rho L/A$. Doubling L doubles R; halving A doubles it again. Both push the same way, so the effects multiply.	Unit 3.1
6	Conservation of charge. Charge cannot accumulate at or vanish from a junction, so current in equals current out.	Unit 3.4
7	Zero. $F = qvB \sin \theta$ and $\theta = 0$, so $\sin \theta = 0$. A magnetic field acts only on the component of velocity across it.	Unit 4.1
8	Zero. The flux is not changing, so $d\Phi/dt = 0$ and Faraday's law gives no emf. Size of flux is irrelevant; only its rate of change matters.	Unit 4.3
9	emf is the potential difference of the source when no current flows. Terminal voltage is the voltage measured across the terminals while current flows, and is always less, because $V = E - Ir$.	Unit 4.4
10	The induced current flows in the direction that opposes the change in flux that produced it. It follows from conservation of energy.	Unit 4.3

Section B

QUESTION 11 · ELECTROSTATICS

(a) The force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them: $F = kq_1q_2/r^2$, where F is force in N, q are the charges in C, r is the separation in m, and $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$.

1 (b) Convert: $r = 20 \text{ cm} = \mathbf{0.20 \text{ m}}$, $q_1 = 3.0 \times 10^{-6} \text{ C}$, $q_2 = 5.0 \times 10^{-6} \text{ C}$ (magnitudes).

The centimetre conversion. Always the first thing to do.

2 $F = (8.99 \times 10^9)(3.0 \times 10^{-6})(5.0 \times 10^{-6}) / (0.20)^2$. Numerator: $(8.99 \times 10^9)(1.5 \times 10^{-11}) = 1.349 \times 10^{-1}$.

The two charges multiply to 1.5×10^{-11} .

3 Denominator: $(0.20)^2 = 0.04$. So $F = 0.1349 / 0.04 = \mathbf{3.37 \text{ N}}$.

0.2 squared is 0.04, not 0.4.

4 The charges have *opposite* signs, so the force is **attractive**.

5 (c) Potential is a *scalar*, so simply add the two contributions with their signs. The midpoint is 0.10 m from each.

No vectors here. This is why potential is easier than field.

6 $V_1 = kq_1/r = (8.99 \times 10^9)(+3.0 \times 10^{-6})/0.10 = +2.697 \times 10^5 \text{ V}$.

7 $V_2 = (8.99 \times 10^9)(-5.0 \times 10^{-6})/0.10 = -4.495 \times 10^5 \text{ V}$.

Keep the negative sign. This is where the marks are.

8 $V = V_1 + V_2 = 2.697 \times 10^5 - 4.495 \times 10^5$.

(b) $F \approx \mathbf{3.37 \text{ N}}$, **attractive** · (c) $V \approx \mathbf{-1.80 \times 10^5 \text{ V}}$

Common error: using magnitudes in part (c). Potential carries a sign, and here the negative charge dominates because it is larger, so the midpoint potential is negative. Field would need vector addition; potential does not.

QUESTION 12 · CAPACITANCE

Recap. (a) Plates of area 0.02 m^2 , separated by 1.0 mm of air: find C . (b) A dielectric of $\epsilon_r = 5$ fills the gap: find the new C . (c) Charged to 12 V : find the charge and the energy stored. (d) Why does a dielectric increase capacitance?

1 (a) $C_0 = \epsilon_0 A/d$, with $d = 1.0 \text{ mm} = 1.0 \times 10^{-3} \text{ m}$.

Millimetres to metres. A factor of 1000.

2 $C_0 = (8.85 \times 10^{-12})(0.02) / (1.0 \times 10^{-3}) = (1.77 \times 10^{-13}) / (10^{-3}) = 1.77 \times 10^{-10} \text{ F}$.

3 (b) With the dielectric: $C = \epsilon_r C_0 = 5 \times 1.77 \times 10^{-10} = 8.85 \times 10^{-10} \text{ F}$.

Bigger, as a dielectric must always make it.

4 (c) Charge: $Q = CV = (8.85 \times 10^{-10})(12) = 1.06 \times 10^{-8} \text{ C}$.

5 Energy: $W = \frac{1}{2}CV^2 = \frac{1}{2}(8.85 \times 10^{-10})(144) = 6.37 \times 10^{-8} \text{ J}$.

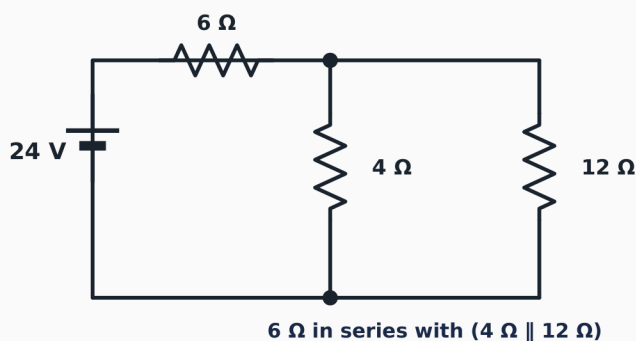
$12^2 = 144$. Square the voltage before multiplying.

6 (d) The dielectric **polarises**: its molecular dipoles align and partially cancel the field between the plates. The voltage therefore falls for the same stored charge, and since $C = Q/V$, a smaller V means a larger C . (a) $1.77 \times 10^{-10} \text{ F}$ · (b) $8.85 \times 10^{-10} \text{ F}$ · (c) $Q = 1.06 \times 10^{-8} \text{ C}$, $W = 6.37 \times 10^{-8} \text{ J}$

Cross-check the energy using $W = \frac{1}{2}QV = \frac{1}{2}(1.06 \times 10^{-8})(12) = 6.37 \times 10^{-8} \text{ J}$. The two forms agree, as they must.

QUESTION 13 · CIRCUIT ANALYSIS

Recap. A 24 V battery drives a 6 Ω resistor in series with a parallel combination of 4 Ω and 12 Ω . Find (a) the total resistance, (b) the total current, (c) the current through the 4 Ω .



Practice exam Q13: collapse the parallel pair first, then add the series 6 Ω .

- 1 (a)** Collapse the parallel pair first. Two items, so product over sum: $R_p = (4 \times 12)/(4 + 12) = 48/16 = \mathbf{3 \Omega}$.
Check: 3 Ω is less than 4 Ω , the smaller of the pair. Correct.
- 2** That 3 Ω is in *series* with the 6 Ω , so they add: $R_{\text{total}} = 6 + 3 = \mathbf{9 \Omega}$.
- 3 (b)** Total current: $I = V/R = 24/9 = \mathbf{2.67 \text{ A}}$.
- 4 (c)** All 2.67 A passes through the 6 Ω resistor, so the voltage dropped across it is $V = IR = (2.67)(6) = 16 \text{ V}$.
The 6 Ω is in series, so it carries the whole current.
- 5** The voltage left for the parallel section is $24 - 16 = \mathbf{8 \text{ V}}$.
Or directly: $V = I \times R_p = 2.67 \times 3 = 8 \text{ V}$. Same answer.
- 6** Both parallel branches share that 8 V, so the current in the 4 Ω is $I = V/R = 8/4 = \mathbf{2.0 \text{ A}}$.

(a) 9 Ω · (b) 2.67 A · (c) 2.0 A through the 4 Ω

Check: the 12 Ω branch carries $8/12 = 0.67 \text{ A}$. The two branch currents are $2.0 + 0.67 = 2.67 \text{ A}$, which equals the total current. It balances, and the smaller resistor carries the larger current, exactly as it must.

QUESTION 14 · MAGNETISM AND INDUCTION

Recap. (a) An electron enters a 0.20 T field at right angles at 3.0×10^6 m/s; find the force and radius. (b) Why does the field do no work? (c) A 150-turn coil: flux 0.05 to 0.01 Wb in 0.20 s. Find the induced emf.

- 1 (a) At right angles, $\sin 90^\circ = 1$, so $F = qvB = (1.6 \times 10^{-19})(3.0 \times 10^6)(0.20)$.

Fronts: $1.6 \times 3 = 4.8$, times 0.2 gives 0.96. Exponents: $-19 + 6 = -13$.

- 2 $F = 9.6 \times 10^{-14}$ N.

- 3 Radius: $r = mv/(qB) = (9.11 \times 10^{-31})(3.0 \times 10^6) / [(1.6 \times 10^{-19})(0.20)]$.

Electron mass here, not proton. Numerator: 2.733×10^{-24} . Denominator: 3.2×10^{-20} .

- 4 $r = 8.54 \times 10^{-5}$ m.

- 5 (b) The magnetic force is always **perpendicular** to the velocity. Work is $W = Fd \cos \theta$, and here $\theta = 90^\circ$, so $\cos 90^\circ = 0$ and $W = 0$. Therefore the **kinetic energy**, and hence the **speed**, remains constant. Only the direction changes.

- 6 (c) $\Delta\Phi = 0.05 - 0.01 = 0.04$ Wb. $\varepsilon = N \Delta\Phi/\Delta t = (150)(0.04)/(0.20) = 6/0.20$.

The change in flux, not either value alone.

- 7 $\varepsilon = 30$ V in magnitude.

(a) $F = 9.6 \times 10^{-14}$ N, $r \approx 8.54 \times 10^{-5}$ m · (b) No work, so speed and kinetic energy are constant · (c) 30 V

Note: in (c) the minus sign from Lenz's law is dropped because the question asks for the *magnitude*.

Mention that the sign indicates opposition to the change, and you will secure the mark.

M.5

Module 5 mock: Maxwell & EM waves

Six one-mark questions and one structured question, in the style of your CBTs. Constants: $k = 8.99 \times 10^9$, $\varepsilon_0 = 8.85 \times 10^{-12}$, $\mu_0 = 4\pi \times 10^{-7}$, $c = 3.0 \times 10^8$.

Section A · multiple choice (6 marks)

Q1. Maxwell completed classical electromagnetism by adding a term to which law?

a. Gauss's law · b. Faraday's law · c. Ampere's law · d. Lenz's law · e. Coulomb's law

Q2. A capacitor is charged by a conduction current of 0.50 A. The displacement current between its plates is

a. 0 A · b. 0.25 A · c. 0.50 A · d. 1.0 A · e. depends on plate area

Q3. An electromagnetic wave in vacuum has wavelength 3.0 m. Its frequency is

a. 1.0×10^8 Hz · b. 9.0×10^8 Hz · c. 1.0×10^9 Hz · d. 3.0×10^8 Hz · e. 1.0×10^{-8} Hz

Q4. An EM wave has electric field amplitude 200 V/m. Its magnetic field amplitude is

a. 6.0×10^{10} T · b. 6.67×10^{-7} T · c. 2.0×10^2 T · d. 1.5×10^6 T · e. 3.0×10^8 T

Q5. In vacuum, an X-ray travels at a speed that is

a. faster than a radio wave · b. slower than a radio wave · c. equal to a radio wave · d. equal to sound · e. zero

Q6. The energy density of an EM wave at a point where $E = 100$ V/m (use $u = \varepsilon_0 E^2$) is

a. 8.85×10^{-8} J/m³ · b. 8.85×10^{-10} J/m³ · c. 1.77×10^{-7} J/m³ · d. 8.85×10^{-12} J/m³ · e. 100 J/m³

Section B · structured question (9 marks)

- Q7.** (a) State the definition of the displacement current, and write the Ampere-Maxwell law. [3]
 (b) A laser emits 10 mW of power into a beam of cross-sectional area $2.0 \times 10^{-6} \text{ m}^2$. Calculate the intensity, the electric field amplitude E_0 , and the magnetic field amplitude B_0 of the beam. [5]
 (c) State the direction of the Poynting vector relative to the direction the wave travels. [1]

Worked answer key · every answer independently computed

Q	ANSWER	WORKING	§
1	c. Ampere's law	The displacement current term $\epsilon_0 d\Phi_E/dt$ was added to Ampere's law.	5.1
2	c. 0.50 A	Charge conservation forces $I_d = I_{\text{conduction}}$, independent of area.	5.1
3	a. $1.0 \times 10^8 \text{ Hz}$	$f = c/\lambda = (3.0 \times 10^8)/3.0 = 1.0 \times 10^8 \text{ Hz}$.	5.2
4	b. $6.67 \times 10^{-7} \text{ T}$	$B_0 = E_0/c = 200/(3.0 \times 10^8) = 6.67 \times 10^{-7} \text{ T}$.	5.2
5	c. equal to a radio wave	All EM waves travel at c in vacuum, whatever the frequency.	5.2
6	a. $8.85 \times 10^{-8} \text{ J/m}^3$	$u = \epsilon_0 E^2 = (8.85 \times 10^{-12})(100)^2 = 8.85 \times 10^{-8} \text{ J/m}^3$.	5.2

Q7 (a). The displacement current is $I_d = \epsilon_0 d\Phi_E/dt$, the magnetic effect of a changing electric flux. The Ampere-Maxwell law is $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0(I_{\text{enc}} + \epsilon_0 d\Phi_E/dt)$. [3]

- 1** Intensity: $I = P/A = (10 \times 10^{-3})/(2.0 \times 10^{-6}) = 5.0 \times 10^3 = 5000 \text{ W/m}^2$.
- 2** Electric field amplitude: $E_0 = \sqrt{2I/(\epsilon_0 c)} = \sqrt{(2(5000) / [(8.85 \times 10^{-12})(3.0 \times 10^8)])} = \sqrt{(3.77 \times 10^6)} = 1941 \text{ V/m}$.
- 3** Magnetic field amplitude: $B_0 = E_0/c = 1941/(3.0 \times 10^8) = 6.47 \times 10^{-6} \text{ T}$.

$$I = 5000 \text{ W/m}^2 \quad E_0 = 1941 \text{ V/m} \quad B_0 = 6.47 \times 10^{-6} \text{ T}$$

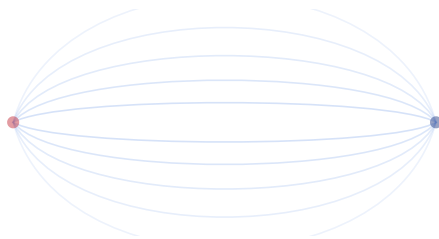
Q7 (c). The Poynting vector $\mathbf{S}$ points in the **same direction as the wave travels** (perpendicular to both $\mathbf{E}$ and $\mathbf{B}$). [1]

Now redo it: $I = 5000 \text{ W/m}^2$, $E_0 = 1940.7 \text{ V/m}$, $B_0 = 6.469 \times 10^{-6} \text{ T}$, all confirmed. If you lost a mark, return to §5.2, redo its checkpoint, then move on.

PART IX

Reference

The formula sheet, the memorise list, a general problem-solving method, and a glossary. Use the formula sheet while you practise; use the memorise list in the final week.



Complete formula and constant sheet

Constants

CONSTANT	SYMBOL	VALUE
Coulomb constant	k	$8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Permittivity of free space	ϵ_0	$8.85 \times 10^{-12} \text{ F m}^{-1}$
Permeability of free space	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Elementary charge	e	$1.602 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67 \times 10^{-27} \text{ kg}$

Module 1: Electrostatics

QUANTITY	FORMULA	NOTES
Quantisation of charge	$q = ne$	n is a whole number
Coulomb's law	$F = kq_1q_2/r^2$	Inverse square
Electric field	$E = F/q = kQ/r^2$	Vector. Unit N/C or V/m
Electric potential	$V = kQ/r$	Scalar. Note plain r
Potential energy	$U = kQq/r = qV$	Joules
Work done	$W = qV$	Charge times voltage
Field from potential	$E = -dV/dr$, or $E = V/d$	Uniform field between plates
Electric flux	$\Phi = EA \cos \theta$	θ from the normal
Gauss's law	$\Phi = Q_{\text{enc}}/\epsilon_0$	Enclosed charge only
Line charge	$E = \lambda/(2\pi\epsilon_0 r)$	Falls as $1/r$
Plane sheet	$E = \sigma/(2\epsilon_0)$	Independent of distance
Inside a hollow conductor	$E = 0$	No enclosed charge

Module 2: Circuits and materials

QUANTITY	FORMULA	NOTES
Dipole moment	$p = qd$	Unit C m
Torque on a dipole	$\tau = pE \sin \theta$	Zero when aligned
Dipole energy	$U = -pE \cos \theta$	Minimum when aligned
Capacitance	$C = Q/V$	Unit farad (F)
Sphere	$C = 4\pi\epsilon_0 R$	Geometry only
Parallel plate	$C = \epsilon_r \epsilon_0 A/d$	$\epsilon_r = 1$ in vacuum

QUANTITY	FORMULA	NOTES
Dielectric	$C = \epsilon_r C_0$	Always increases C
Capacitors in parallel	$C_{eq} = C_1 + C_2 + \dots$	They add
Capacitors in series	$1/C_{eq} = 1/C_1 + 1/C_2 + \dots$	Less than the smallest
Energy stored	$W = Q^2/2C = \frac{1}{2}CV^2 = \frac{1}{2}QV$	Three equivalent forms

Module 3: Electricity and circuit analysis

QUANTITY	FORMULA	NOTES
Electric current	$I = \Delta Q/\Delta t$	1 A = 1 C/s
Work by a battery	$W = (\Delta Q)\mathcal{E}$	Joules
Ohm's law	$V = IR$	1 Ω = 1 V/A
Resistivity	$R = \rho L/A$	Long and thin means high R
Resistors in series	$R_{eq} = R_1 + R_2 + \dots$	They add
Resistors in parallel	$1/R_{eq} = 1/R_1 + 1/R_2 + \dots$	Less than the smallest
Two in parallel	$R_{eq} = R_1 R_2 / (R_1 + R_2)$	Product over sum
Voltage divider	$V_1 = [R_1/R_T] \times V$	Wanted resistor on top
Current divider	$I_1 = [R_2/(R_1 + R_2)] \times I_T$	Other resistor on top
Kirchhoff's current law	$\sum I_{in} = \sum I_{out}$	Conservation of charge
Kirchhoff's voltage law	$\sum V = 0$ around a loop	Conservation of energy

Module 4: Magnetism and induction

QUANTITY	FORMULA	NOTES
Lorentz force	$F = qvB \sin \theta$	Zero if $v = 0$ or $\theta = 0$
Full Lorentz force	$F = q(E + v \times B)$	Both fields present
Radius of circular path	$r = mv/(qB)$	Use $v \sin \theta$ if at an angle
Cyclotron frequency	$f = qB/(2\pi m)$	Independent of speed
Period	$T = 2\pi m/(qB)$	Also independent of speed
Pitch of a helix	$p = (v \cos \theta) \times T$	Cos for pitch, sin for radius
Force on a conductor	$F = BIL \sin \theta$	Maximum at 90°
Biot-Savart law	$dB = (\mu_0/4\pi)(I dl \times \hat{r})/r^2$	General but laborious
Straight wire	$B = \mu_0 I/(2\pi r)$	Falls as $1/r$
Centre of a loop	$B = \mu_0 NI/(2R)$	N turns
Ampere's law	$\oint B \cdot dl = \mu_0 I_{enc}$	Use when symmetric
Solenoid	$B = \mu_0 nI$	n = turns per metre. Uniform inside
Toroid	$B = \mu_0 NI/(2\pi r)$	N = total turns

QUANTITY	FORMULA	NOTES
Magnetic flux	$\Phi = BA \cos \theta$	Unit weber (Wb)
Faraday's law	$\varepsilon = -N d\Phi/dt$	Rate of change matters
Motional emf	$\varepsilon = BLv$	Conductor moving through B
Transformer	$V_s/V_p = N_s/N_p$	AC only
emf and internal resistance	$E = I(R + r), V = E - Ir$	V is always less than E
Self-induced emf	$\varepsilon = -L dI/dt$	Unit 4.5. The minus sign is Lenz again
Energy in an inductor	$W = \frac{1}{2}LI^2$	Unit 4.5. Mirrors $\frac{1}{2}CV^2$ for a capacitor
Magnetic dipole moment	$m = NIA$	Unit 4.6. N turns, current I, area A
Torque on a current loop	$\tau = mB \sin \theta = NIAB \sin \theta$	Unit 4.6. Zero when m is along B

REFERENCE R.2

The must-memorise list

If you memorise nothing else, memorise these. Each carries the hook that makes it stick.

ITEM	THE HOOK
Like repels, unlike attracts	Opposites attract. Everything electrostatic starts here.
$q = ne$	Charge comes in coins, not in liquid. Never a fraction of an electron.
$F = kq_1q_2/r^2$	Inverse square. Triple the distance, force falls to a ninth.
$E = kQ/r^2$ but $V = kQ/r$	Field is squared, potential is not. Field is a vector, potential is a scalar.
Superposition	One at a time, then add. Works for forces, fields and potentials.
Flux out = charge in / ε_0	Gauss. External charges contribute nothing.
$E = 0$ inside a conductor	No charge inside means no field inside. The Faraday cage.
$C = Q/V$	Charge per volt. Fixed by geometry, not by Q or V.
$W = \frac{1}{2}CV^2$	The half is there because voltage climbed from zero as you charged it.
Capacitors add in parallel	Backwards from resistors. Remember one, flip for the other.
$V = IR$	Cover the one you want and the formula rearranges itself.
$R = \rho L/A$	Long and thin resists; short and fat does not.
Parallel R is less than the smallest	Free check on every parallel answer. If not, you forgot to flip the reciprocal.
KCL is charge, KVL is energy	Junctions conserve charge; loops conserve energy.
$F = qvB \sin \theta$	A stationary charge feels nothing: $v = 0$. Along the field, $\sin 0 = 0$.
Magnetic fields do no work	They steer but never speed up. Kinetic energy is constant.
$f = qB/2\pi m$	No v in it. The cyclotron principle.
$\varepsilon = -N d\Phi/dt$	CHANGE is the word. A steady flux induces nothing.

ITEM	THE HOOK
Lenz opposes	Conservation of energy. Nature always fights back.
$V = E - Ir$	Why headlights dim when you start the engine.

REFERENCE R.3

A general problem-solving method

● MUST-MEMORISE · THE SEVEN STEPS

1. **Read the question twice.** Underline what is *asked for*. Many marks are lost by answering a question that was not set.
2. **List what you know**, with symbols and units. Write the unknown with a question mark.
3. **Convert everything to SI immediately.** Centimetres to metres, milliamps to amps, microfarads to farads. Do this before anything else.
4. **Choose the formula** that contains what you know and what you want. If none does, you need an intermediate step.
5. **Rearrange in symbols first**, and substitute numbers only at the end.
6. **Compute**, keeping track of powers of ten separately from the front numbers.
7. **Check.** Is the size believable? Does it have the right unit? Does the sanity check apply (parallel less than smallest, voltages summing to the source)?

MEMORY HOOK: read, list, convert, choose, rearrange, compute, check. Step 3 and step 7 are the two that save the most marks, and they are the two that are most often skipped.

REFERENCE R.4

Glossary

TERM	MEANING
Ampere (A)	Unit of current. One coulomb per second.
Amperian loop	An imaginary closed path chosen to make Ampere's law easy.
Branch	A single element in a circuit, such as a resistor or a source.
Capacitance	Charge stored per volt applied. Unit: farad.
Conductor	A material in which charge moves freely.
Coulomb (C)	Unit of electric charge.
Cyclotron frequency	Revolutions per second of a charge in a magnetic field. Independent of speed.
Dielectric	An insulator between capacitor plates. Increases capacitance.
Dipole	Two equal and opposite charges separated by a small distance.
Doping	Adding impurities to a semiconductor to make it conduct.
Eddy current	Loops of current induced inside a solid conductor by changing flux.
emf	Energy per unit charge given by a source. Measured in volts, despite the name.
Equipotential	A surface on which the potential is the same everywhere.
Farad (F)	Unit of capacitance. One coulomb per volt.

TERM	MEANING
Gaussian surface	An imaginary closed surface chosen to make Gauss's law easy.
Insulator	A material in which charge is bound and cannot move freely.
Internal resistance	The resistance of a source itself. Causes terminal voltage to fall below emf.
Junction (node)	A point where two or more conductors meet.
Loop	A closed conducting path in a circuit.
Ohm (Ω)	Unit of resistance. One volt per amp.
Ohmic	Obeying Ohm's law, with resistance constant at fixed temperature.
Permittivity	How readily a material permits an electric field. Symbol ϵ .
Permeability	How readily a material permits a magnetic field. Symbol μ .
Polarisation	Creation of induced dipoles in an insulator by an applied field.
Quantisation	Charge exists only in whole multiples of e .
Resistivity	A material's intrinsic opposition to current. Symbol ρ . Unit $\Omega \text{ m}$.
Semiconductor	A material between conductor and insulator, whose resistance falls as it heats.
Short circuit	A path of zero resistance, which carries all the current.
Solenoid	A long coil of wire, producing a uniform field inside.
Superposition	Compute each contribution separately, then add.
Terminal voltage	The voltage across a source while current flows. Always less than the emf.
Tesla (T)	Unit of magnetic flux density.
Toroid	A solenoid bent into a ring.
Transformer	Two coils coupled by mutual induction. Changes voltage. AC only.
Weber (Wb)	Unit of magnetic flux. One tesla square metre.

REFERENCE R.1 · ADDITION

Module 5 formulas

QUANTITY	FORMULA	NOTES
Displacement current	$I_d = \epsilon_0 d\Phi_E/dt$	Effect of a changing E field
Ampere-Maxwell law	$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0(I_{\text{enc}} + \epsilon_0 d\Phi_E/dt)$	Old Ampere + new term
Gauss (E)	$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}}/\epsilon_0$	Charges make E
Gauss (B)	$\oint \mathbf{B} \cdot d\mathbf{A} = 0$	No monopoles
Faraday	$\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$	Changing B makes E
Speed of light	$c = 1/\sqrt{\mu_0\epsilon_0}$	$= 3.0 \times 10^8 \text{ m/s}$ in vacuum
Wave relations	$E = cB \cdot c = f\lambda$	Fields locked; speed $= f \times \lambda$
Plane wave	$E = E_0 \sin(kx - \omega t)$	$k = 2\pi/\lambda$, $\omega = 2\pi f$

QUANTITY	FORMULA	NOTES
Energy density	$u = \epsilon_0 E^2 = B^2/\mu_0$	Electric & magnetic halves equal
Poynting / intensity	$S = (1/\mu_0)E \times B$; $I = \frac{1}{2}\epsilon_0 c E_0^2$	Energy flow per unit area

● **Must-memorise · Module 5 additions to the list**

ITEM	THE HOOK
$I_d = \epsilon_0 d\Phi_E/dt$	A changing E counts as a current. Charge need not move.
Maxwell fixed Ampere	One added term; it makes light possible.
$c = 1/\sqrt{(\mu_0\epsilon_0)}$	Two lab constants give the speed of light. No optics needed.
$E = cB$	E is always the bigger number, by a factor of c.
EM waves are transverse	$E \perp B \perp$ direction of travel, all in phase.
Same spectrum, one speed	Radio to gamma: all light, all at c, differing only in f.